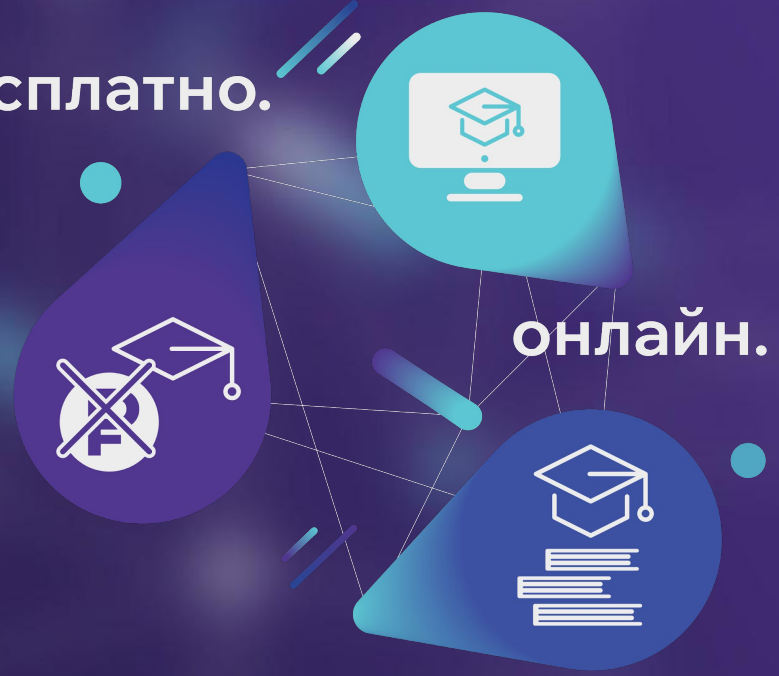
Deep Learning School



DLS



Nina Konovalova

@AfeliaN

@ReadingAI

Diffusion interpretability

What lives inside diffusion and how to understand it?

DLS

Plan: four steps into the model

04

Attention

`look`

where does the model look?

05

Steering

`push`

can we move it where we want?

06

SAEs

`name`

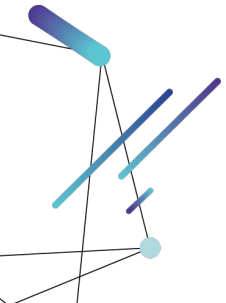
which features are inside?

07

Transcoders

`explain`

how are features connected?



DLS

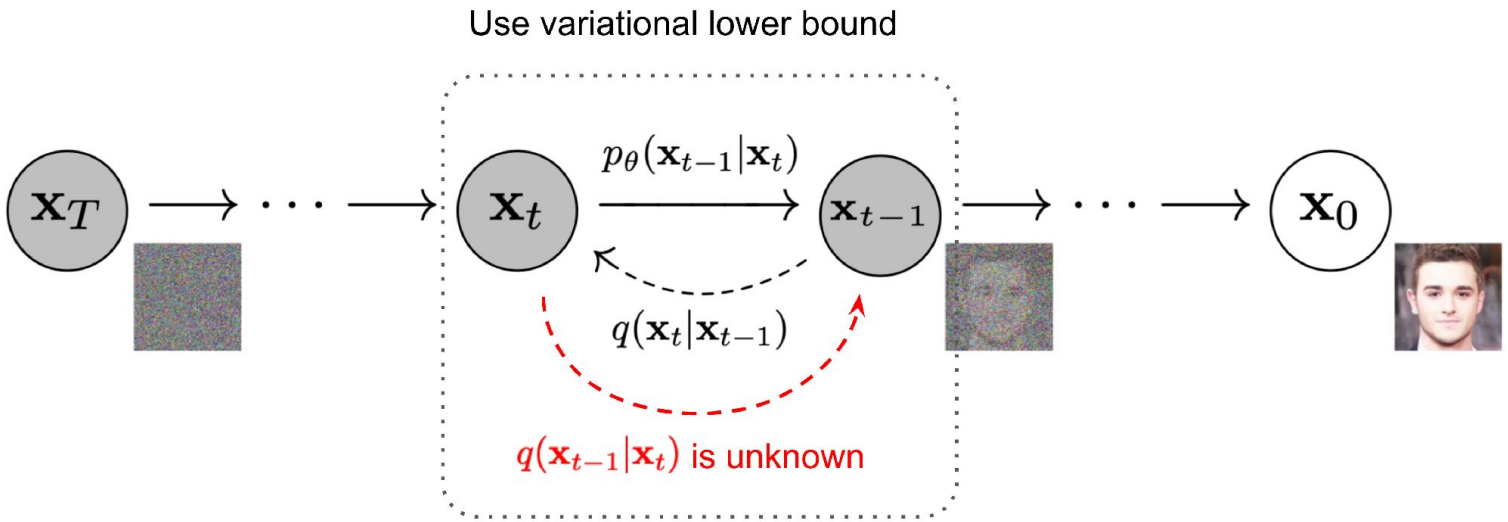
Recap & motivation

01 · why look inside

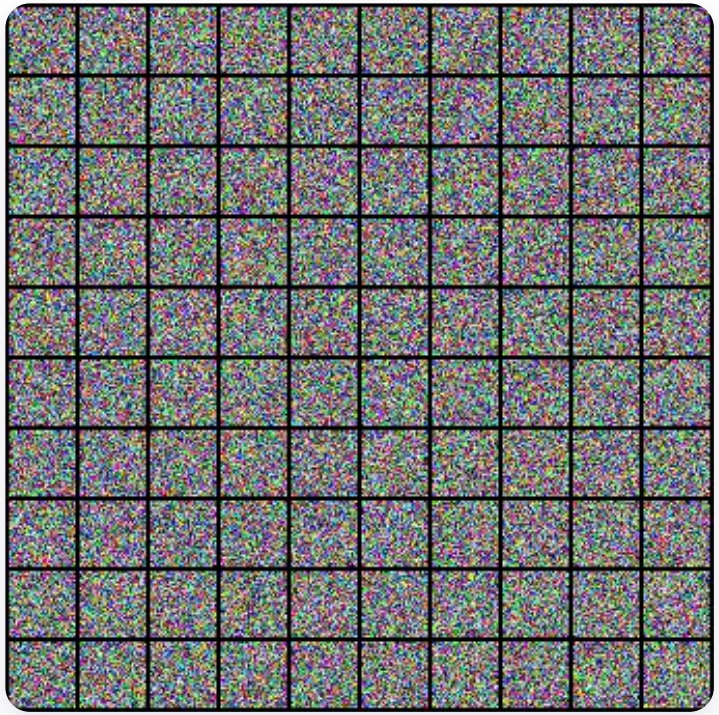
DDPM · 2020 →

DLS

2D diffusion



$$L_{simple,t} = \mathbb{E}_{x_0,\epsilon}\left[\|\epsilon_t - \epsilon_\theta(x_t, t)\|^2\right]$$



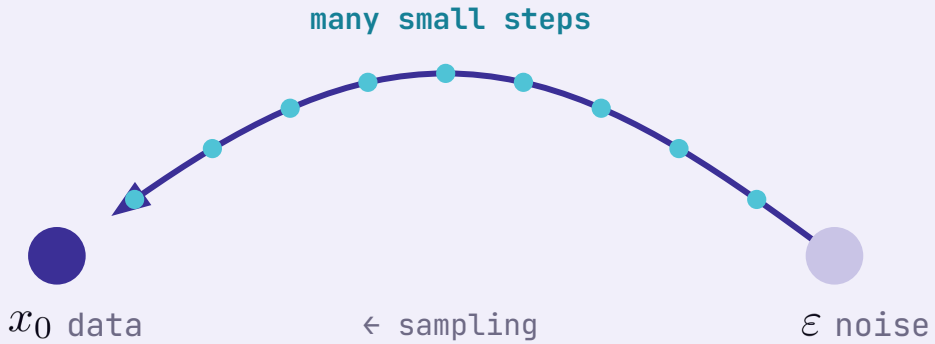
DDPM: learn to reverse a fixed noising process with a simple denoising loss

DLS

Recap: DDPM vs flow

DDPM

SDXL, SD 1.5



$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon$$

predicts $\varepsilon_\theta(x_t, t) \approx \varepsilon$

Flow

Flux, SD3



$$x_t = (1-t)\, x_0 + t\, \varepsilon$$

predicts $v_\theta(x_t, t) \approx \varepsilon - x_0$

Same game, noise → data: curved path + predict noise vs straight path + predict direction

DLS

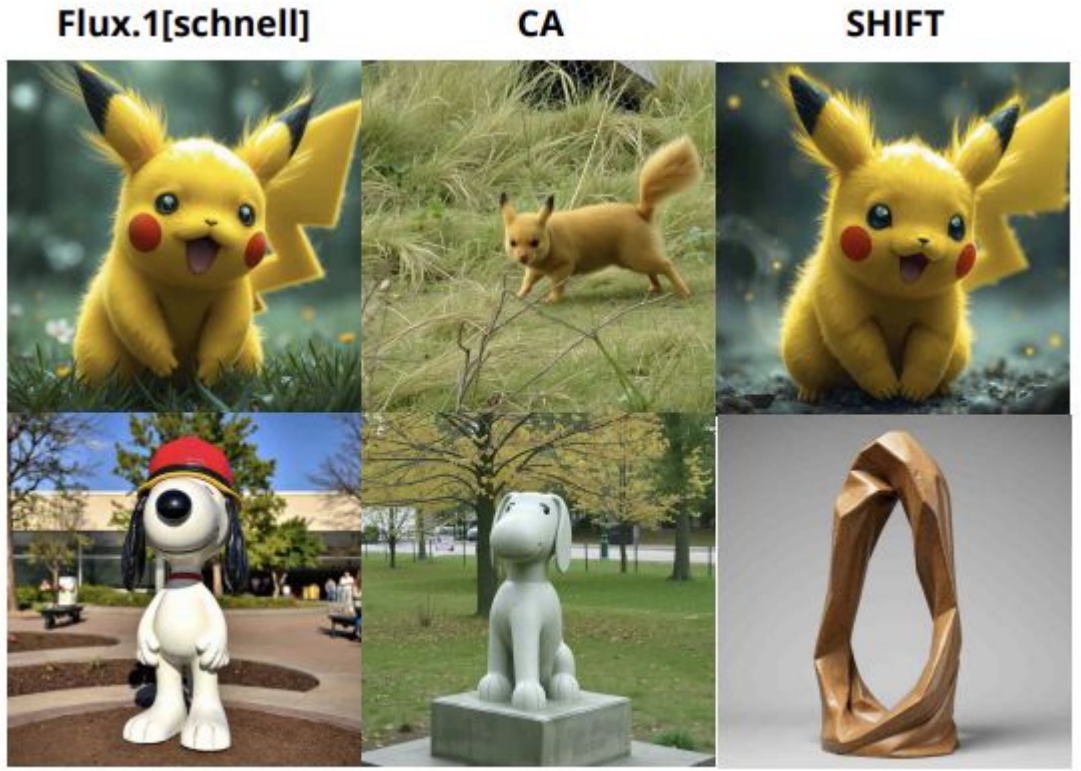
Why interpret? Safety

SAFETY

Erase concepts — and verify they are gone

- nudity, artist styles, characters
- adversarial prompts bring them back

Keep other concepts as in origin model



Interpretability question
Which features encode the concept?

Concept Sliders · 2024 →

Prompt-to-Prompt · 2022 →

DLS

Why interpret? Control

CONTROL

Precise edits without retraining

- continuous sliders: age, smile, weather
- change one object, keep the scene

Interpretability question

Which direction is "age"?



Beyond Surface Stats · 2023 →

DIFT · 2023 →

DLS

Why interpret? Science

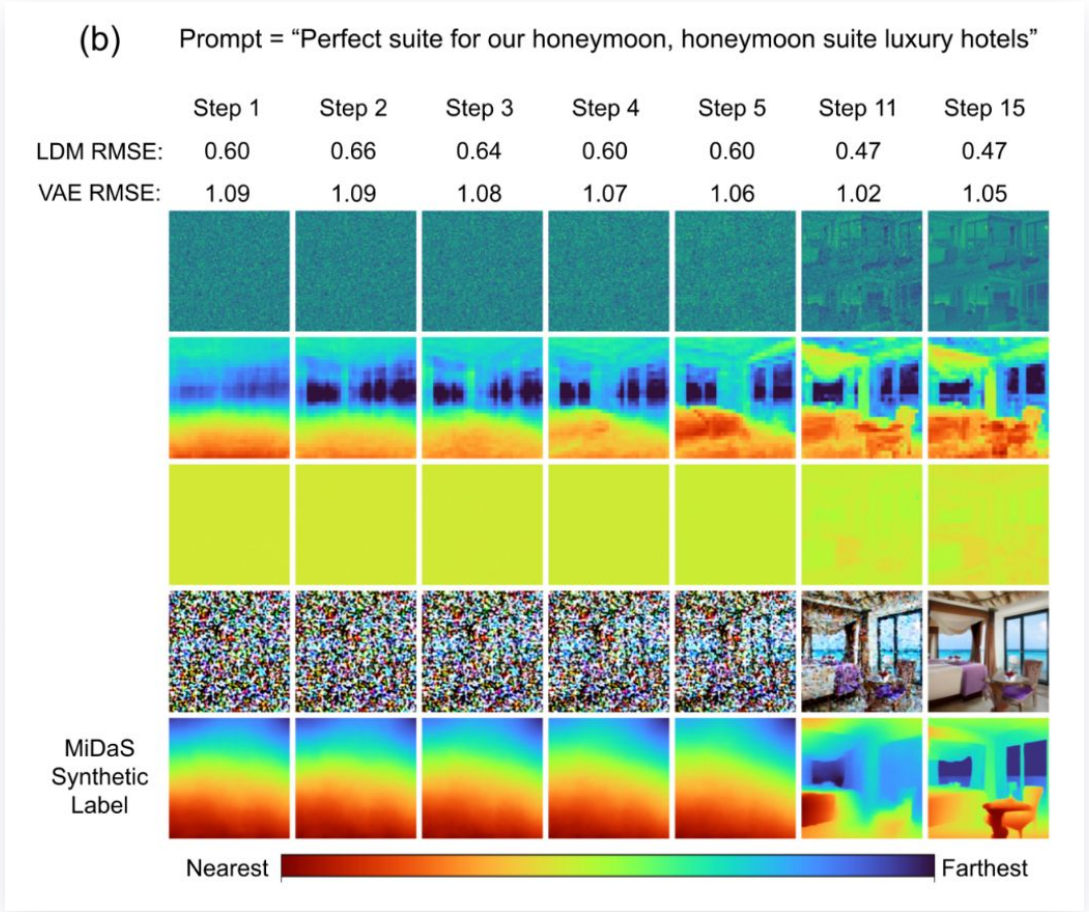
SCIENCE

What does the model know about the world?

- linear probes read depth from activations
- early — before the image is visible

Interpretability question

A 3D scene, or just pixels?



Diffract · 2026 →

Demystifying · 2025 →

DLS

Why interpret? Debugging

DEBUGGING

Why does generation fail?

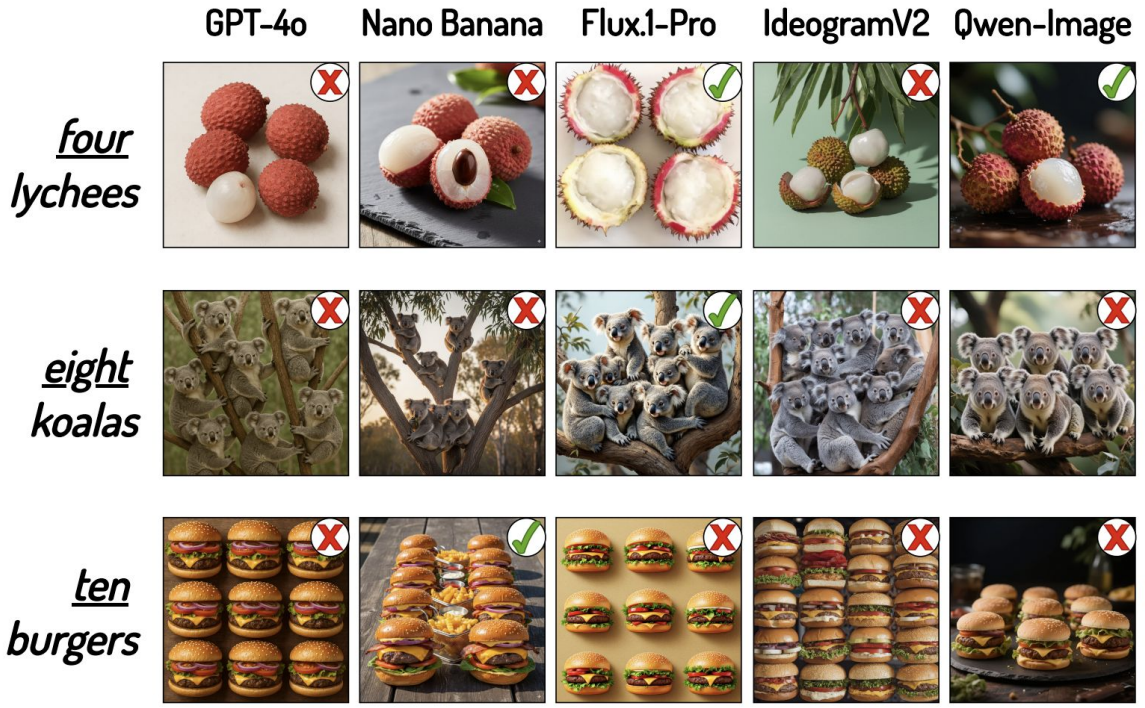
- binding, counting, spatial relations
- colors, forms

Interpretability question

Where does the failure first appear?

white stop sign





DLS

Anatomy: layers and time

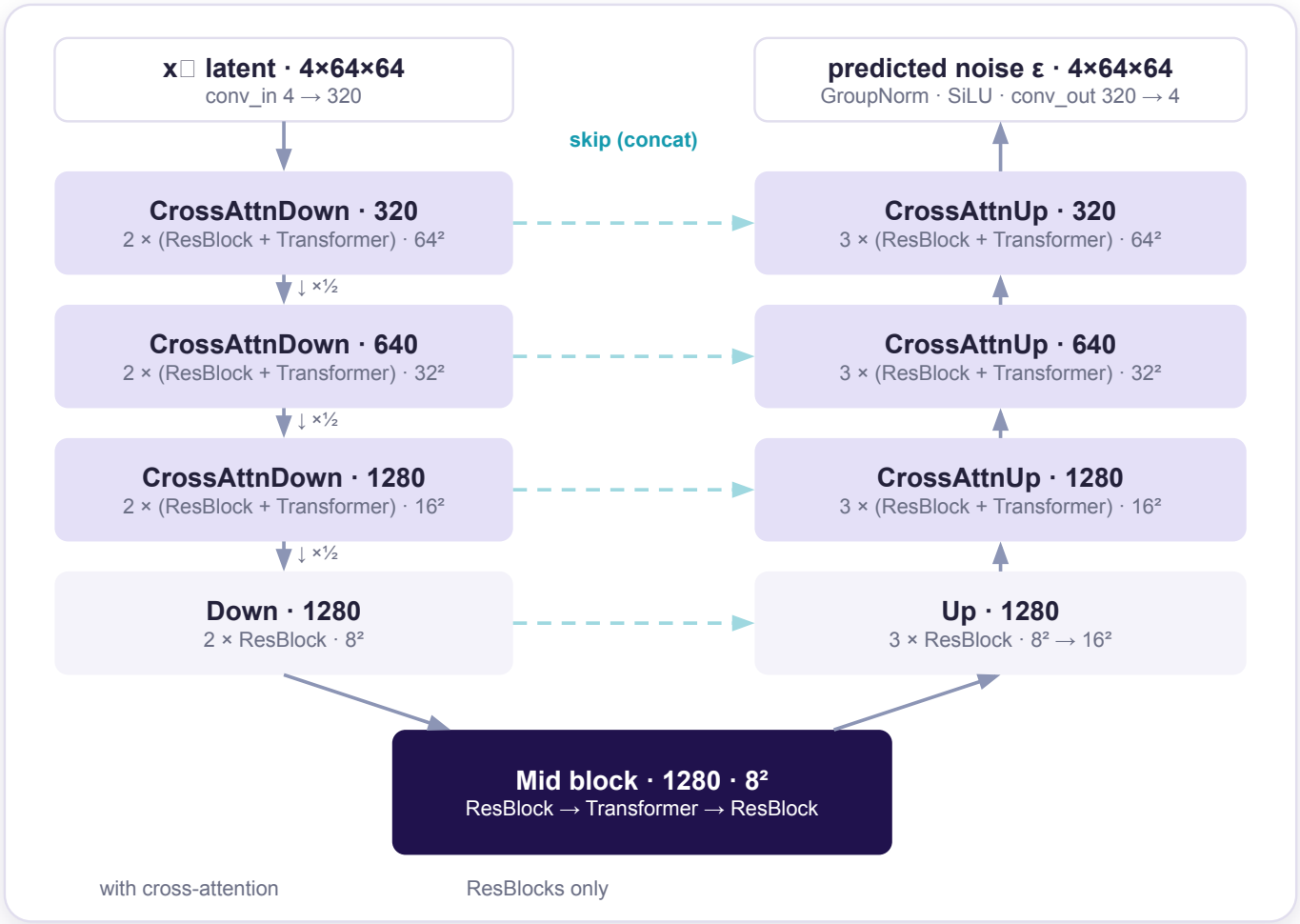
03 · where to look



SDXL: U-Net architecture

- 1024 px image → 128 × 128 latent
- U-Net, 2.6B;

- Difference between up, mid, down?

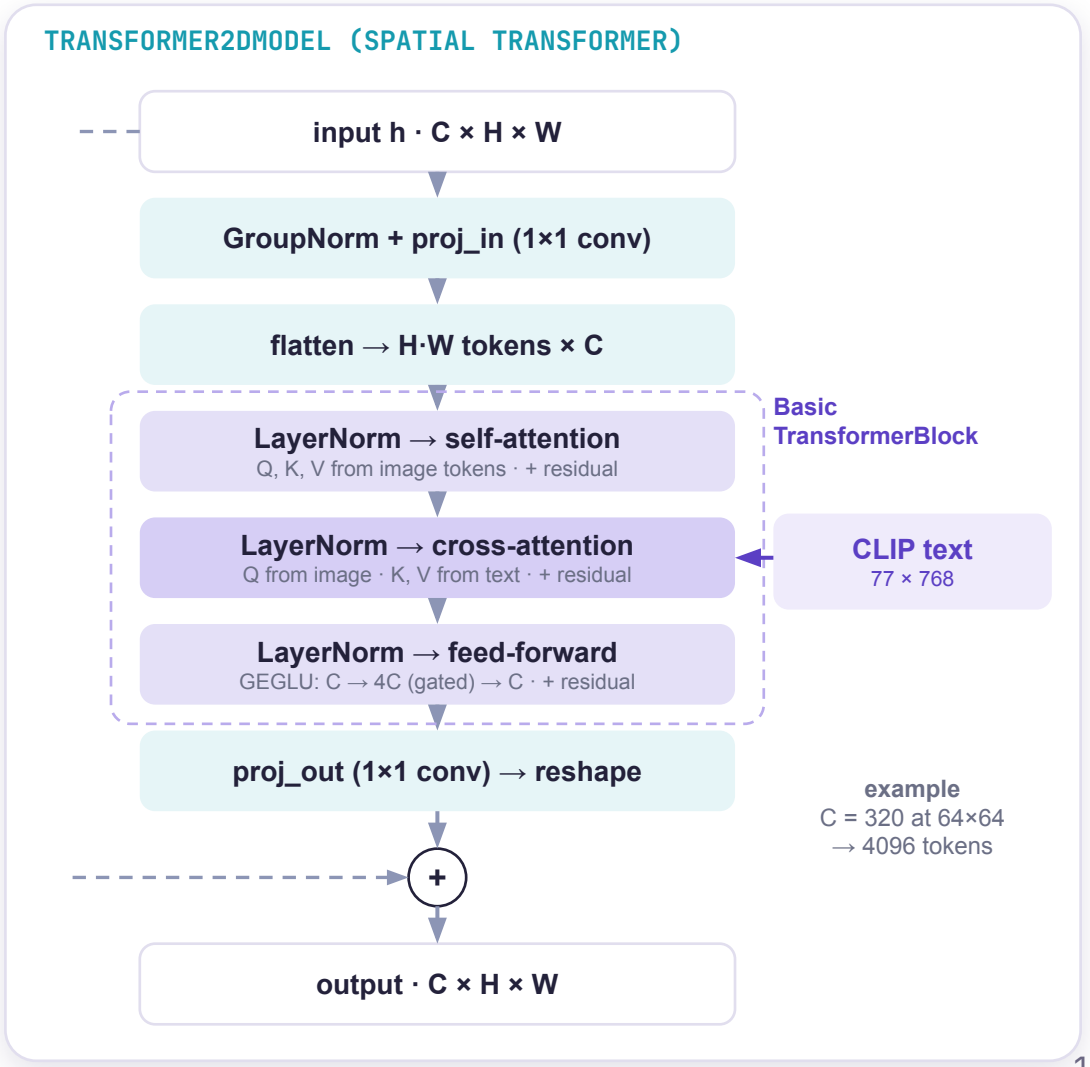


DLS

SDXL: U-Net architecture

- 1024 px image → 128 × 128 latent
- U-Net, 2.6B;

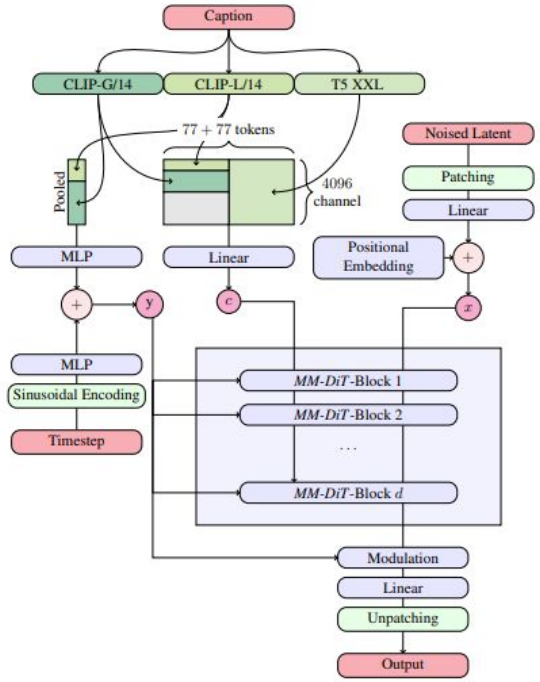
- Difference between self-, cross-,ff layers?



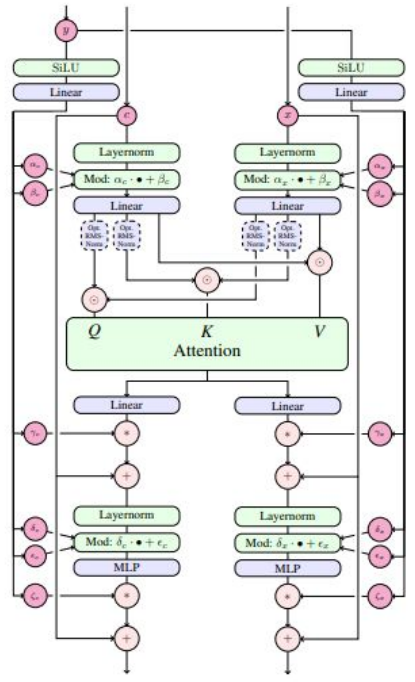
DLS

MM-DiT: SD3 and FLUX

- Transformer architecture
- Different language encoders
- timestep → adaLN (scale, shift, gate)
- different block types



(a) Overview of all components.



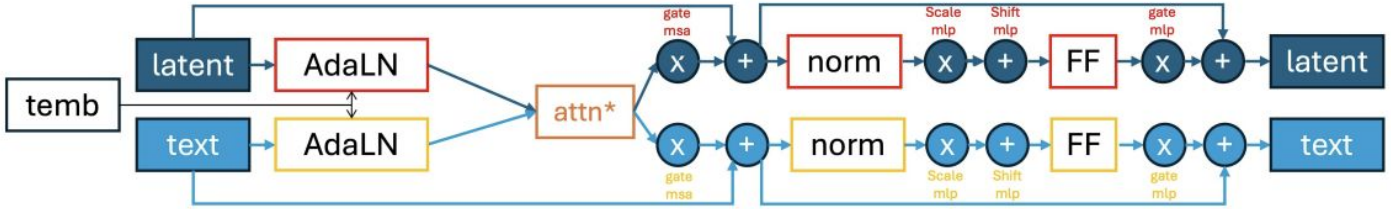
(b) One *MM-DiT* block

DLS

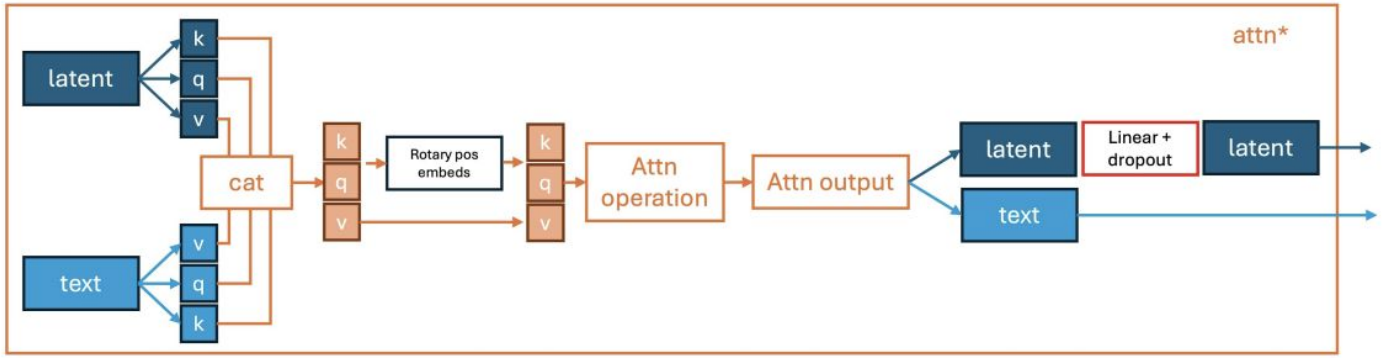
MM-DiT: SD3 and FLUX

- Transformer architecture
- Different language encoders
- timestep → adaLN (scale, shift, gate)
- different block types

What lives in **double blocks**?



(a) Double-Stream Attention Block: latent and prompt embeddings are processed separately, in a standard multi-modal attention scheme.

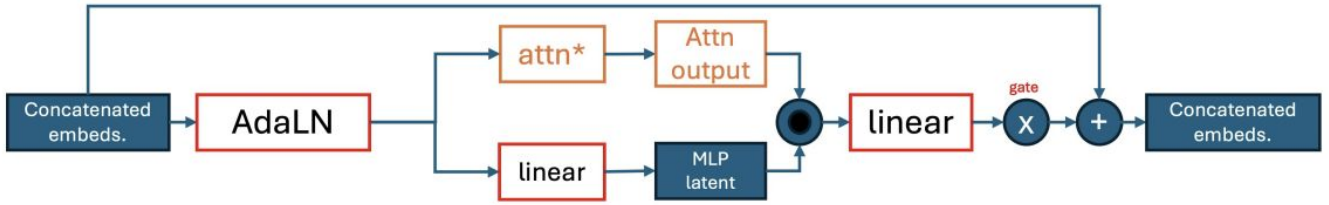


DLS

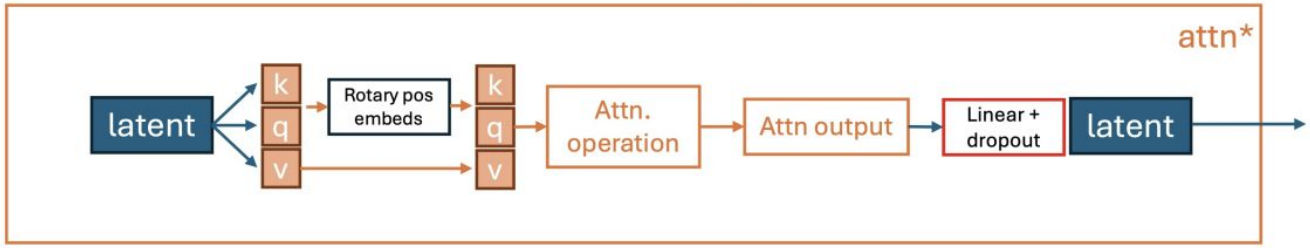
MM-DiT: SD3 and FLUX

- Transformer architecture
- Different language encoders
- timestep → adaLN (scale, shift, gate)
- different block types

What lives in **single blocks**?



(a) Single-Stream Attention Block: latent and prompt embeddings are processed together, in a simultaneous attention scheme.

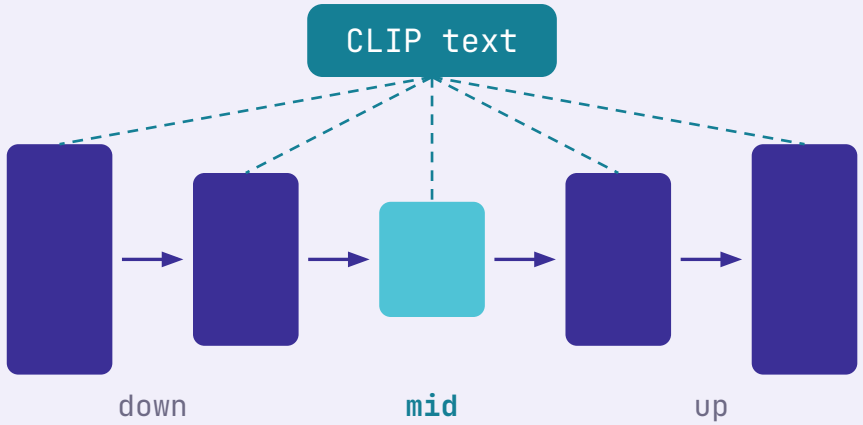


(b) The K, Q and V metrices are computed directly from the concatenated representation.

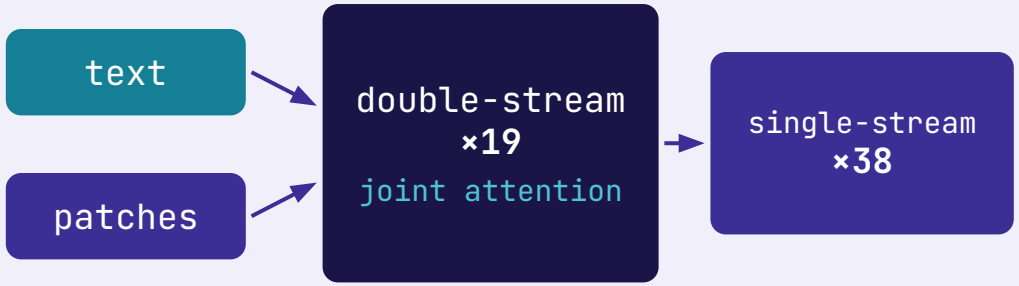
DLS

U-Net vs MM-DiT at a glance

U-Net SD 1.5 · SDXL



MM-DiT SD3 · FLUX



Intervention points: text encoder | attention | residual stream | MLP | smth else?

DLS

Timesteps split the work



$t = T$ `noise` — `image` $t = 0$

Early steps

composition, layout

Middle steps

objects, style

Late steps

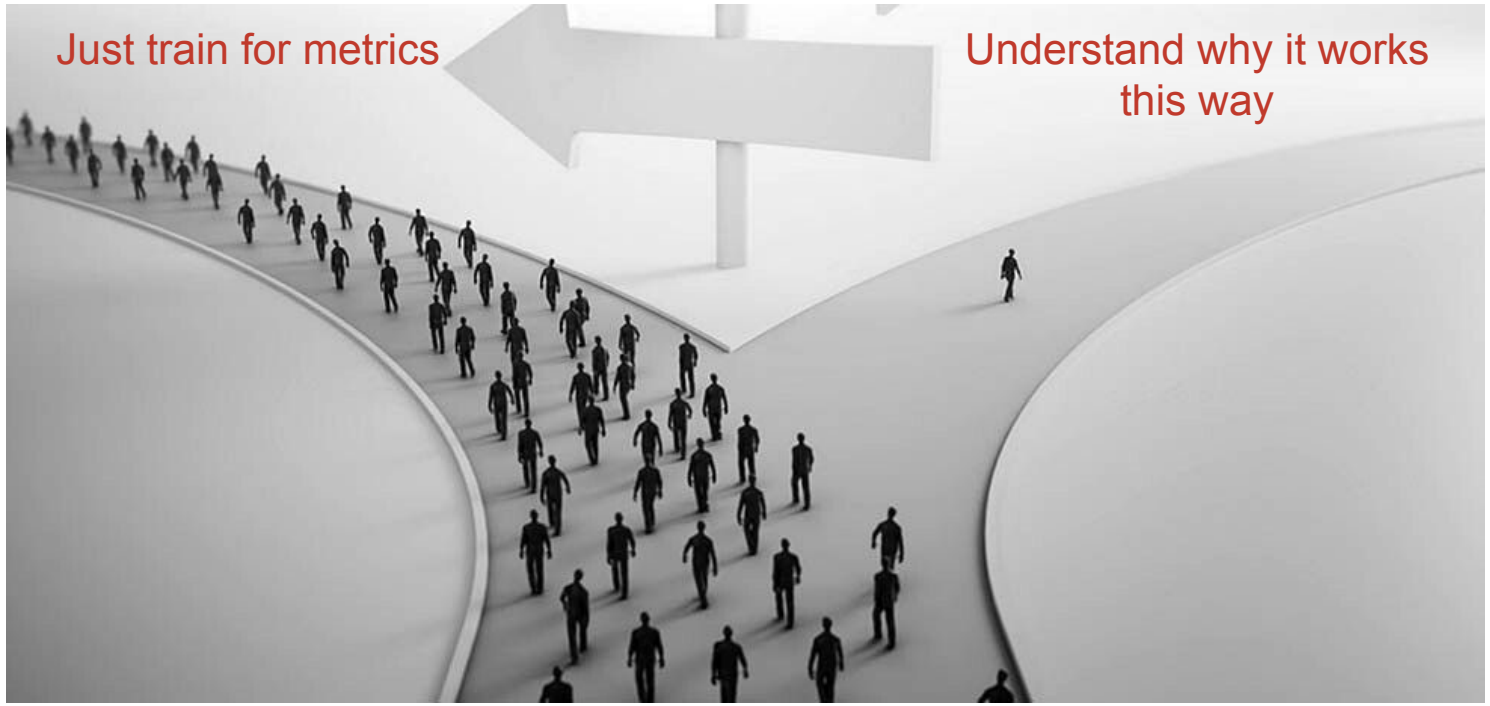
texture, details

Question: which step should be used for different interventions purpose?

DLS

Overall questions

- What information different blocks brings?
- What information different timesteps brings?
- How to find **exact** direction for concept?
- How does the space of activations looks like?



DLS

Attention analysis

04 · where the model looks

DLS

U-Net: three questions

1 What?

what does attention carry?

cross: what · self: where and shape

DAAM · P2P · DiffSeg

→

2 Where?

which block to touch?

layout, content, style sit in different blocks

InstantStyle · B-LoRA · P+

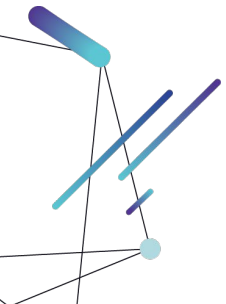
→

3 When?

which steps?

composition → style → texture

Tinaz et al. · PnP · MasaCtrl



DAAM · 2023 →

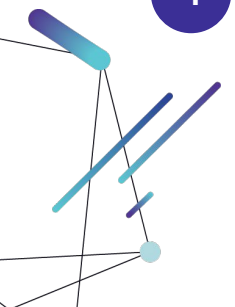
Prompt-to-Prompt · 2022 →

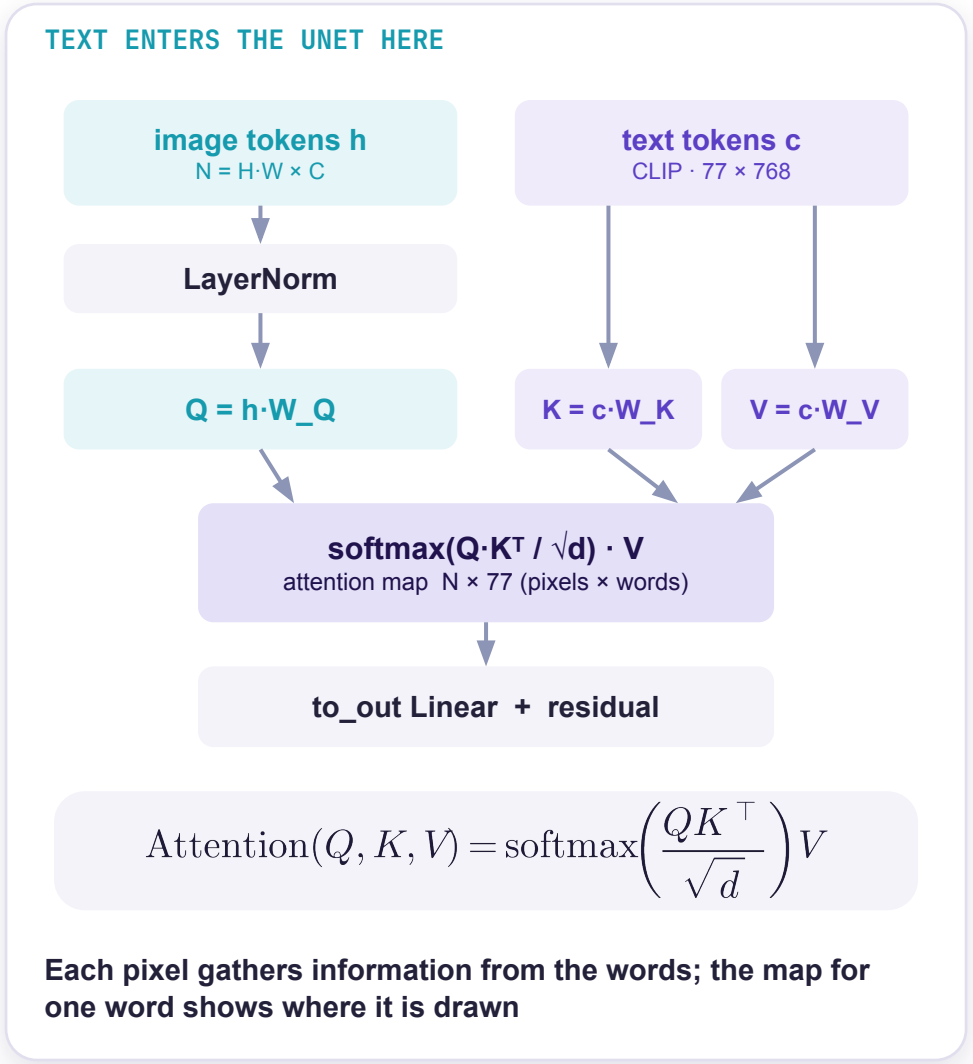
DLS

Cross-attention maps

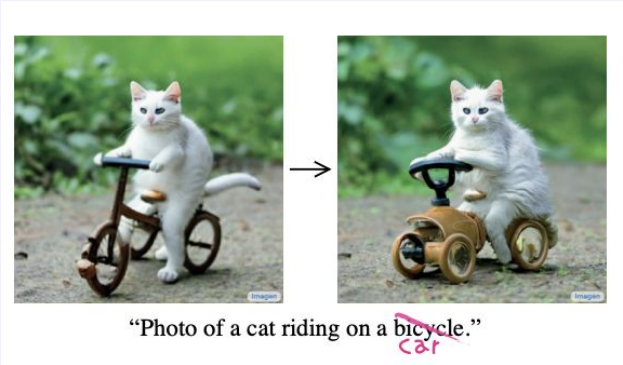
$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)$$

1. Q: one query per patch
2. K, V: prompt tokens
3. column j → map of token j
4. average heads, layers, steps





What people do with the maps



Prompt-to-Prompt

Editing



DAAM

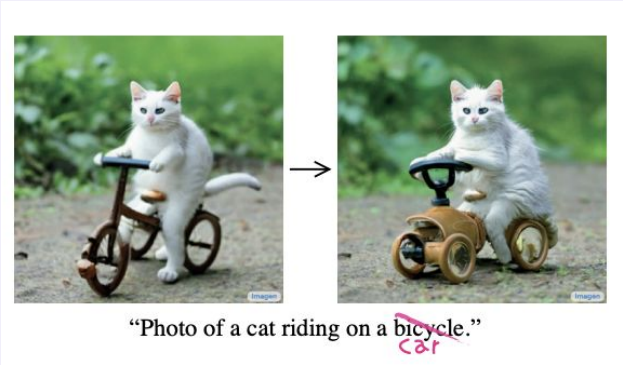
adjectives spread → analyse



Attend-and-Excite

fixing problems

What people do with the maps



Prompt-to-Prompt

Editing



DAAM

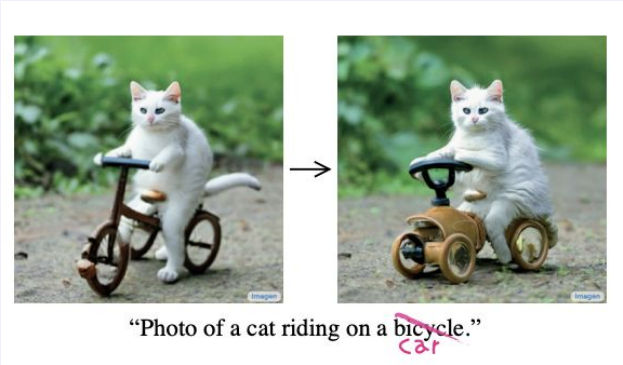
adjectives spread → analyse



Attend-and-Excite

fixing problems

What people do with the maps



Prompt-to-Prompt

Editing



DAAM

adjectives spread → analyse



Attend-and-Excite

fixing problems

What information does cross-attention carry?

DAAM · 2023 →

code →

DLS

DAAM: a heatmap for every word

cross-attention $A_{:,k}^{(t,l,h)}$ → **upsample** $\mathrm{up}_{512}(\cdot)$ → **sum** $\sum_{t,\,l,\,h}$ → **threshold** $\geq \tau \max D_k$

$$D_k = \sum_t \sum_l \sum_h \mathrm{up}_{512}\left(A_{:,k}^{(t,l,h)}\right)$$

$$\mathrm{mask}_k = \mathbf{1}\left[D_k \geq \tau \max D_k\right], \quad \tau = 0.4$$



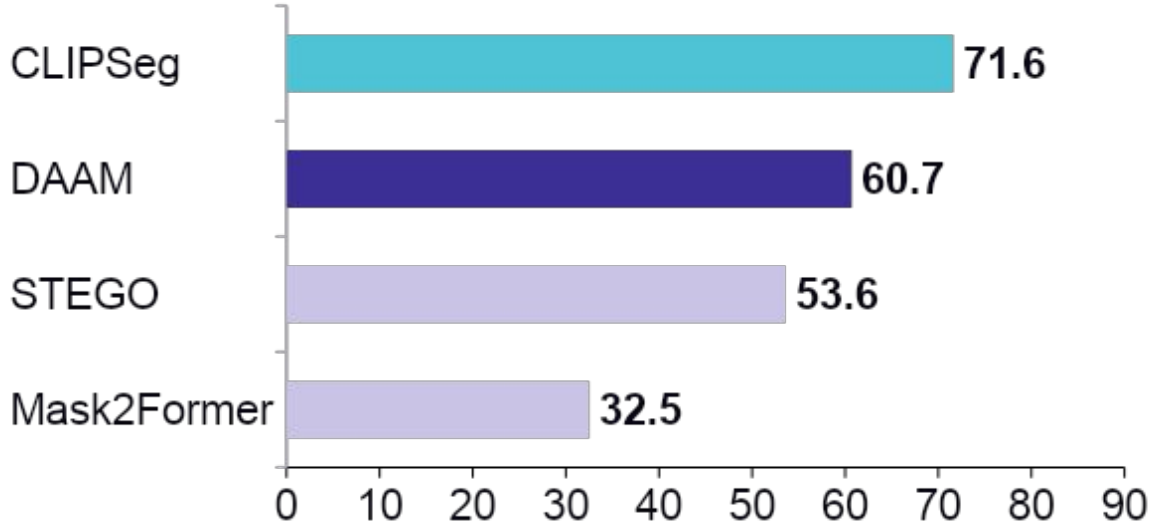
no training · any word · SD 2.0

DAAM · 2023 →

DLS

DAAM: are the maps right?

noun segmentation, mIoU

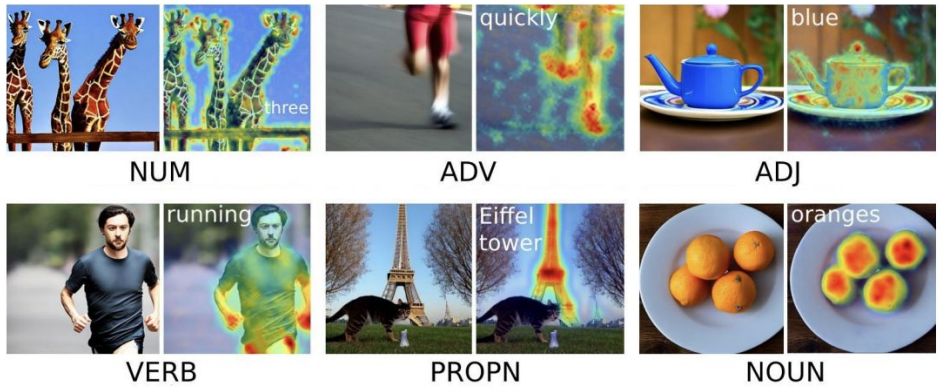


80 COCO classes only: Mask2Former 84 vs DAAM 65

human ratings

good nouns, verbs, adjectives

fair numerals, adverbs

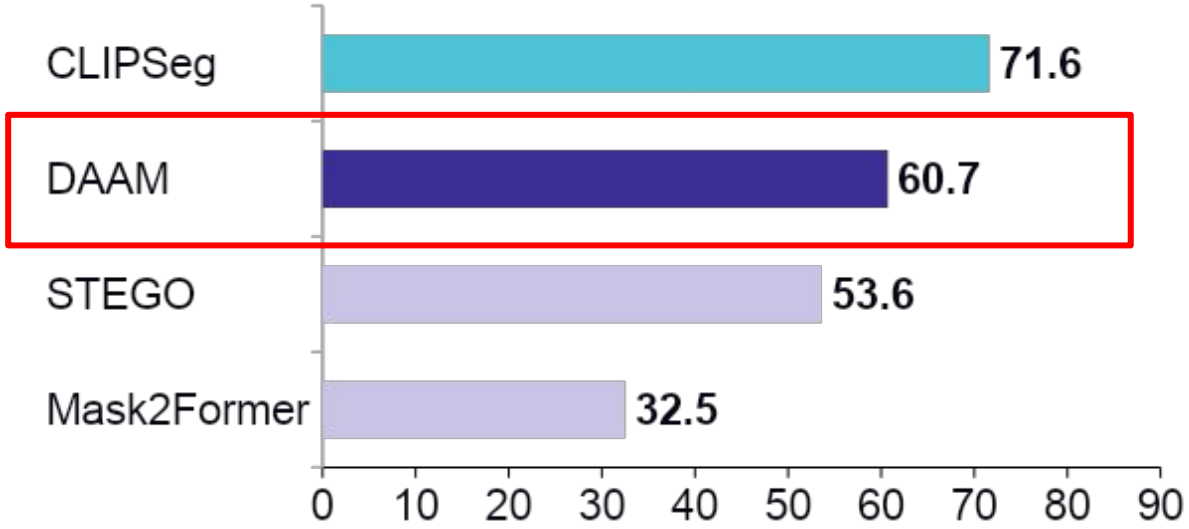


DAAM · 2023 →

DLS

DAAM: are the maps right?

noun segmentation, mIoU

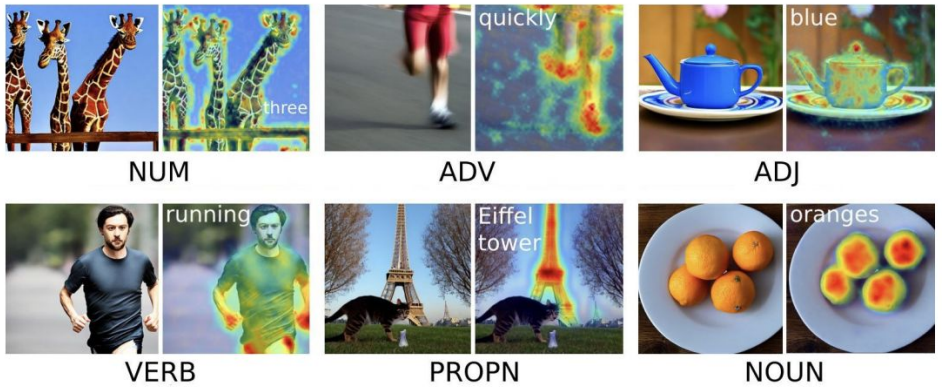


80 COCO classes only: Mask2Former 84 vs DAAM 65

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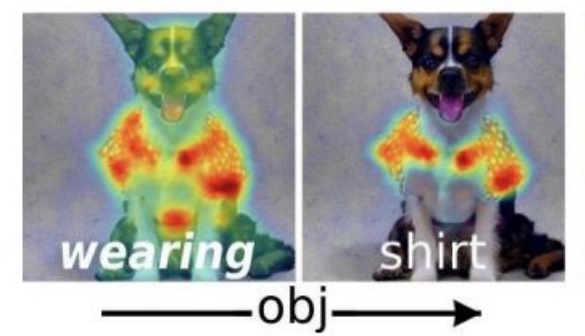
fair numerals, adverbs



DAAM · 2023 →

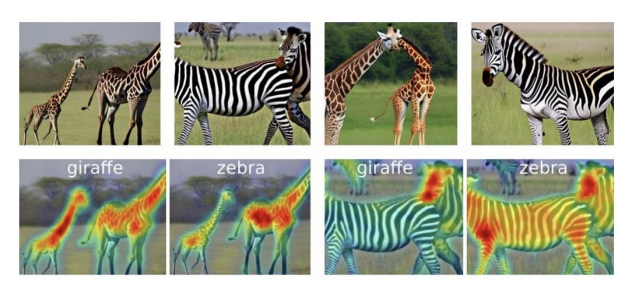
DLS

DAAM: what the maps reveal



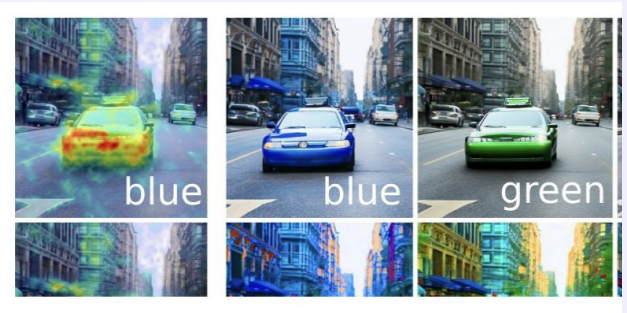
Syntax in pixels

a head word covers its dependent



Objects merge

one of two similar objects disappears



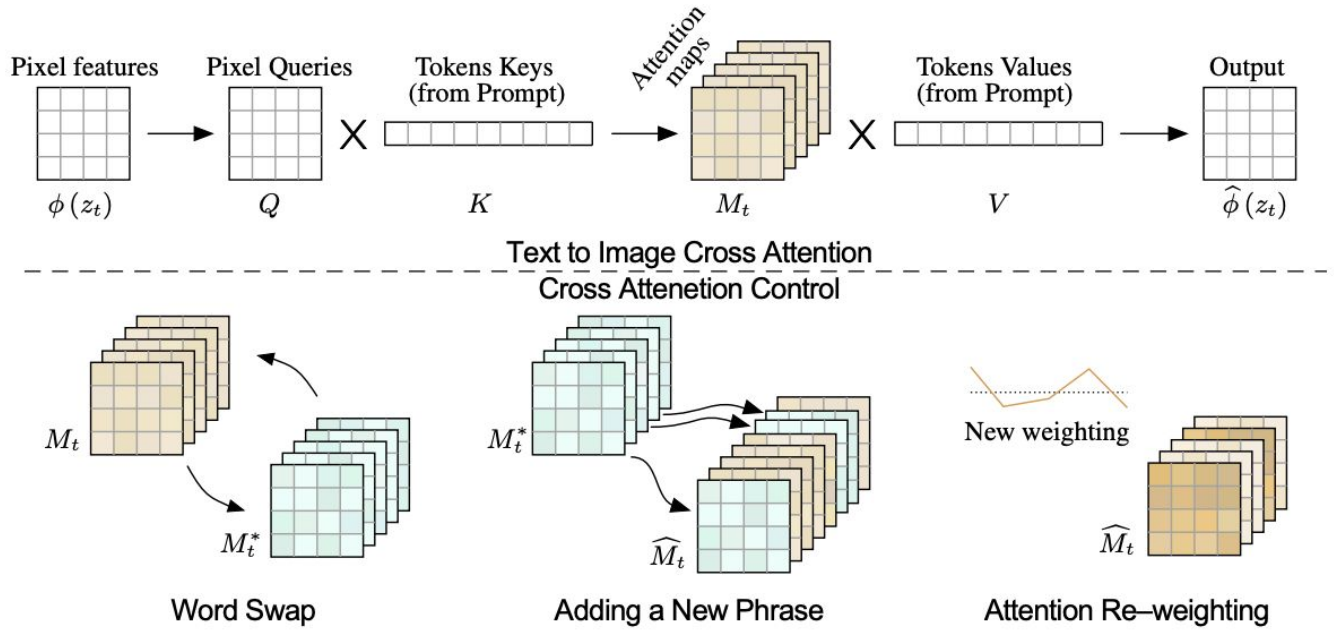
Adjectives spread

colour leaks into the scene

[Prompt-to-Prompt · 2022 →]

Prompt-to-Prompt





Self-attention and heads

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)$$

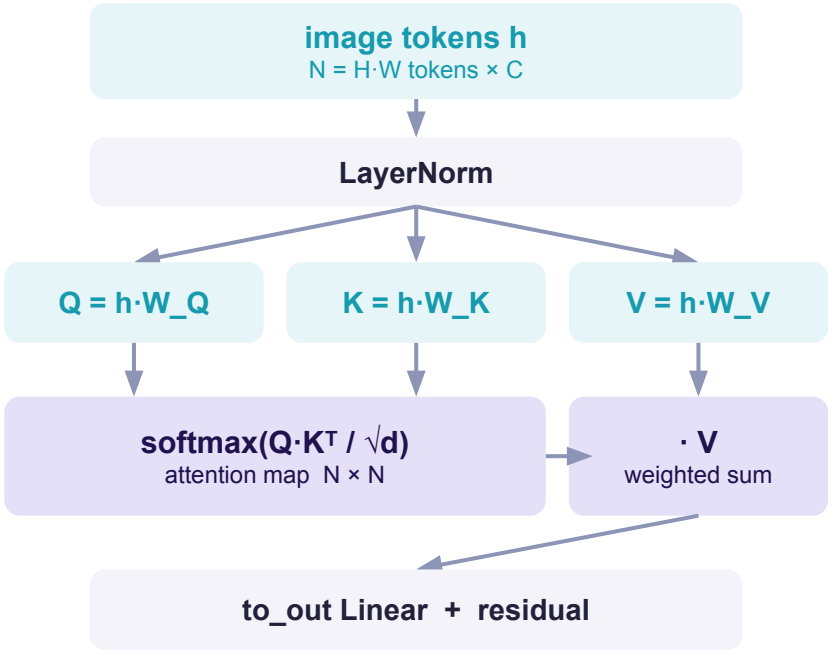


generated image



token grid 32×32 = 1024 tokens

INSIDE THE SPATIAL TRANSFORMER



$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V$$

Every pixel can look at every other pixel: global context at the cost of O(N²)

DiffSeg · CVPR 2024 →

DiffSeg: segmentation from self-attention

collect self-attention
one pass · t = 300 · no prompt

→

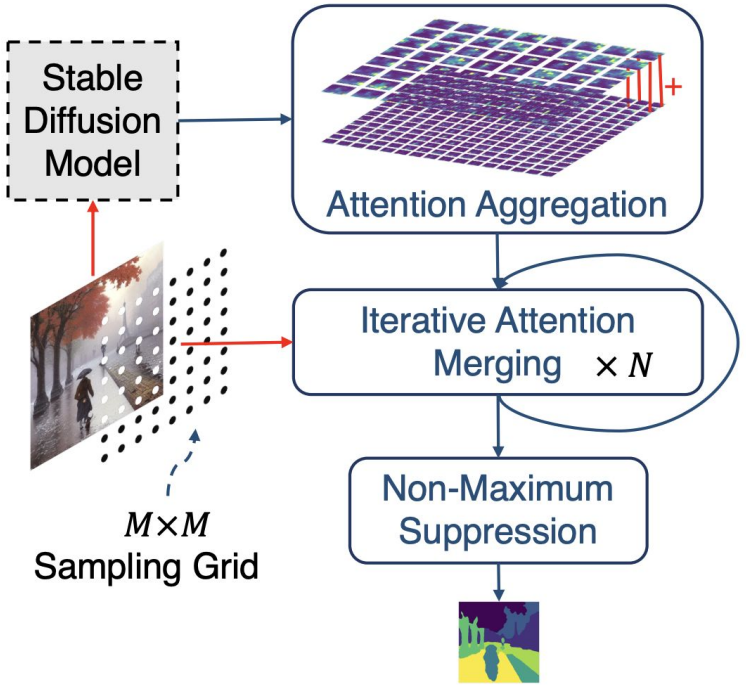
merge similar maps
anchor grid · KL distance

→

assign pixels
pixel → strongest map = mask

Why it works

- a pixel's map lights up its own object
- pixels of one object have similar maps

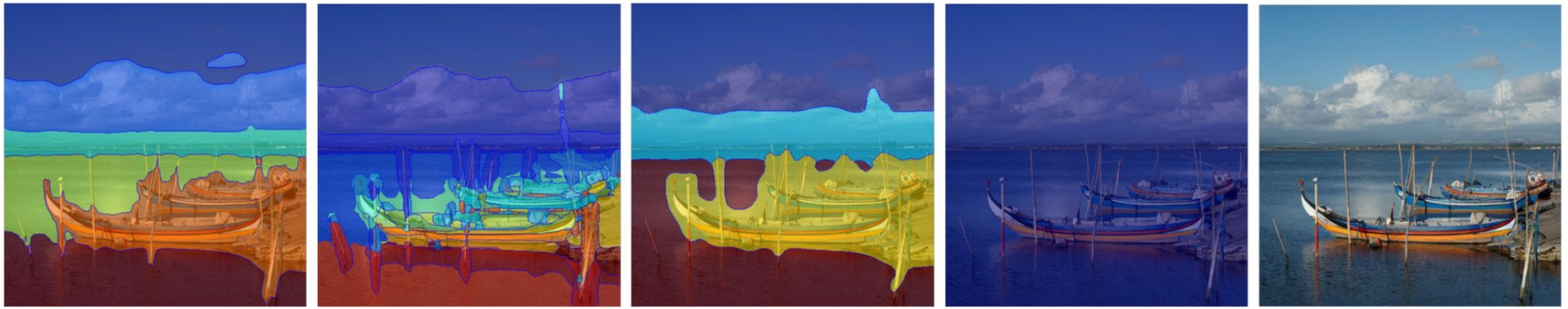


DiffSeg · CVPR 2024 →

DAAM · 2023 →

DLS

DiffSeg: segmentation from self-attention



(a) Proportional (b) Only 64 × 64 maps (c) Only 32 × 32 maps (d) Only 16(8) × 16(8) maps (e) Original image

P+ · 2023 →

InstantStyle · 2024 →

B-LoRA · ECCV 2024 →

DLS

Using block roles: three methods

P+

a prompt per layer

low-res layers → shape,
high-res layers → colour

Voynov et al. · SD 1.4

InstantStyle

image → style block only

style moves, content does not leak

Wang et al. · SDXL

B-LoRA

LoRA on 2 blocks

one image → content and style apart, then mix

Frenkel et al. · SDXL

P+ · 2023 →

InstantStyle · 2024 →

B-LoRA · ECCV 2024 →

DLS

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P+ · 2023 →

DLS

P+

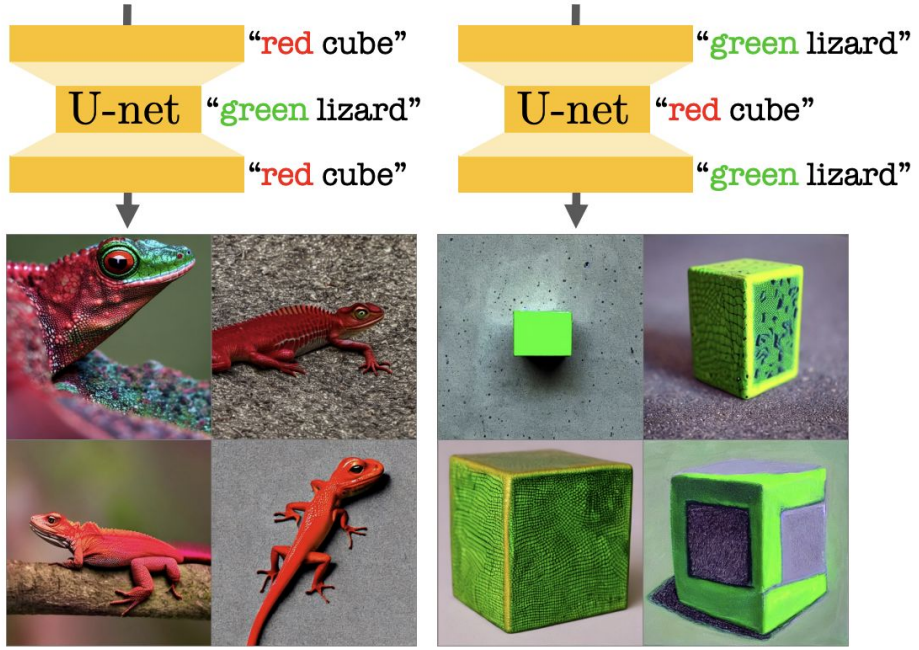
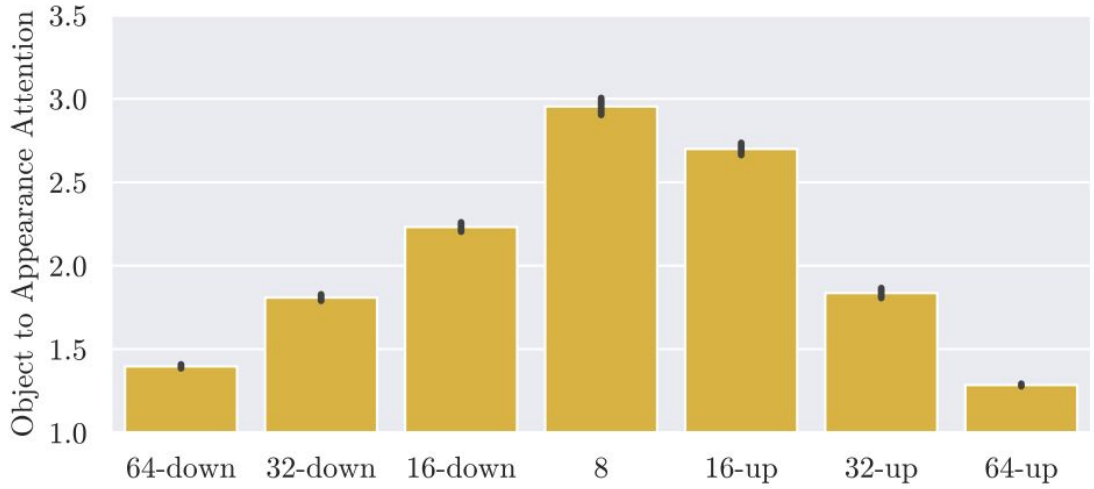
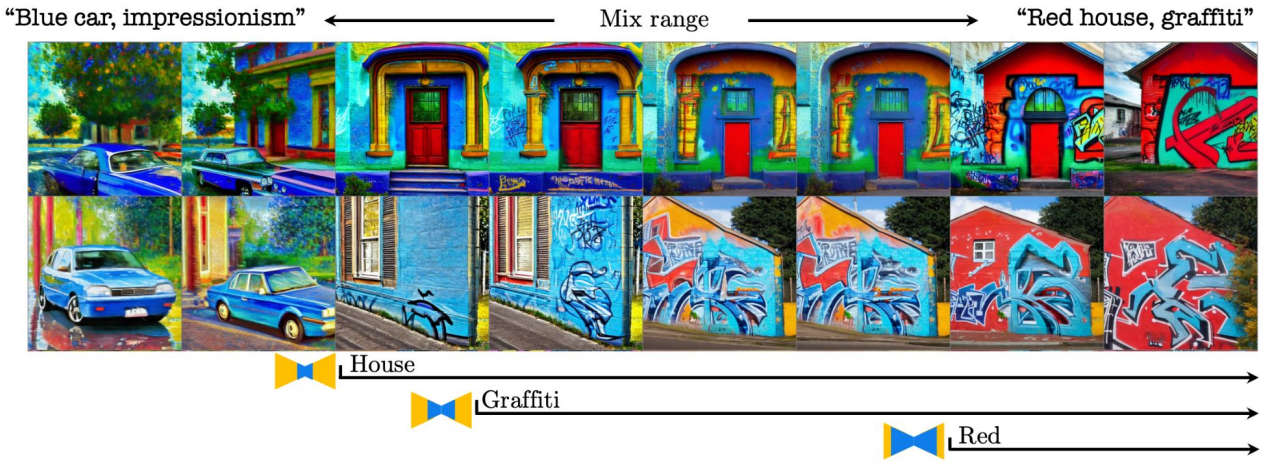


Figure 3. **Per-layer Prompting.** We provide different text prompts (a precursor to $\mathcal{P}+$) to different cross-attention layers in the denoising U-net. We see that color (`"red"`, `"green"`) is determined by the fine outer layers and content (`"cube"`, `"lizard"`) is determined by the coarse inner layers.





P+ · 2023 →

InstantStyle · 2024 →

B-LoRA · ECCV 2024 →

DLS

Using block roles: three methods

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Wang et al. · SDXL

B-LoRA

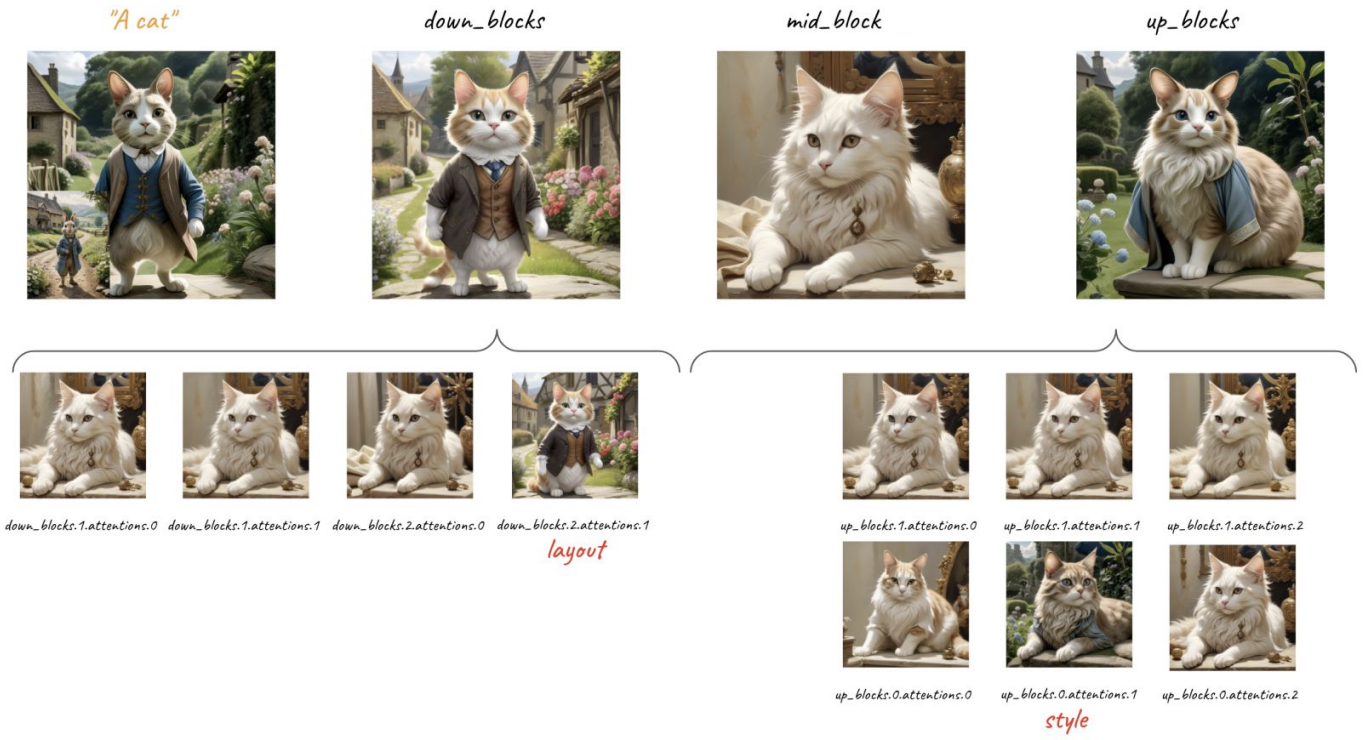
LoRA on 2 blocks

one image → content and style apart, then mix

Frenkel et al. · SDXL

[InstantStyle · 2024 →]

InstantStyle



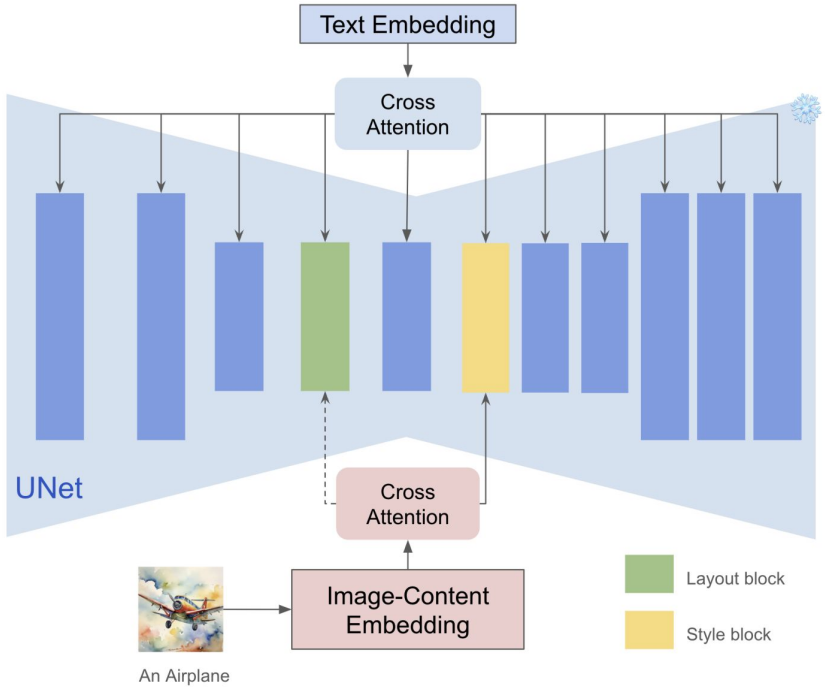
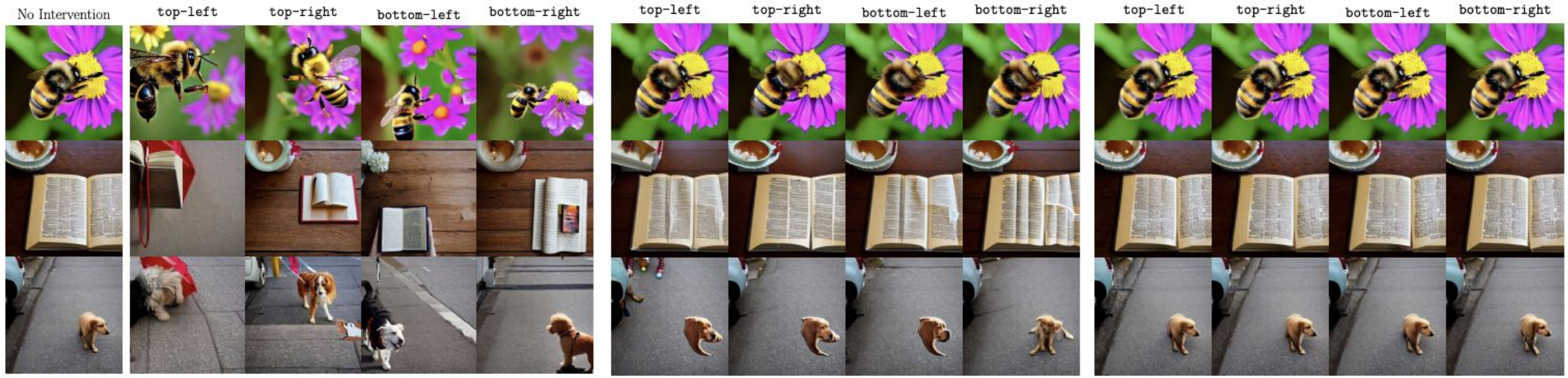


Fig. 7. Fine-grained Analysis over Attention Blocks. We conduct comprehensive experiments on IP-Adapter (SDXL) to analyse the effect of each attention layer for image feature. We are surprised to find that up_blocks.0.attentions.1 and down_blocks.2.attentions.1 are the most representative layers, controlling style and spatial layout respectively. As for whether layout is a style, it varies from person to person.

Tinaz et al. · 2025 →

DLS

When to intervene: by timestep



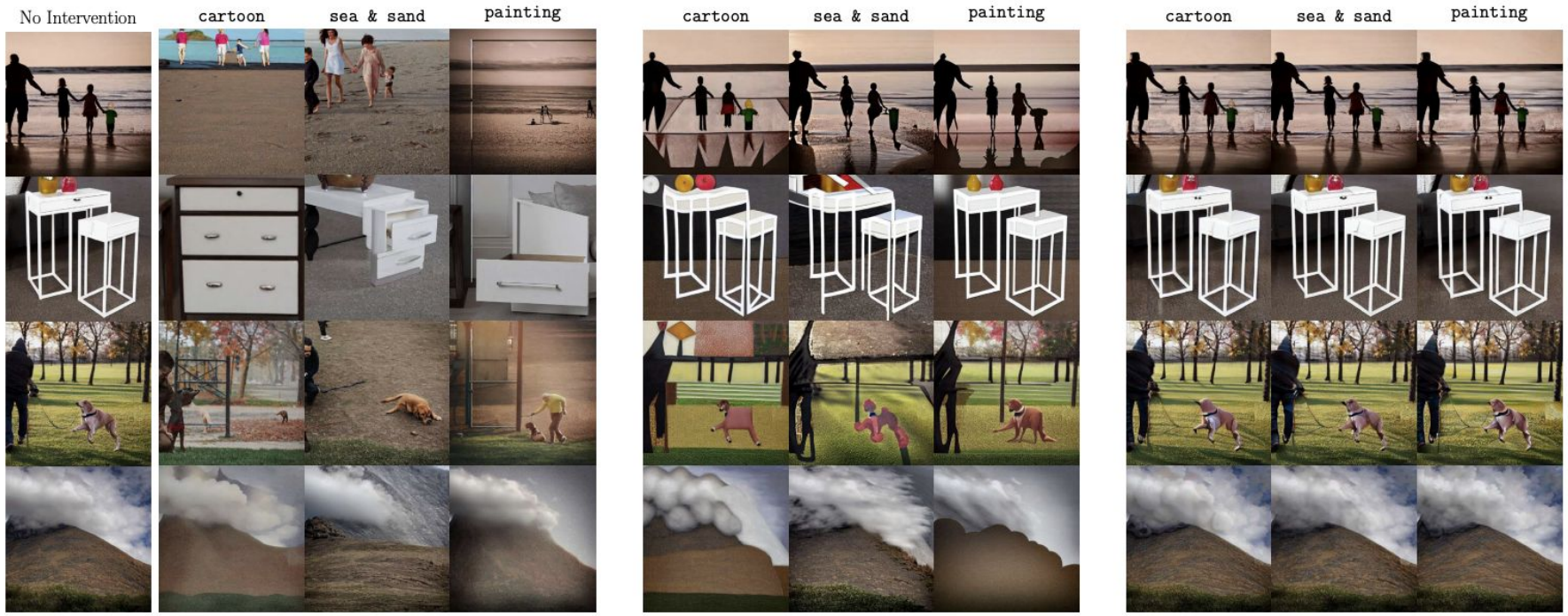
(a) Early-stage intervention. (b) Middle-stage intervention (c) Final-stage intervention

Figure 4: Effect of spatially targeted interventions at different stages of diffusion, aimed at manipulating image layout. We can restrict objects to the specified quadrant of the image when intervened in early stages of diffusion. However, in middle and final stages our interventions are unsuccessful.

Tinaz et al. · 2025 →

DLS

When to intervene: by timestep



(a) Early-stage intervention (b) Middle-stage intervention (c) Final-stage intervention

Figure 5: Effect of global interventions aimed at manipulating image style. Intervening in the early stages of diffusion drastically modifies image composition without imbuing the image with a particular style. In stark contrast, middle-stage global interventions successfully manipulate image style without interfering with image composition. However, in the final stages of diffusion, such global interventions have no effect on style or composition, and only result in minor textural changes.

DLS

MM-DiT: the same three questions

1 What?

what does joint attention carry?

four blocks instead of cross + self

Shin · TACA · ConceptAttention · Seg4Diff

→

2 Where?

which layers and heads?

roles are scattered over depth

Stable Flow · FreeFlux · Zarei · HeadHunter

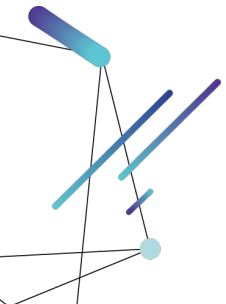
→

3 When?

which steps?

concepts lock in earlier than in U-Net

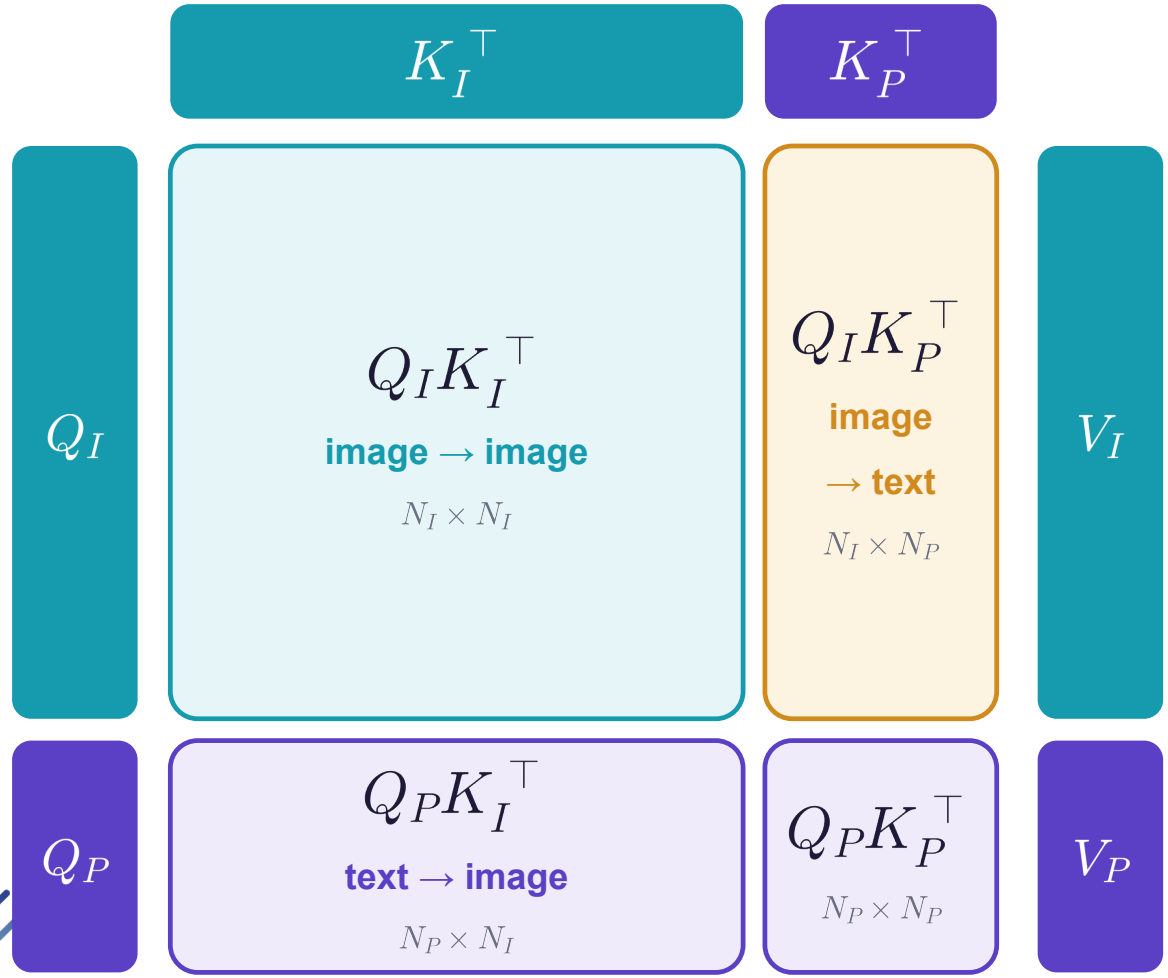
Sammani · Shin · TACA · SHIFT



Shin et al. · 2025 →

DLS

Joint attention: four blocks



self-attention: already in SD 1.x

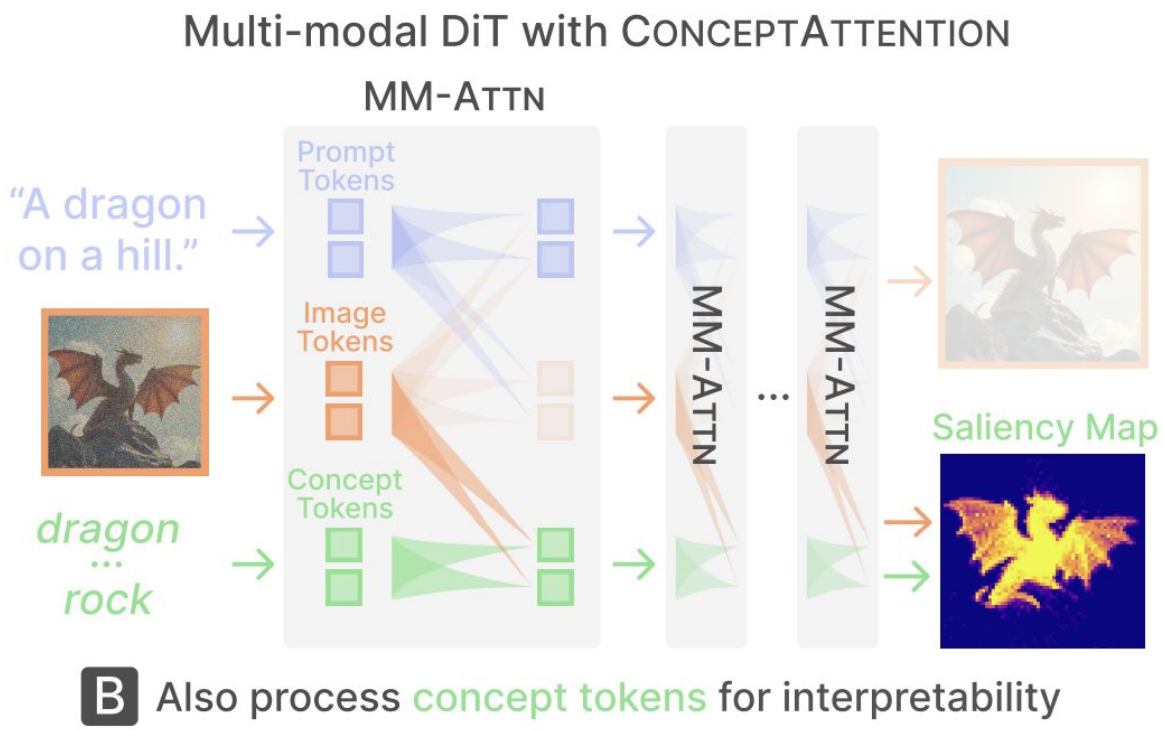
cross-attention: already in SD 1.x

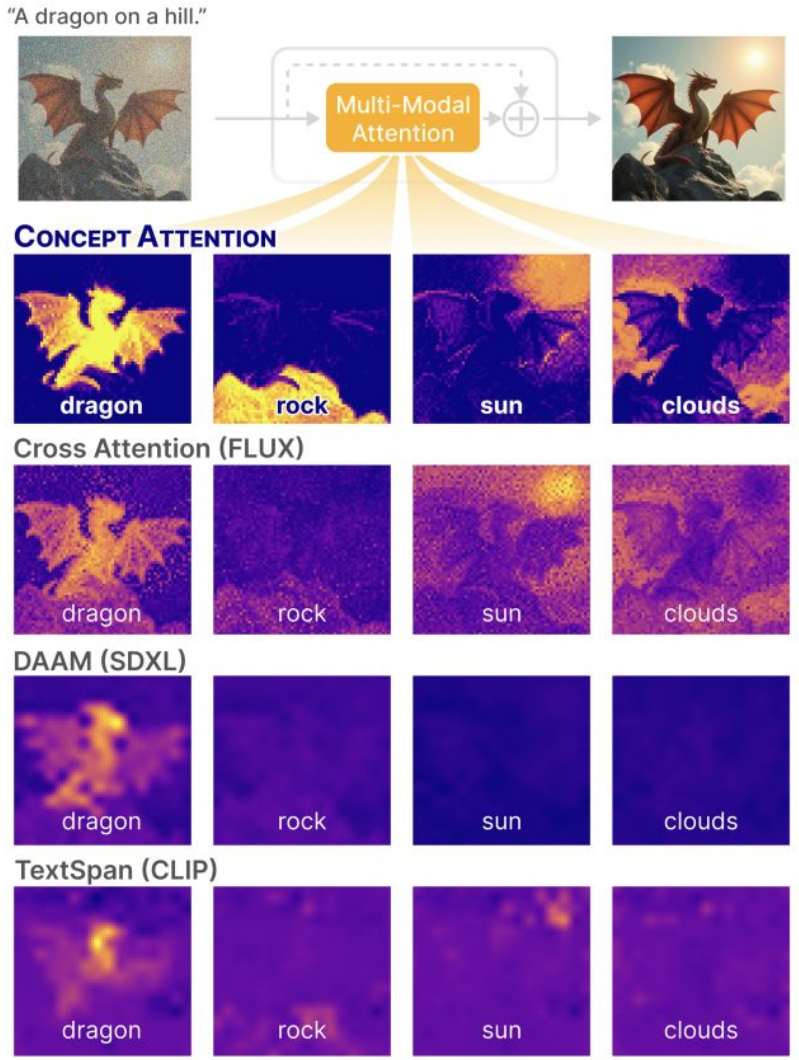
new: text tokens attend to image and text

ConceptAttention · 2025 →

DLS

ConceptAttention – DAAM for DiTs

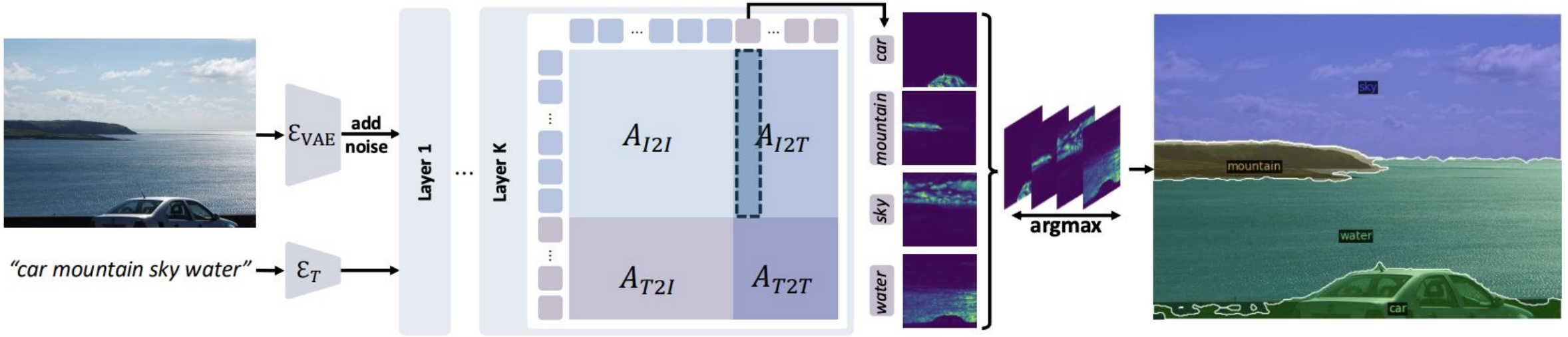




concept tokens read the image, never write

Seg4 Diff · 2025 →

Seg4Diff





Seg4Diff · 2025 →

DLS

Seg4Diff

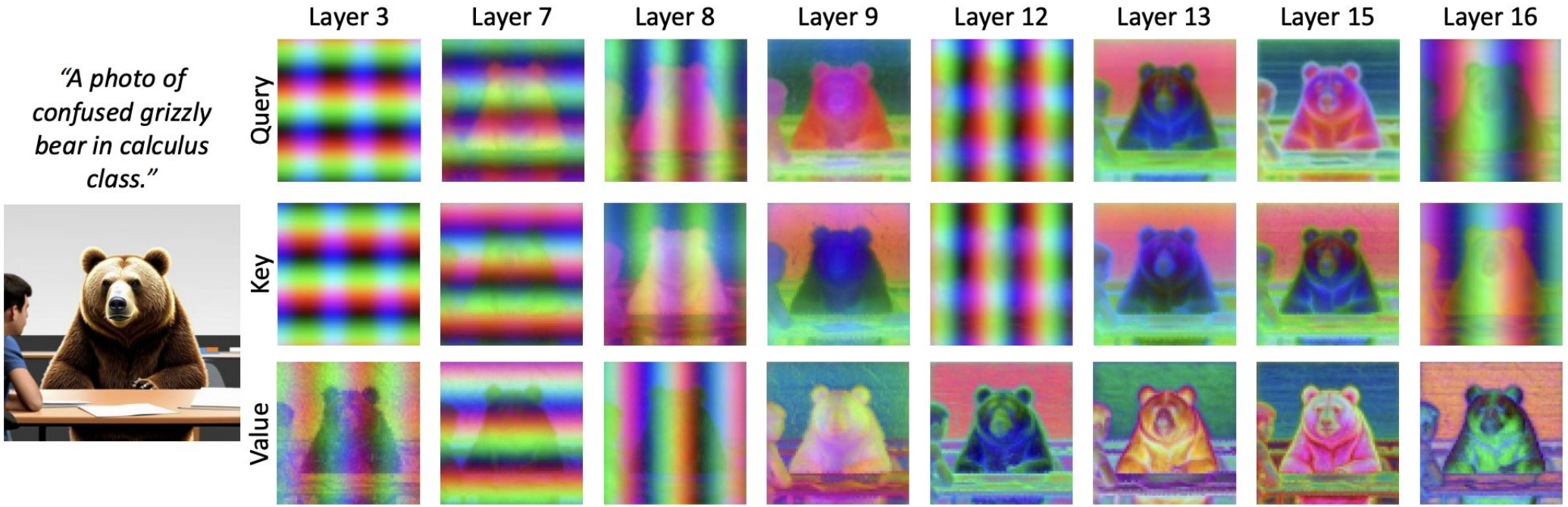


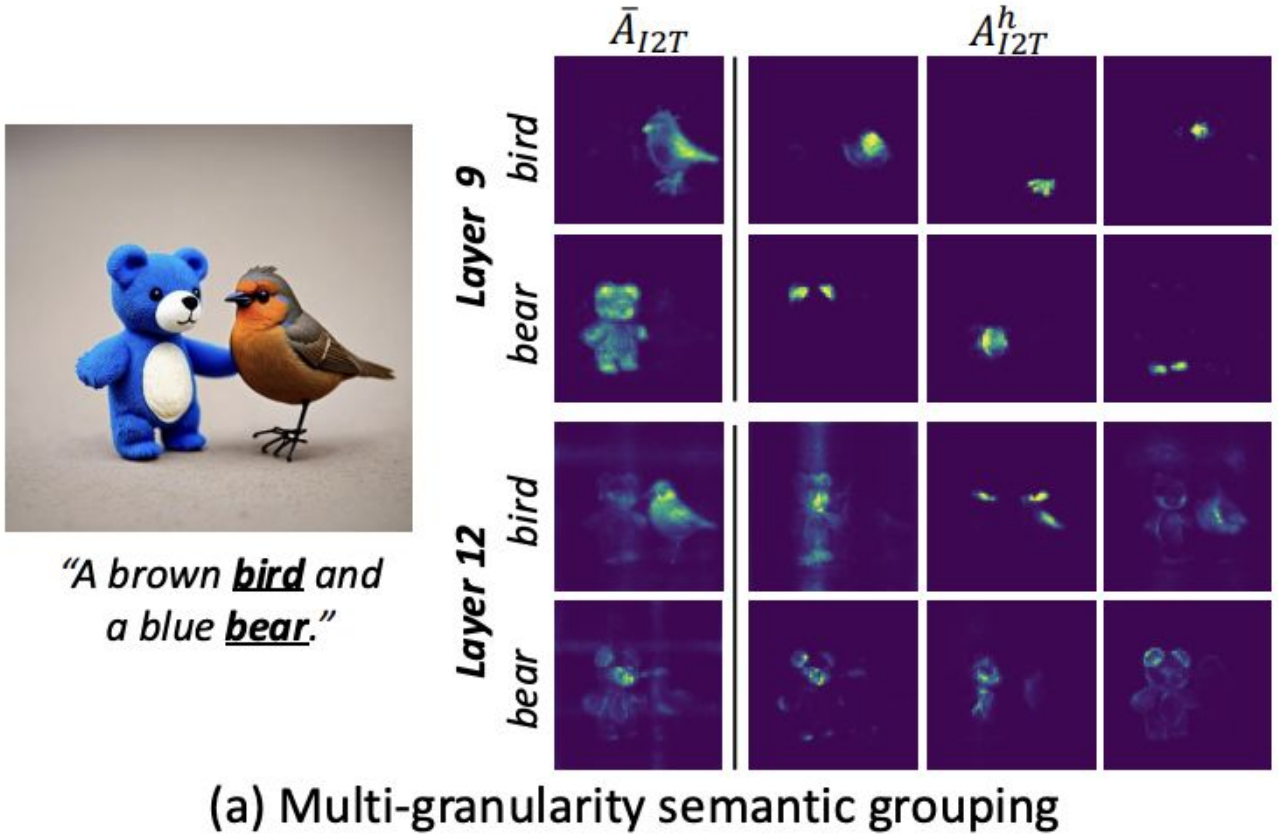
Figure 3: **PCA visualization of query, key and value projections.** PCA results demonstrate that some layers exhibit strong positional bias, whereas some layers show clear semantic groups.

PCA of query, key, value features · colour = top-3 components · SD3

stripes = the layer encodes position (3, 7, 12, 16) · **bear shape** = the layer groups by meaning (9, 13, 15)

Seg4Diff · 2025 →

Seg4Diff



(a) Multi-granularity semantic grouping

What is shown

image-reads-text maps for the words **"bird"** and **"bear"**

I2T in the paper's notation

How to read

Ā = averaged over heads → whole object

A^h = single heads → parts: eyes, belly, feet

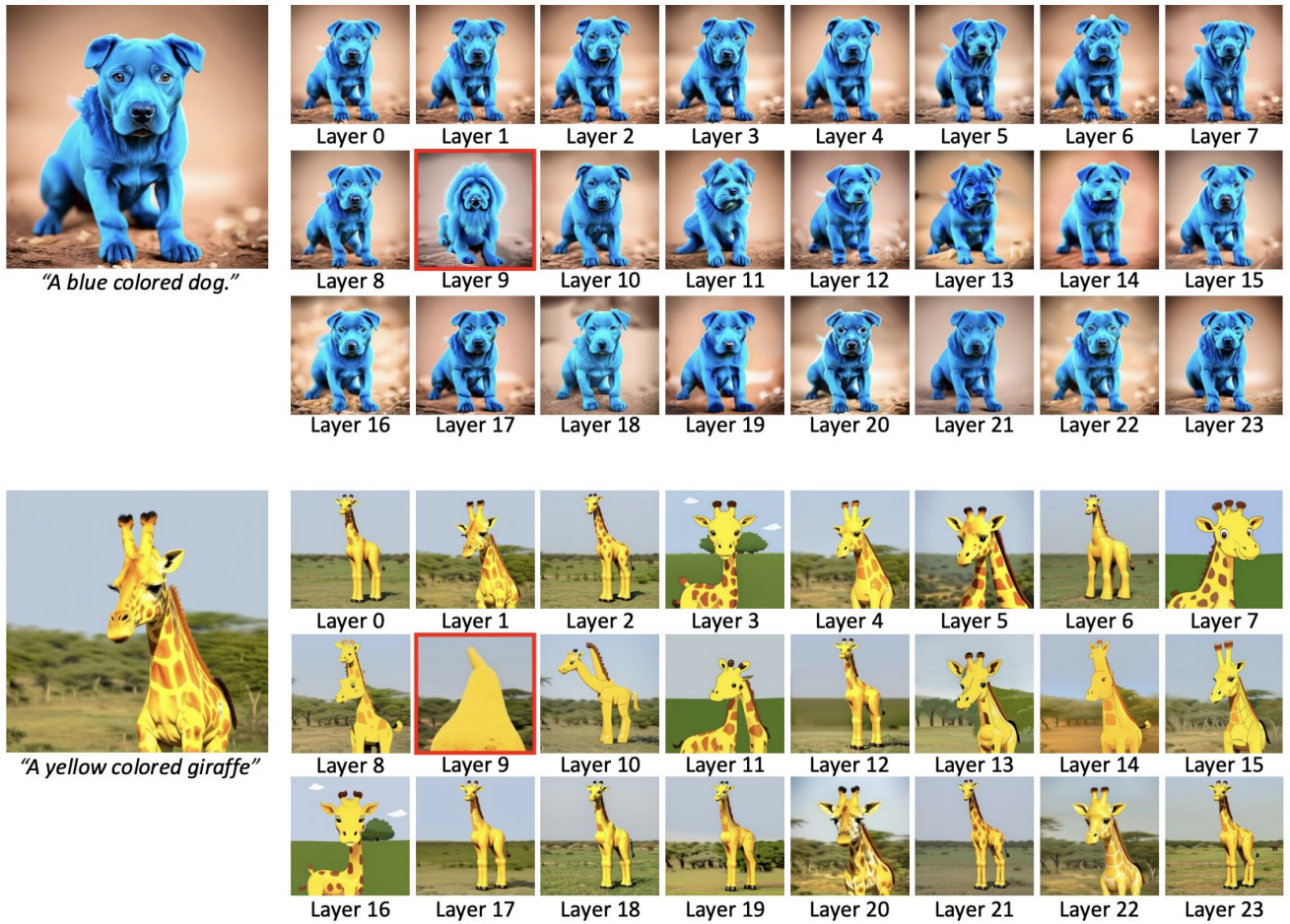
Layers

layer 9: clean object masks

layer 12: noisier, spread out

Seg4Diff · 2025 →

Seg4Diff



Experiment

blur the image-reads-text attention
in one layer, keep the rest
same prompt, same seed

Result

most layers: almost the same image
layer 9 (red frame): the object falls apart
the colour survives, the dog and the giraffe do not

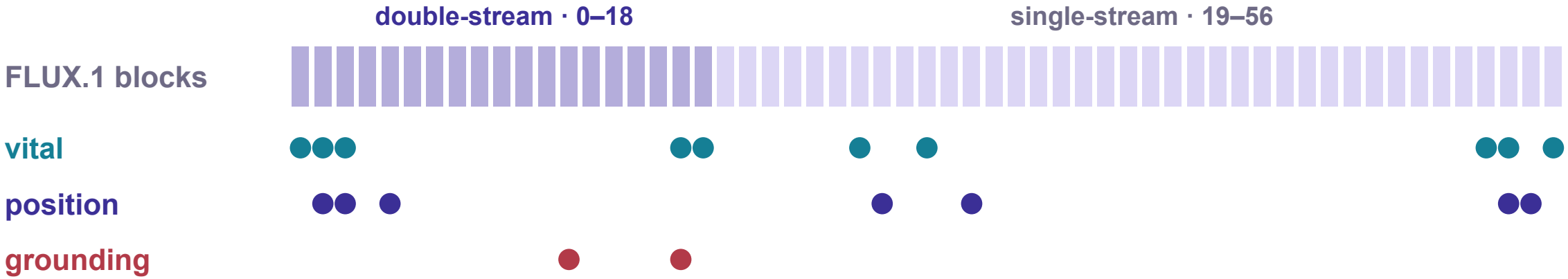
→ **layer 9** ties the words to the image

Stable Flow · CVPR 2025 →

FreeFlux · 2025 →

DLS

MM-DiT layers have roles



vital
skip it → image changes most
Stable Flow

position
shift its RoPE → image breaks
FreeFlux

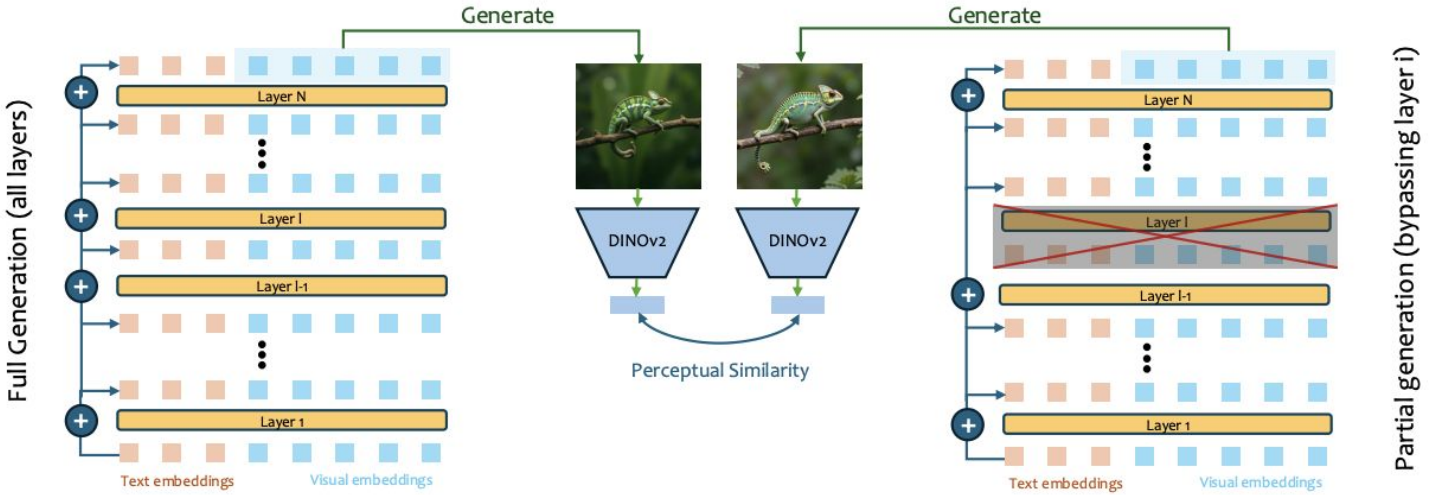
grounding
its maps are masks (SD3: 9)
Seg4Diff

Stable Flow · CVPR 2025 →

DLS

Stable Flow: a few vital layers

1. **skip one layer**
   keep only its residual connection
2. **measure the change**
   DINOv2 similarity over 64 prompts
3. **edit only there**
   inject source image tokens in vital layers



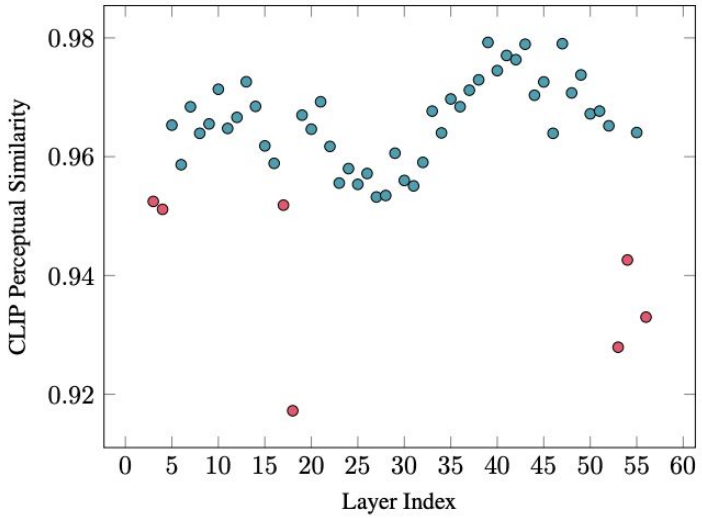
Stable Flow · CVPR 2025 →

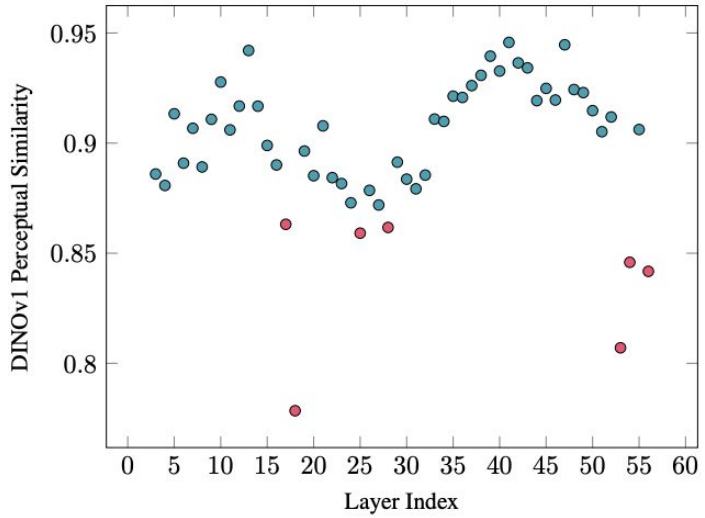
DLS

Stable Flow: a few vital layers

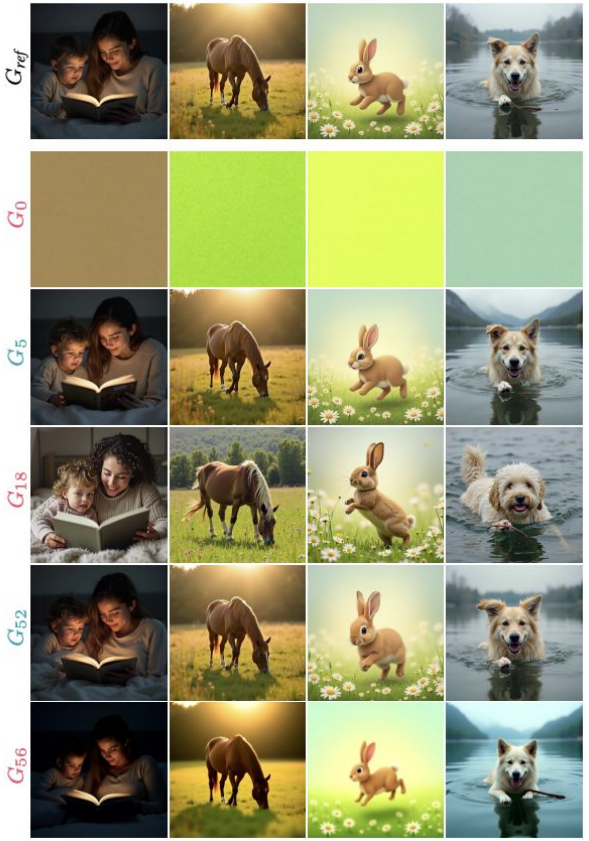
Skip one layer at a time — how much does the image change?

each dot = one layer bypassed · FLUX.1-dev · 64 prompts





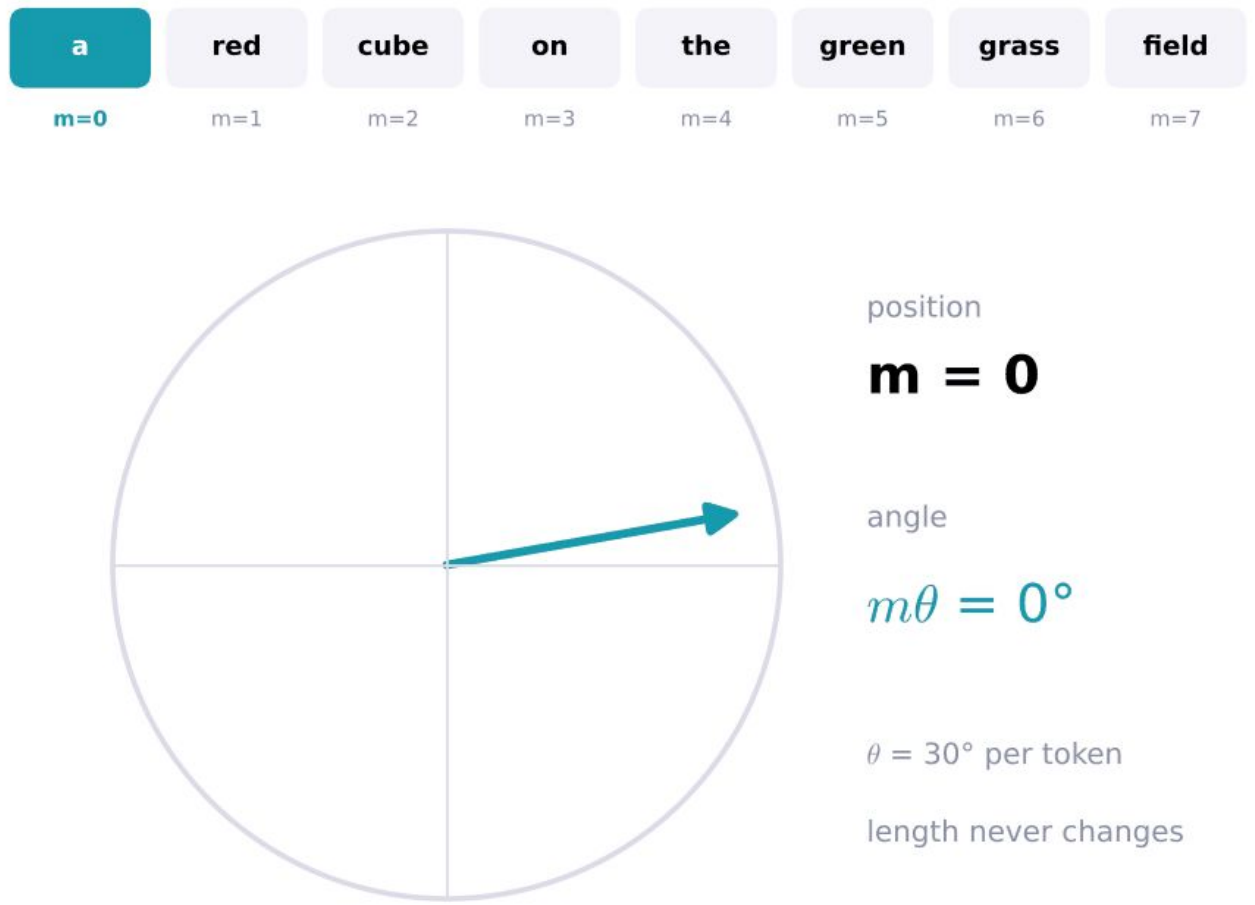
● **most layers:** almost no change ● **vital:** the image changes a lot



top row: reference
Gℓ = layer ℓ skipped
pink = vital · **teal** = not vital

paper →

RoPE in 2D: walking along the sequence



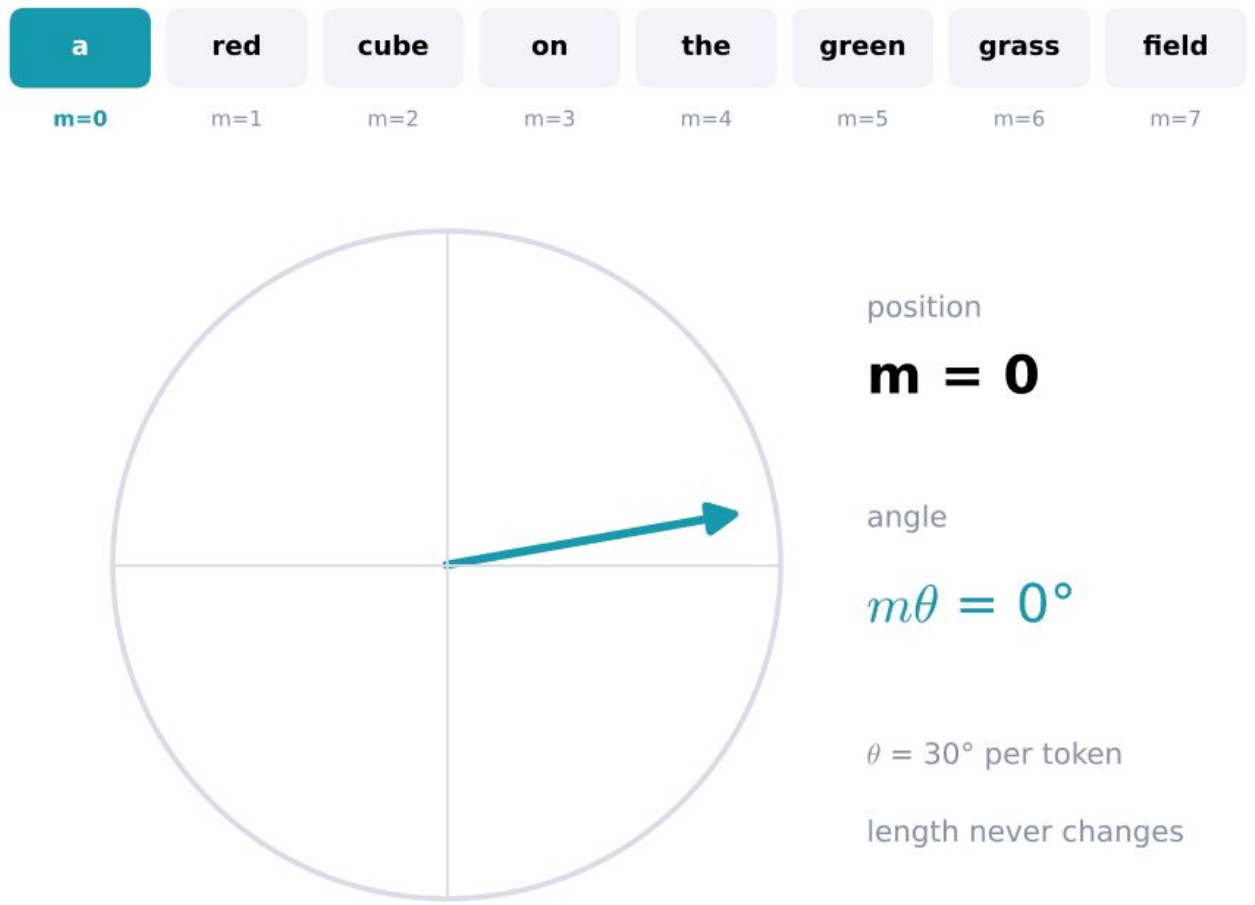
D = 2: ONE PAIR

the whole vector q = (x, y) is just rotated

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos m\theta & -\sin m\theta \\ \sin m\theta & \cos m\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

paper →

RoPE in 2D: walking along the sequence



D = 2: ONE PAIR

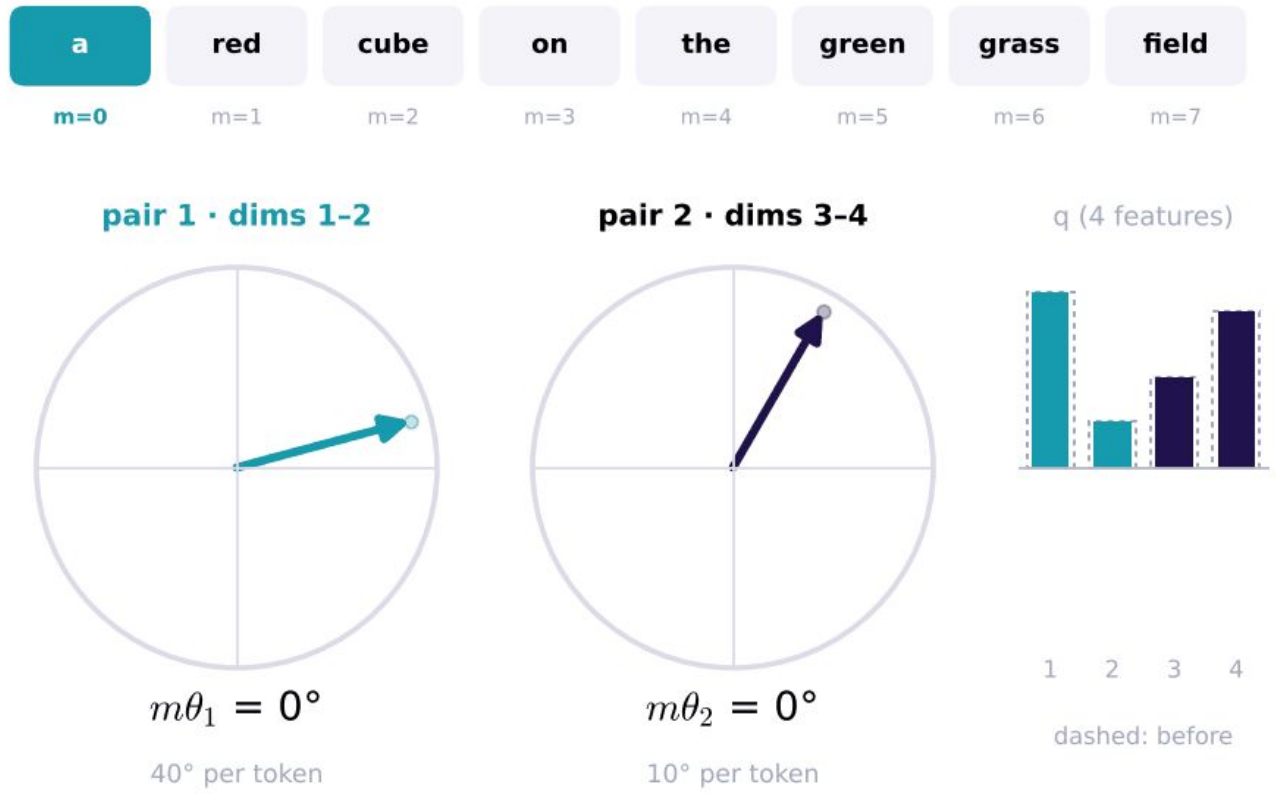
the whole vector q = (x, y) is just rotated

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos m\theta & -\sin m\theta \\ \sin m\theta & \cos m\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- token at position m → angle mθ
- the length never changes
- for two tokens the angle between them is (n − m)θ: this is all attention sees

paper →

RoPE in 4D: two pairs, two speeds



BLOCK-DIAGONAL ROTATION

$$R_m = \begin{bmatrix} R(m\theta_1) & 0 \\ 0 & R(m\theta_2) \end{bmatrix}$$

ATTENTION SCORE

$$(R_m q_m)^\top (R_n k_n) = q_m^\top R_{n-m} k_n$$

depends only on the distance n − m

FreeFlux · ICCV 2025 →

DLS

FreeFlux: position vs content layers

double-stream · 0–18

single-stream · 19–56



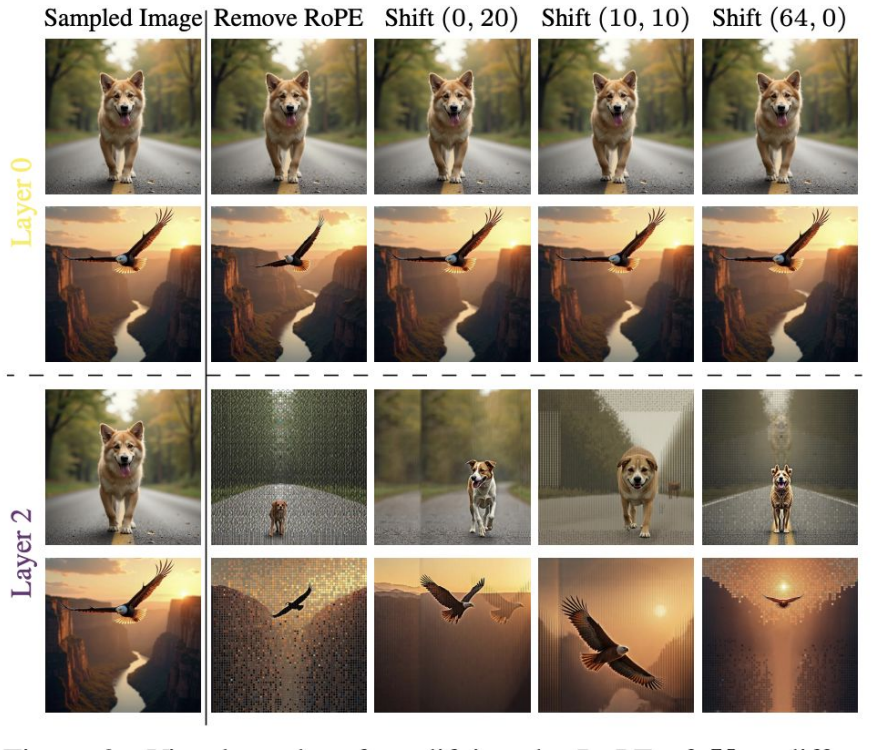
position-dependent

content-dependent

1. **shift key RoPE in one layer**
   queries keep their positions

2. **image breaks → position layer**
   PSNR drop, 50 prompts × 5 seeds

3. **pick layers per edit**
   add object → position · pose → content

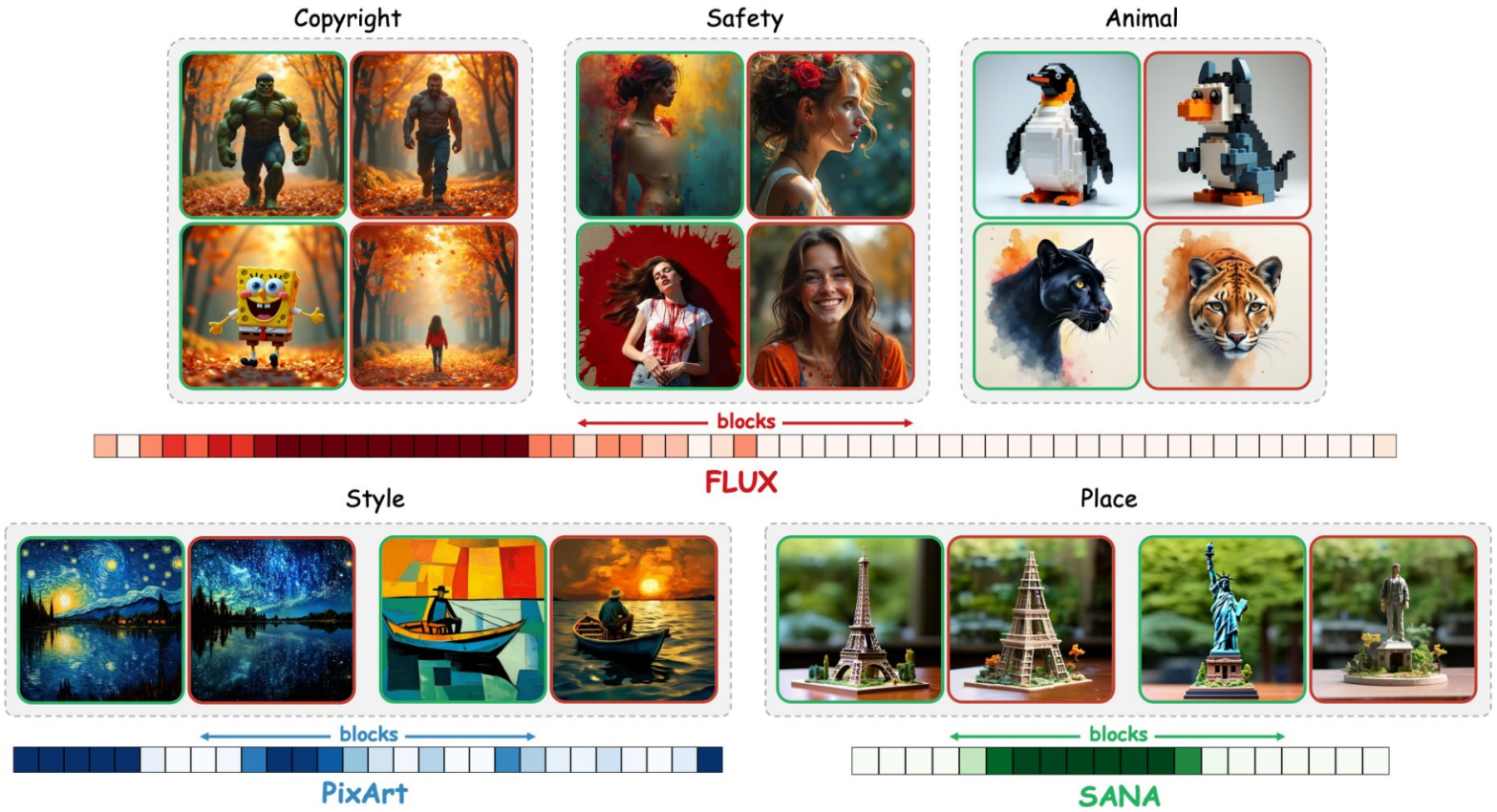


Zarei et al. · NeurIPS 2025 →

Basu et al. · 2024 →

DLS

Localizing knowledge: blocks per concept



How to read

bar = blocks of the model
darker = carries the concept more often

green frame: normal
red frame: concept withheld
in the localized blocks only

What it shows

a few blocks hold a concept

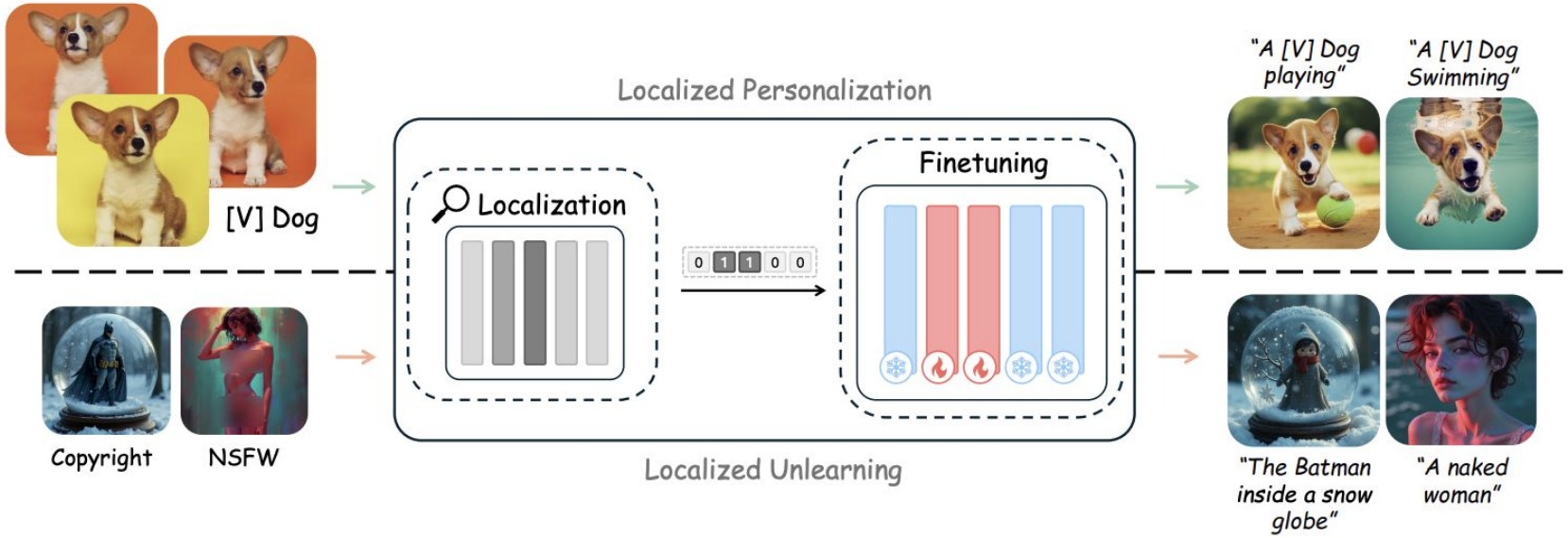
FLUX: early blocks
SANA: a narrow band
PixArt: spread out

Zarei et al. · NeurIPS 2025 →

Localizing knowledge: train only those blocks

Find the blocks once → fine-tune only them

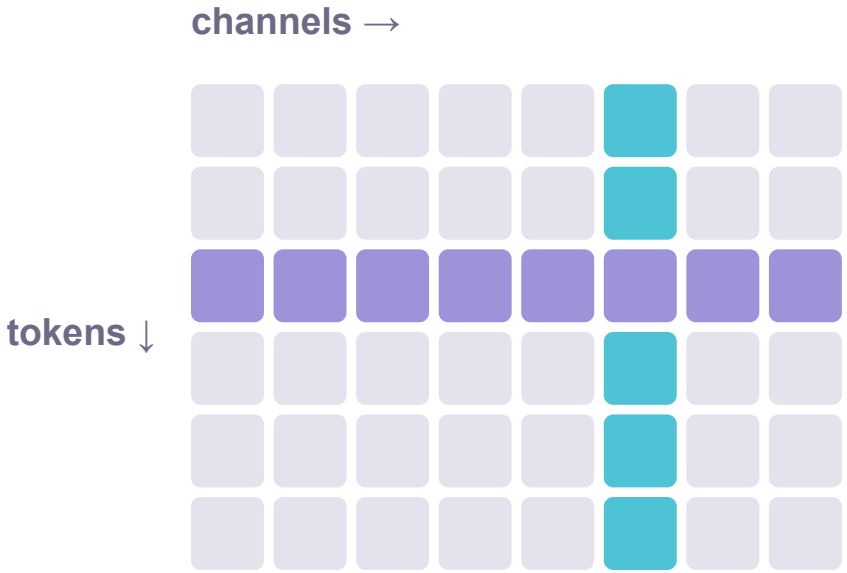
red = trained · **blue** = frozen



top: DreamBooth on 9 of 28 PixArt blocks beats full fine-tuning
bottom: unlearning 30 artist styles on 5 blocks ≈ full fine-tuning, 15–20% faster, ~30% less memory

DLS

Massive activations, registers, sinks



column = huge channel → massive activations
row = huge-norm token → registers

1 Massive activations

LLM: a few huge values on one token
DiT: a few channels, huge on every image token

2 Registers

ViT: huge-norm background patches → extra tokens
DiT: outlier tokens → registers in the denoiser

[Massive Act. · 2024 →]

[StreamingLLM · 2024 →]

DLS

Massive activations in LLMs

A few huge values on one token — and that token becomes the sink

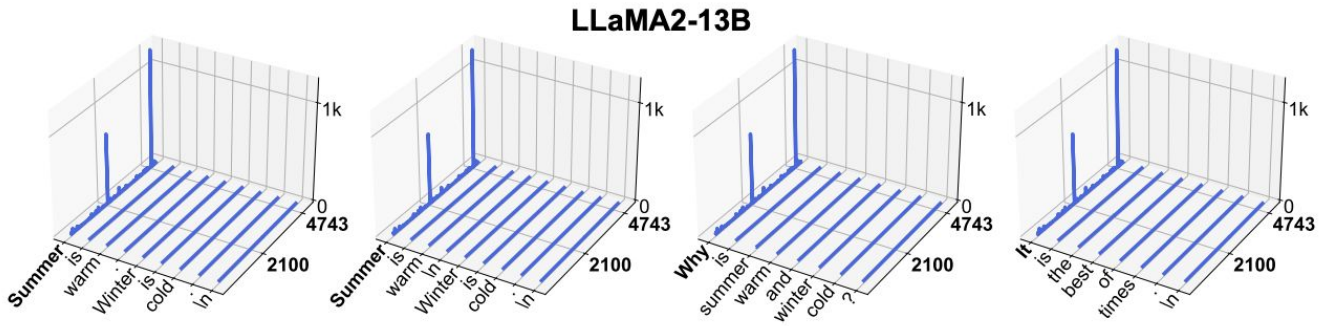


Figure 2: **Massive activations in LLaMA2-13B.** In this model, they appear in two fixed feature dimensions (2100, 4743), and are limited to the starting token.

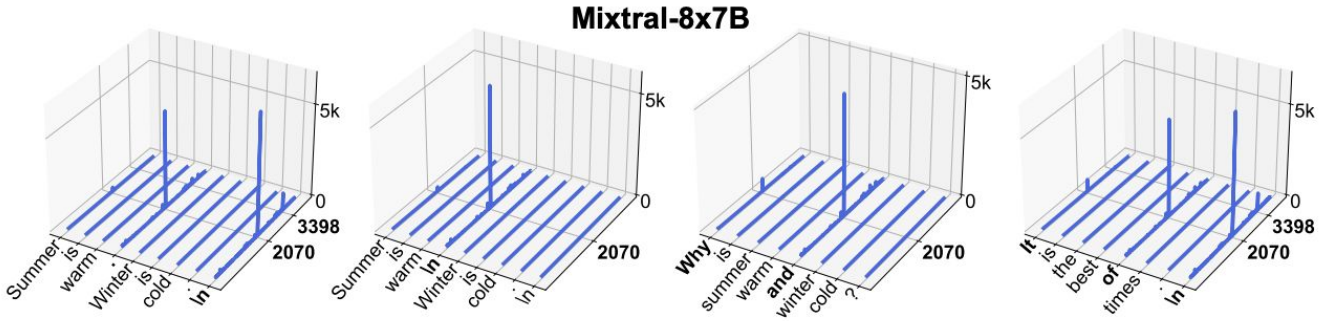


Figure 3: **Massive activations in Mixtral-8x7B.** In this model, they lie in two feature dimensions (2070, 3398), and are found within the starting token, delimiter tokens and certain word tokens ("`and`" and "`of`").

1 **where**
start / delimiter token, a few fixed channels

2 **→ attention sink**
other tokens put their attention there

[Massive Act. · 2024 →] [StreamingLLM · 2024 →]

DLS

Massive activations in LLMs

A few huge values on one token — and that token becomes the sink

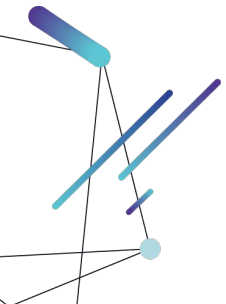
	LLaMA2-7B				LLaMA2-13B			
Intervention	WikiText	C4	PG-19	Mean Zero-Shot	WikiText	C4	PG-19	Mean Zero-Shot
Original	5.47	7.85	8.57	68.95%	4.88	7.22	7.16	71.94%
Set to zero	`inf`	`inf`	`inf`	36.75%	5729	5526	4759	37.50%
Set to mean	5.47	7.86	8.59	68.94%	4.88	7.22	7.16	71.92%

Table 3: Intervention analysis of massive activations in LLaMA2-7B and 13B. We set massive activations to fixed values and evaluate the perplexity (↓) and zero-shot accuracy (%, ↑) of intervened models.

1. **where**
   start / delimiter token, a few fixed channels

2. **→ attention sink**
   other tokens put their attention there

3. **a fixed bias**
   set to mean → no change · zero → model breaks



StreamingLLM · 2024 →

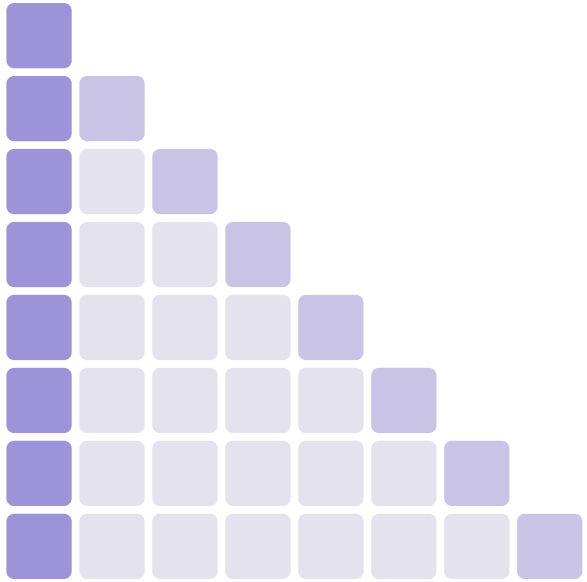
Massive Act. · 2024 →

DLS

Attention sinks: StreamingLLM

The first token soaks up spare attention — evict it from the KV cache and generation falls apart

Attention map



rows = queries, columns = keys
column 0 takes most of it

Window attention: keep the last 8 tokens

first tokens evicted → perplexity explodes

StreamingLLM: 4 sink tokens + the last 8



stable on 4M+ tokens · up to 22× faster than recomputing

sink tokens rolling window evicted

1 Why a sink

softmax sums to 1 — when no key is relevant, the spare attention has to go somewhere

$$\sum_j \mathrm{softmax}\left(q\,k_j^\top/\sqrt{d}\right) = 1$$

2 Position, not meaning

replace the first 4 tokens with “\n” — streaming still works

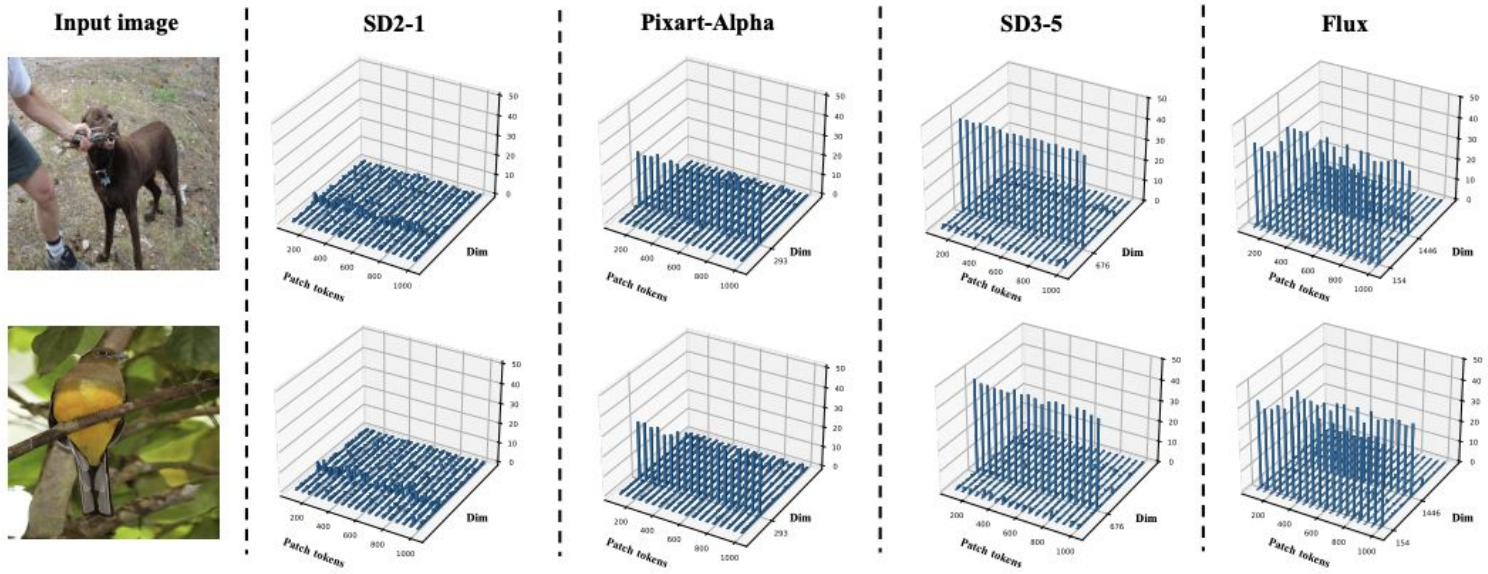
3 A dedicated sink

pretrain with one learnable sink token — then that one token is enough

Gan et al. · 2025a →

Massive activations in DiTs

Not one token: a few fixed channels, huge on every image token



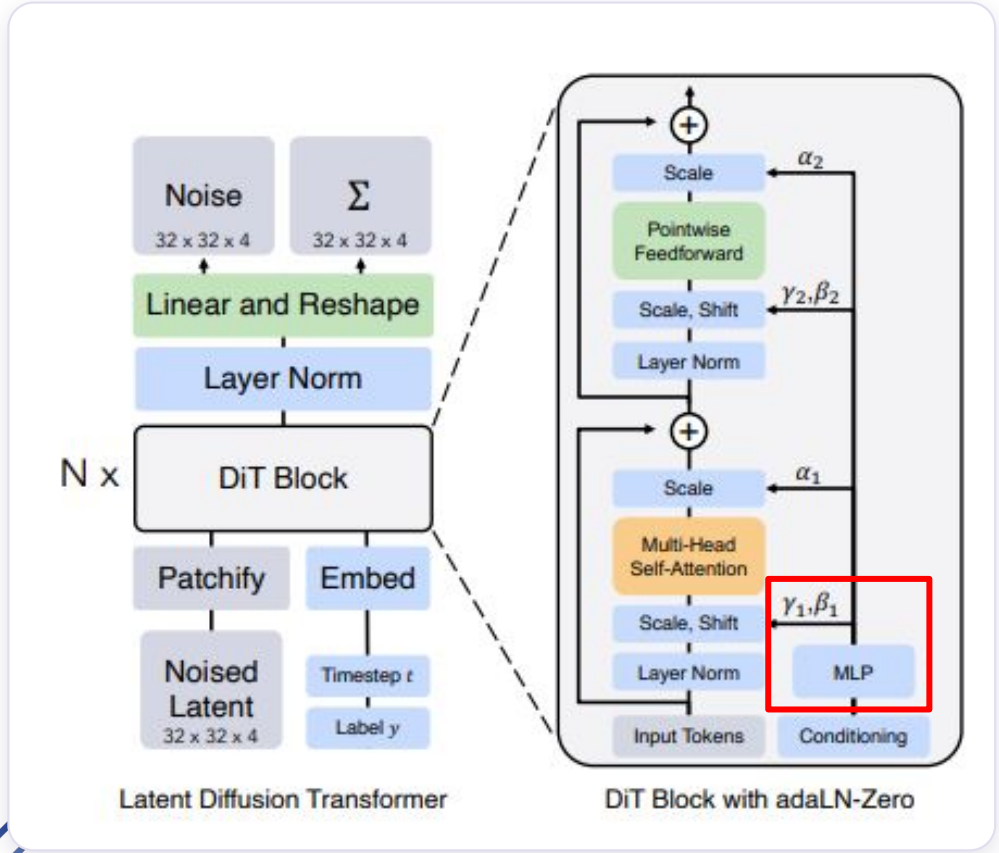
1 **where**

FLUX: dims 154, 1446 · SD3.5: dim 676

Gan et al. · 2025a →

Massive activations in DiTs

Not one token: a few fixed channels, huge on every image token



1 **where**
FLUX: dims 154, 1446 · SD3.5: dim 676

2 **set by**
the adaLN-zero scale, i.e. the timestep

Gan et al. · 2025a →

Massive activations in DiTs

Not one token: a few fixed channels, huge on every image token

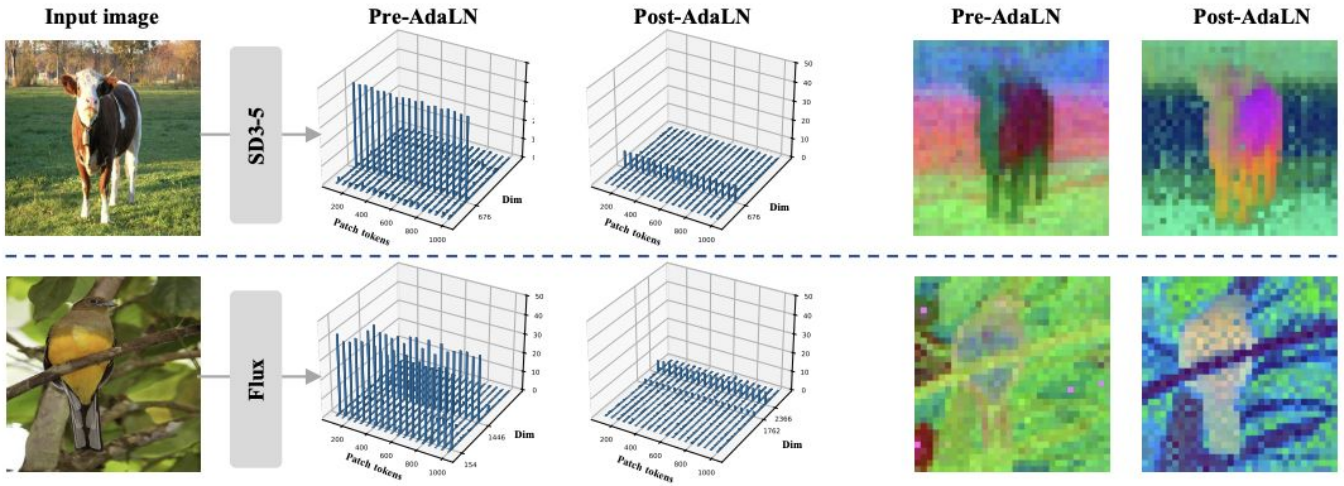


Figure 5: **AdaLN reduces massive activations and enhances feature semantics and discrimination.** We visualize the activation magnitudes and feature maps of both pre-AdaLN and post-AdaLN features in DiT block.

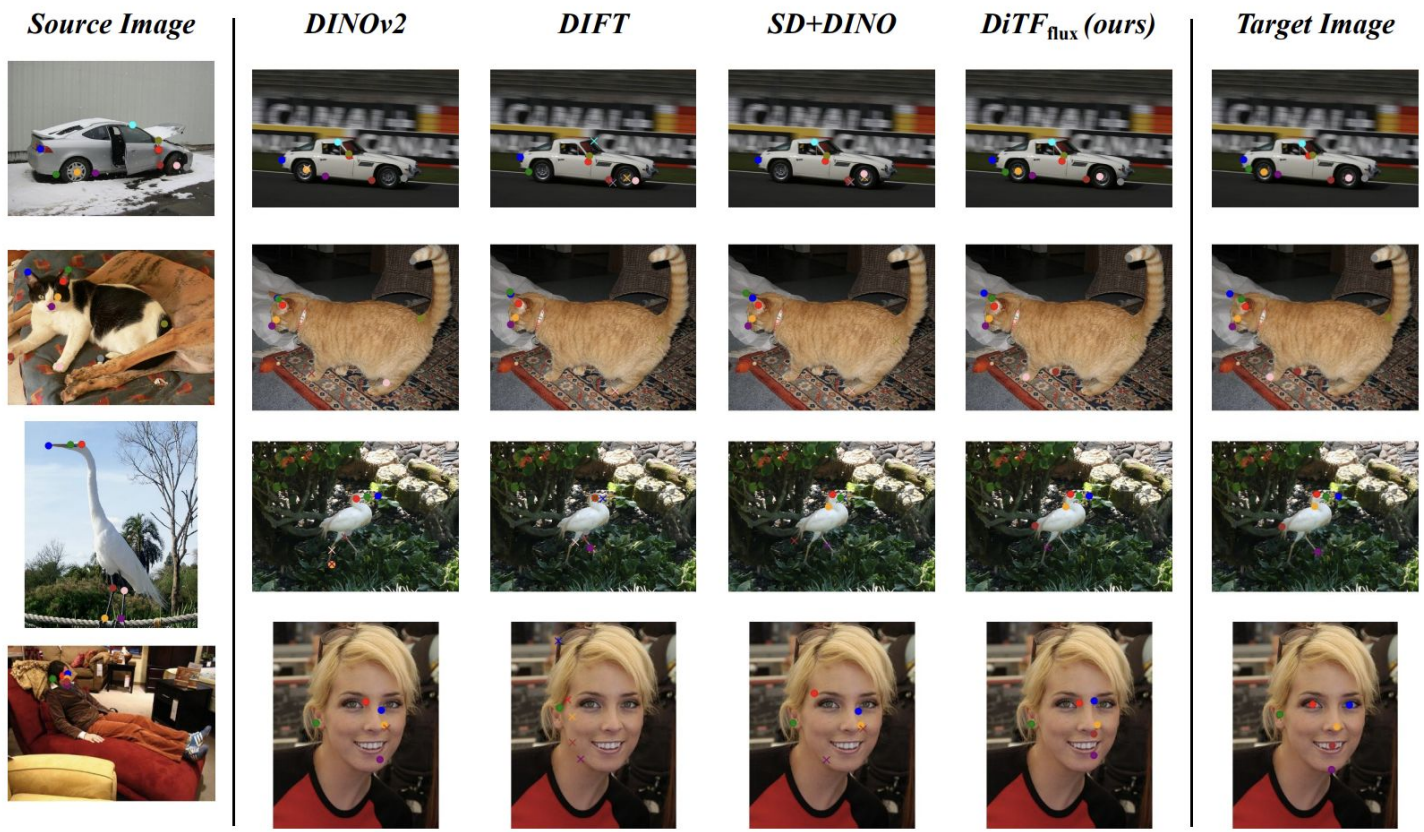
1. **where**
   FLUX: dims 154, 1446 · SD3.5: dim 676

2. **set by**
   the adaLN-zero scale, i.e. the timestep

Gan et al. · 2025a →

Massive activations in DiTs

Not one token: a few fixed channels, huge on every image token



1. **where**
   FLUX: dims 154, 1446 · SD3.5: dim 676

2. **set by**
   the adaLN-zero scale, i.e. the timestep

3. **why care**
   they hide local info, normalise → better matching

Gan et al. · 2025b →

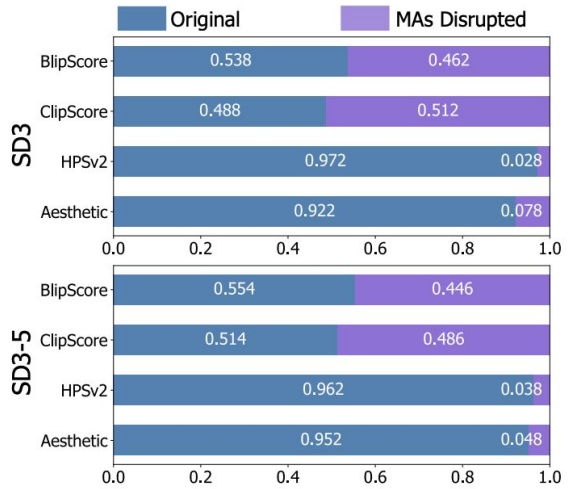
DLS

Massive activations → detail

Switch off 2 channels in one block: the details go, the content stays



(a) Visual comparison



(b) Win probability comparison

What is switched off

blocks



one block (FLUX: block 22)

channels

2 of 3072 (FLUX: 154, 1446)

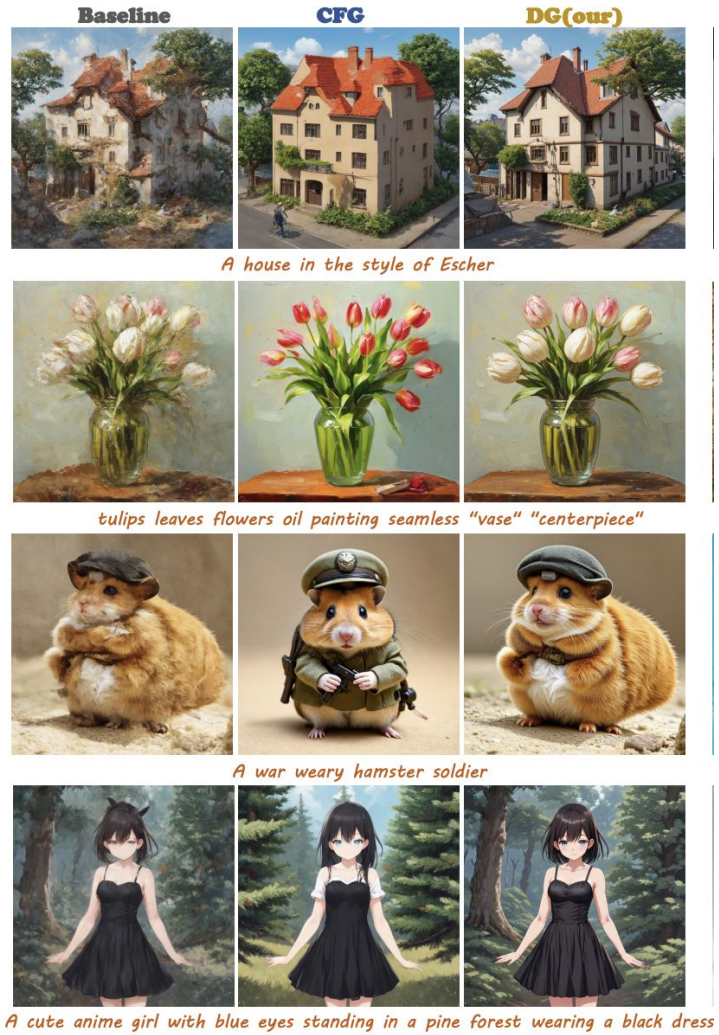
What happens

same content: CLIP ≈ 50 / 50
details collapse: loses 97% on HPS

→ massive activations = detail

Gan et al. · 2025b →

Detail Guidance



Guide away from the model without massive activations

CFG

$$\hat{D} = D(c) + \lambda \left(D(c) - D(\varnothing) \right)$$

negative: the same model, no prompt

Detail Guidance

$$\hat{D} = D(c) + w \left(D(c) - D_{\text{off}}(c) \right)$$

negative: massive activations off

How the negative is made

zero the massive channels at one block, run the rest
FLUX: dims 154, 1446 at block 22

Results on FLUX

	HPSv2.1	Aesthetic
no CFG → DG	19.3 → 27.9	5.50 → 6.13
CFG → CFG + DG	29.2 → 29.3	5.96 → 6.12

Few Channels · 2026 →

DLS

Few Channels Draw the Whole Picture

Three experiments with the largest channels of a DiT



Figure 1: Overview of our findings. (Left) Disrupting the top-k massive activation channels severely degrades generation quality, revealing their functional importance. (Center) These channels exhibit structured spatial organization, which we capture as a mask. (Right) Transplanting the top-k activations within this mask from a source to a target generation enables localized semantic transfer, producing coherent compositions rather than pixel-level blending.

1 switch them off

every block, every step → the image breaks

blocks

every block, every step

channels

top-k: 12 … up to 1024 of 3072

2 cluster on 12 of them

→ subject mask

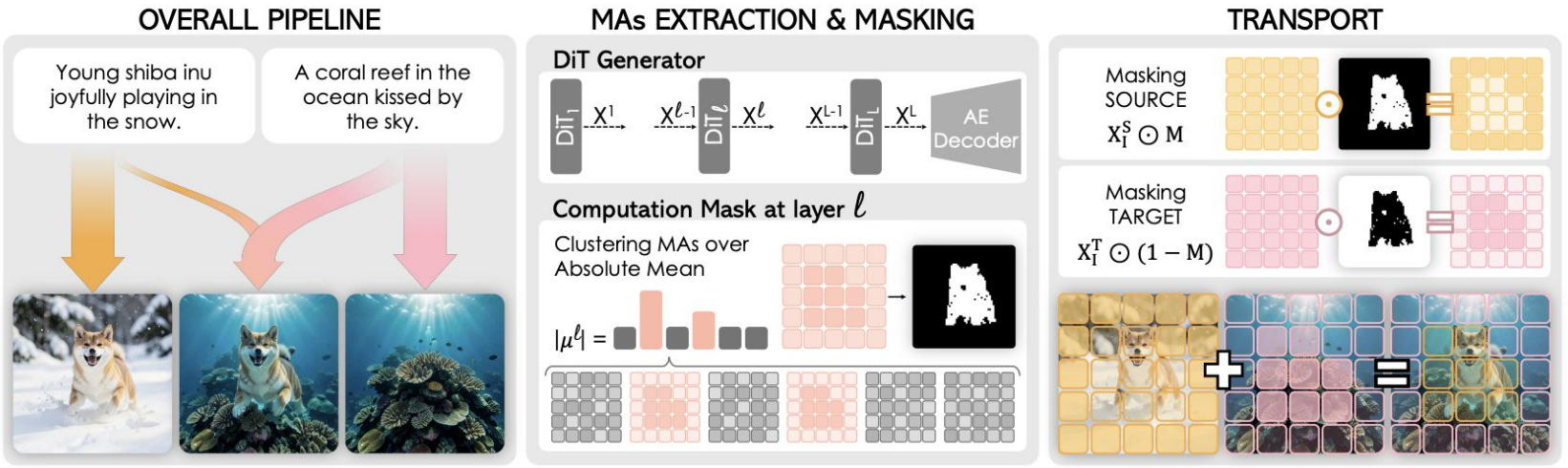
3 copy them inside the mask

→ the corgi moves into the desert

Few Channels · 2026 →

Few Channels: how the transfer works

Two prompts, one noise — mix the top channels inside the subject mask



1 same noise, two prompts
shiba in the snow + coral reef → shiba in the reef

2 find channels and mask
top channels by |mean| → cluster tokens → mask M

3 transport
inside M take the source, outside the target

$$X_I = X_I^S \odot M \; + \; X_I^T \odot (1 - M)$$

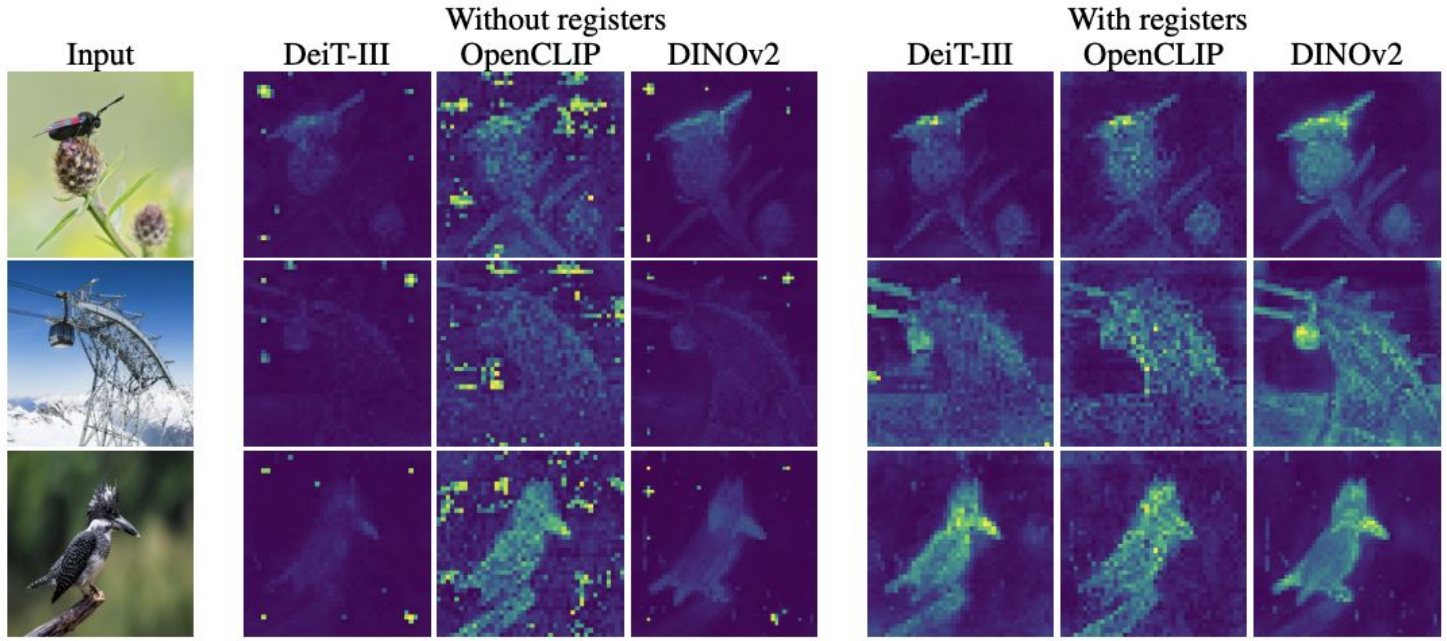
[ViT Registers · ICLR 2024 →]

[Test-time registers · NeurIPS 2025 →]

DLS

Registers in ViTs

Big ViTs recycle ~2% of background patches as scratch space



1. **artifacts**
   a few patches get a huge norm, in the background

2. **what they hold**
   global info, not local: bad at position, good at class

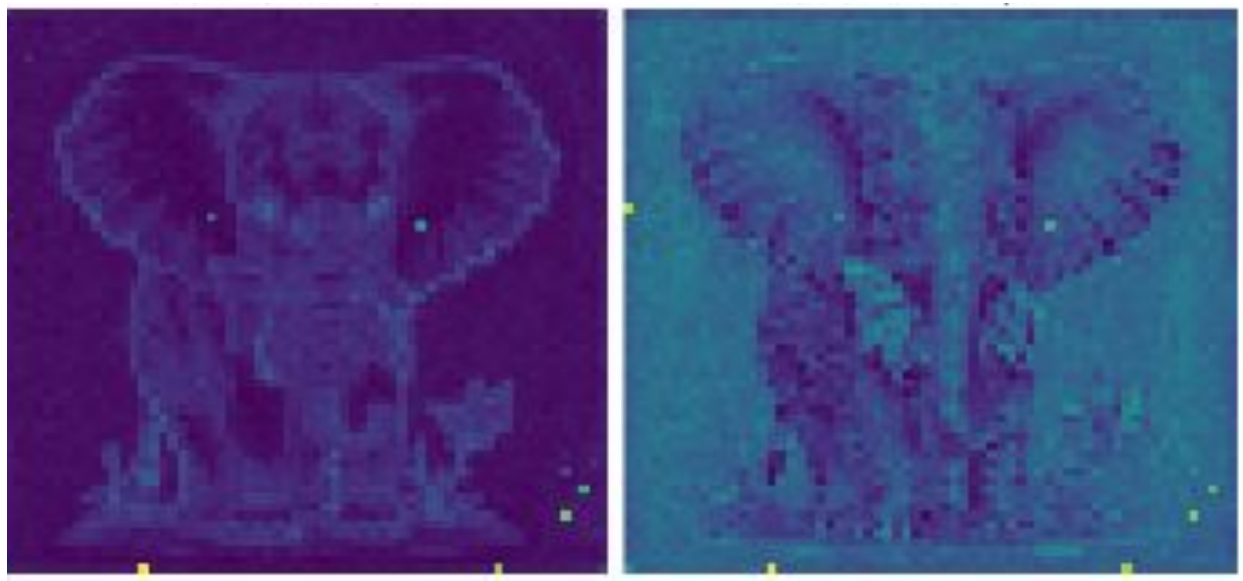
3. **fix: add registers**
   extra tokens, dropped at the output → clean maps

Registers Matter · ICML 2026 →

DLS

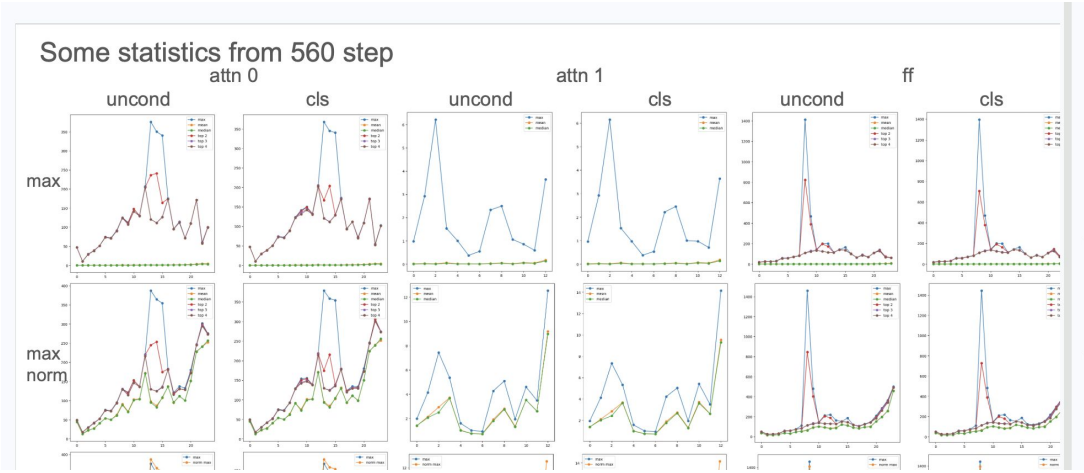
Bright tokens in FLUX

Pretrained FLUX does have bright image tokens



1 **FLUX**
outliers in the text tokens and several bright image tokens

1 **SD3.5**
outliers in the text tokens and several bright image tokens



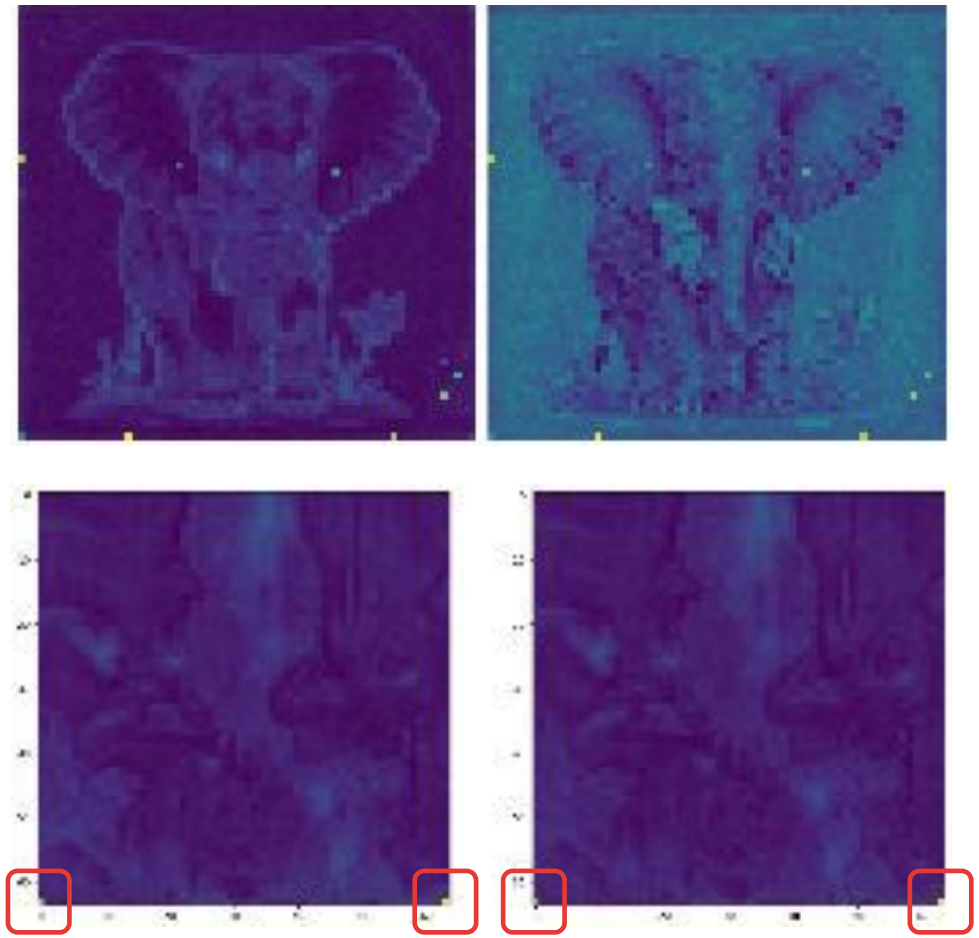
Li et al. · 2026 →

Bright tokens in FLUX and SD3.5

A few image tokens are far brighter than the rest — in both models

FLUX

SD3.5



1. **token norm per patch**
   brighter = larger norm

2. **a handful of outliers**
   isolated tokens with a much larger norm

3. **in both FLUX and SD3.5**
   unlike the small ImageNet DiTs

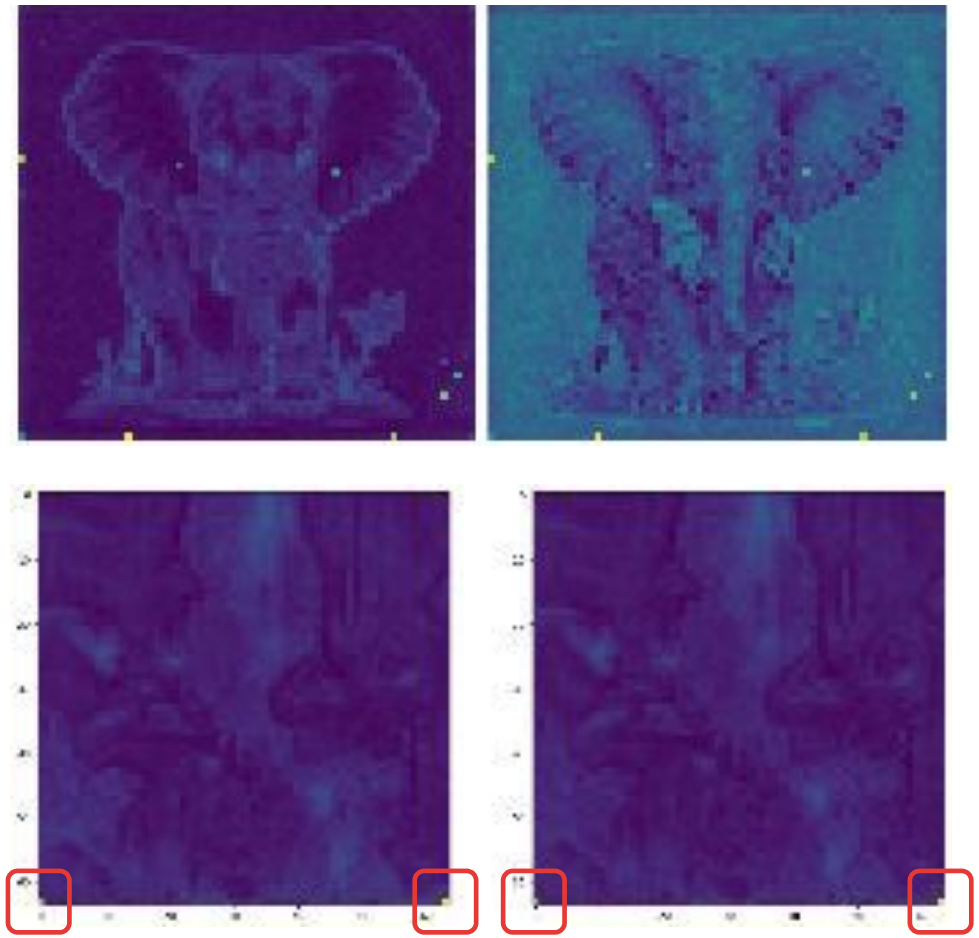
Li et al. · 2026 →

Bright tokens in FLUX and SD3.5

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FLUX

SD3.5



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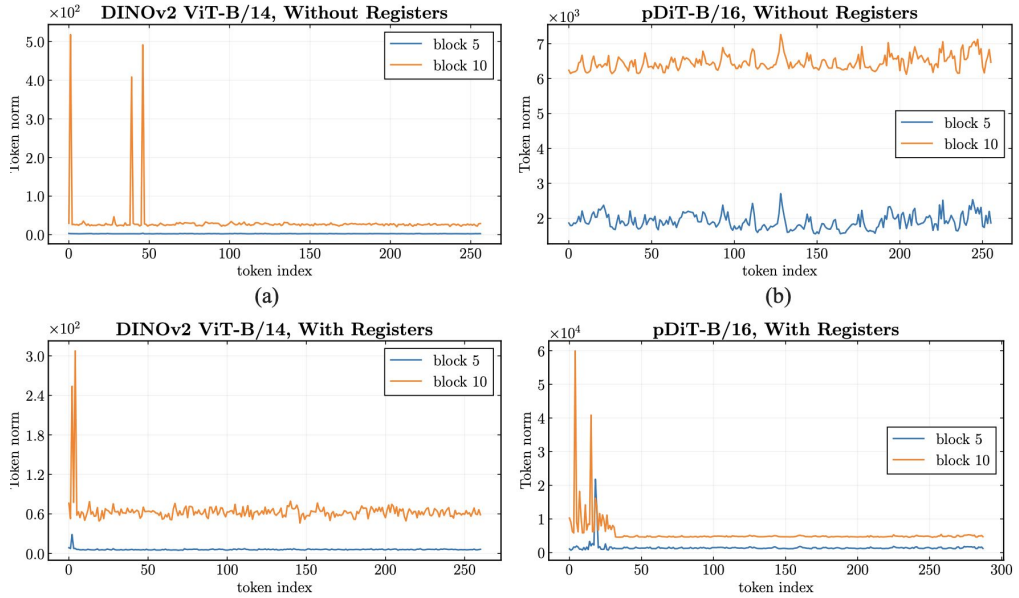
4. **one learnable register in FLUX**
   takes over part of the attention sinks · GenEval 67.1 → 68.9, a modest gain

Registers Matter · ICML 2026 →

DLS

Registers in pixel-space DiTs

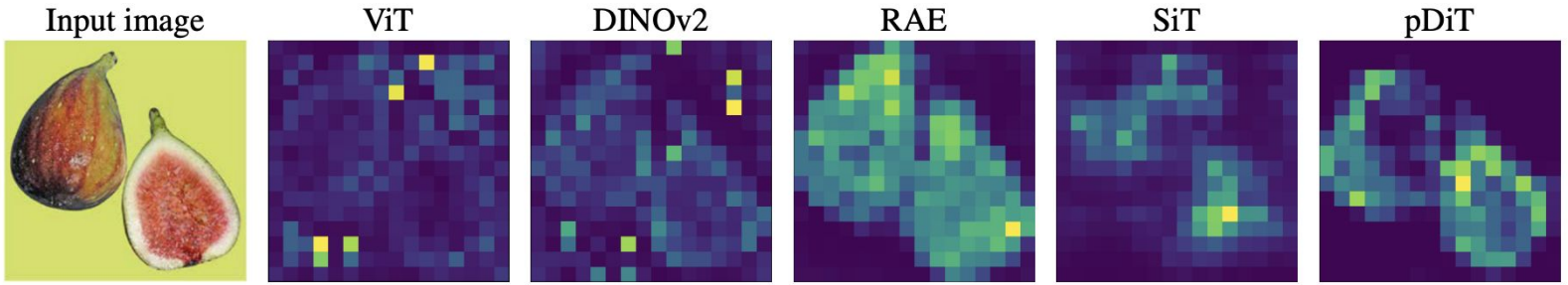
Their ImageNet DiTs show no outlier patches — yet registers still help



1. **no outlier patches**
   in their ImageNet models; every patch is in the loss

2. **add registers**
   the huge norms move into the registers

3. **cleaner features**
   smoother patch features at high noise · 32 registers best

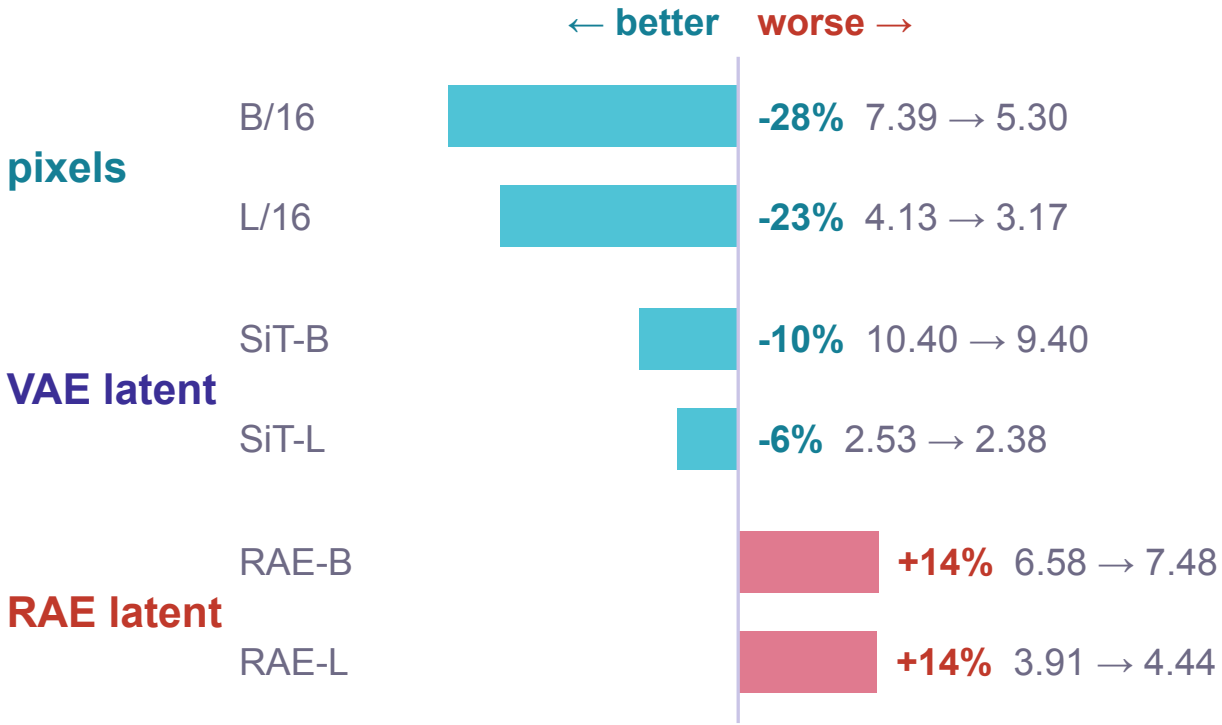


Registers Matter · ICML 2026 →

DLS

Registers: where diffusion happens decides

Big gain in pixels, small in a VAE latent, none in an RAE latent



change of FID with registers · ImageNet 256 · as reported

Why

pixel space is the hardest: largest patch norms, noisiest features

latent spaces are already compact and structured

What it means for FLUX / SD

VAE latent → a small gain expected

but FLUX does have bright image tokens → next slides

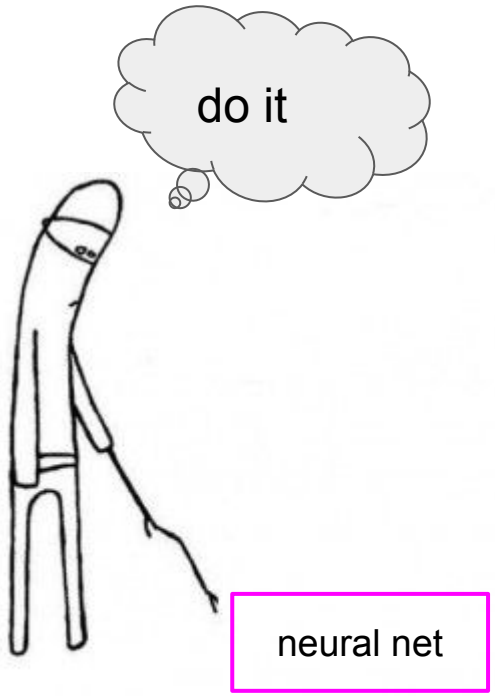
DLS

Steering

05 · control without retraining

DLS

Making a pretrained model do our task

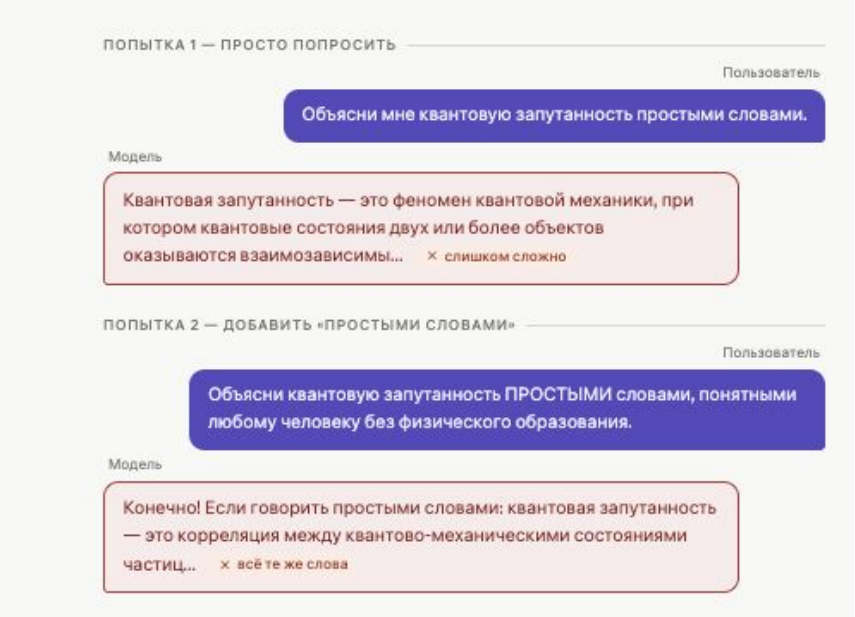


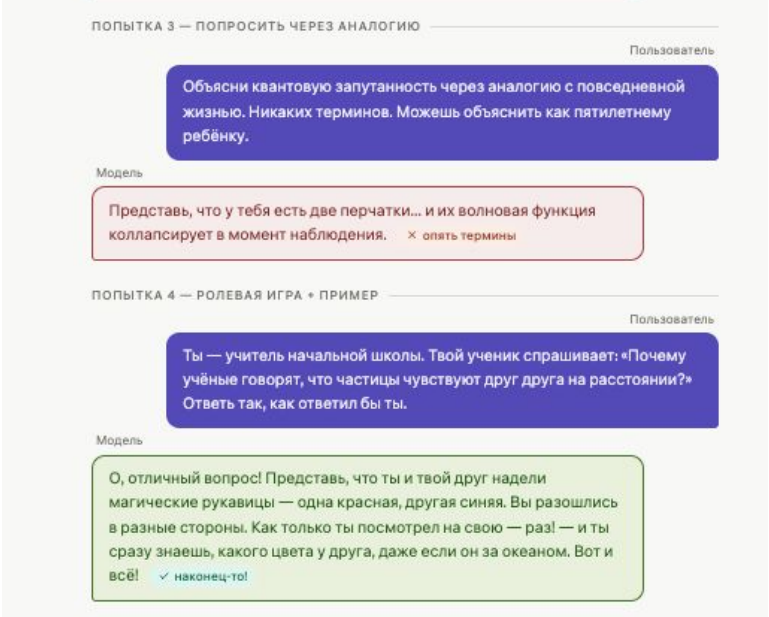
prompting → full fine-tuning → partial fine-tuning → adapters → steering

GPT-3 · 2020 →

DLS

Prompt engineering

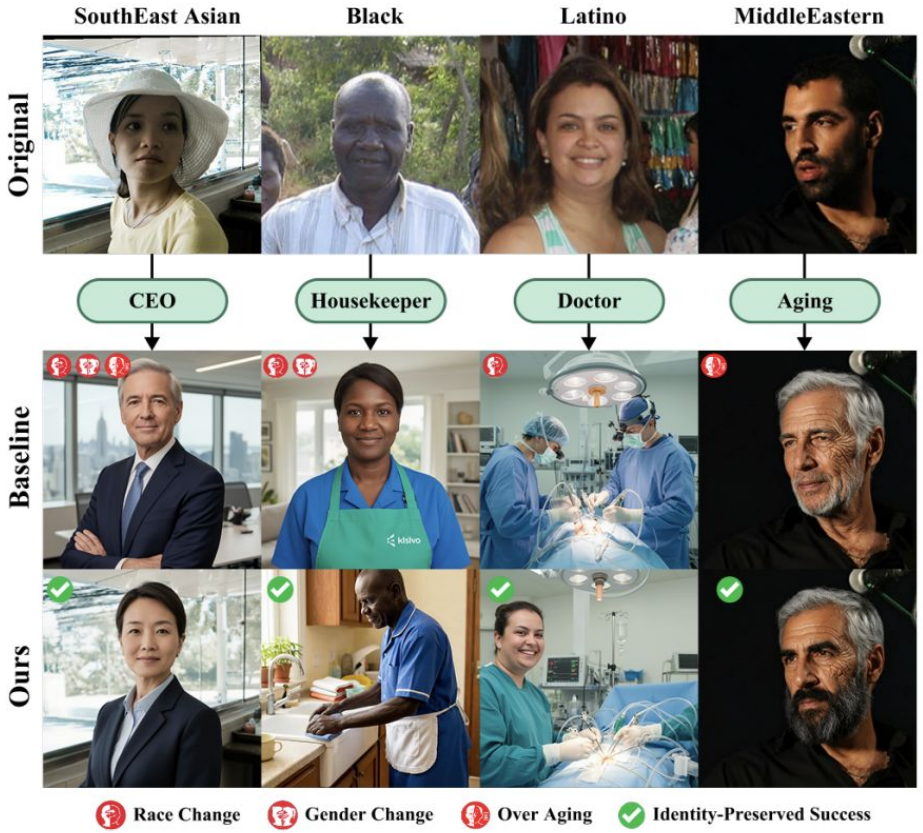


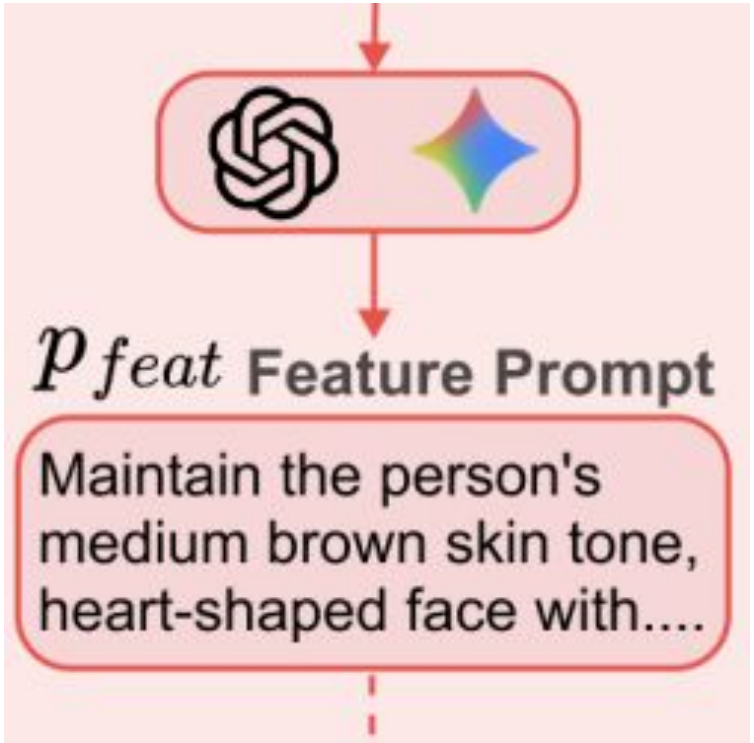


Cheap, but brittle: we only control the input

DLS

Prompt engineering

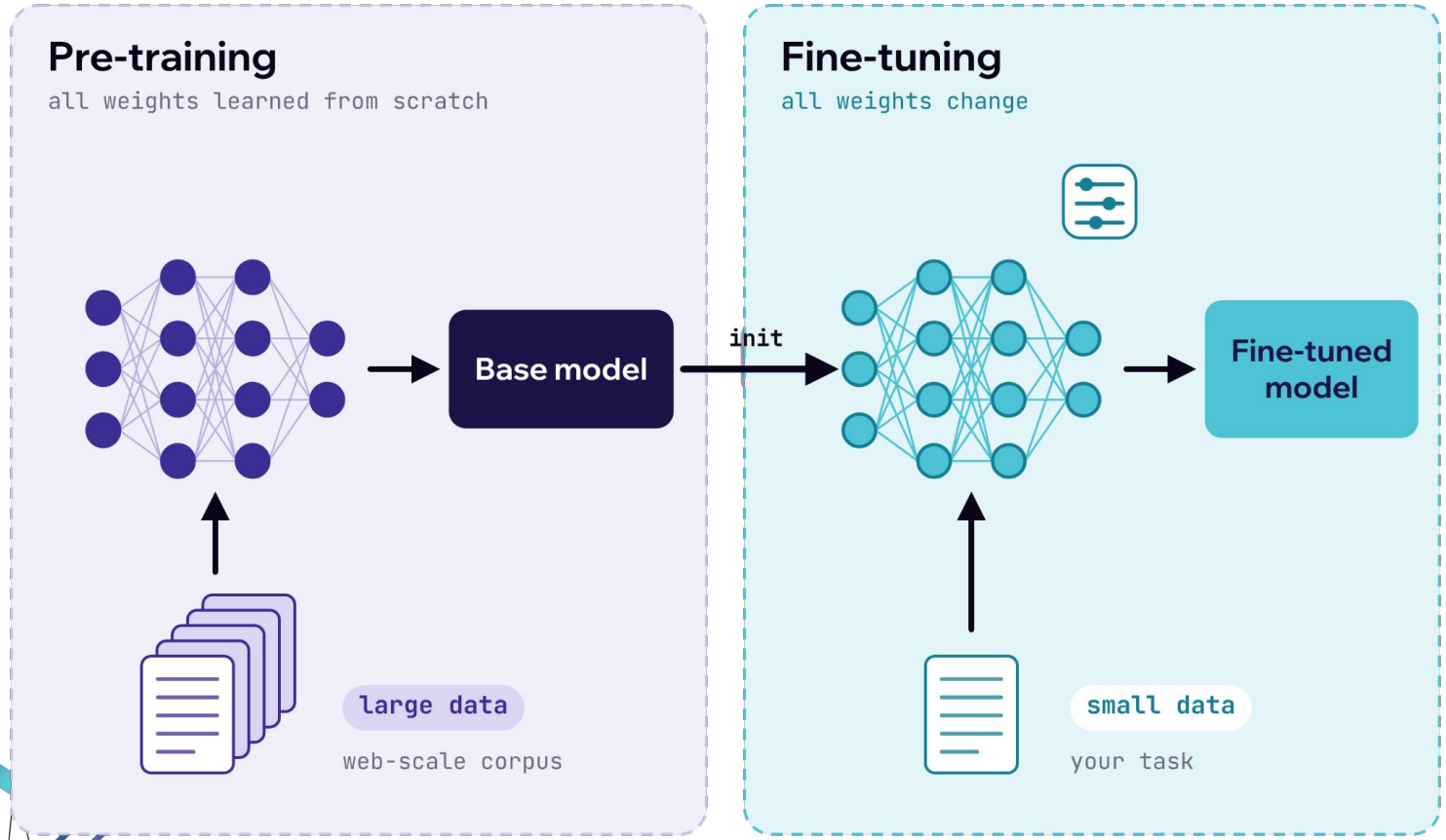




For diffusion also works

DLS

Fine-tuning the whole model

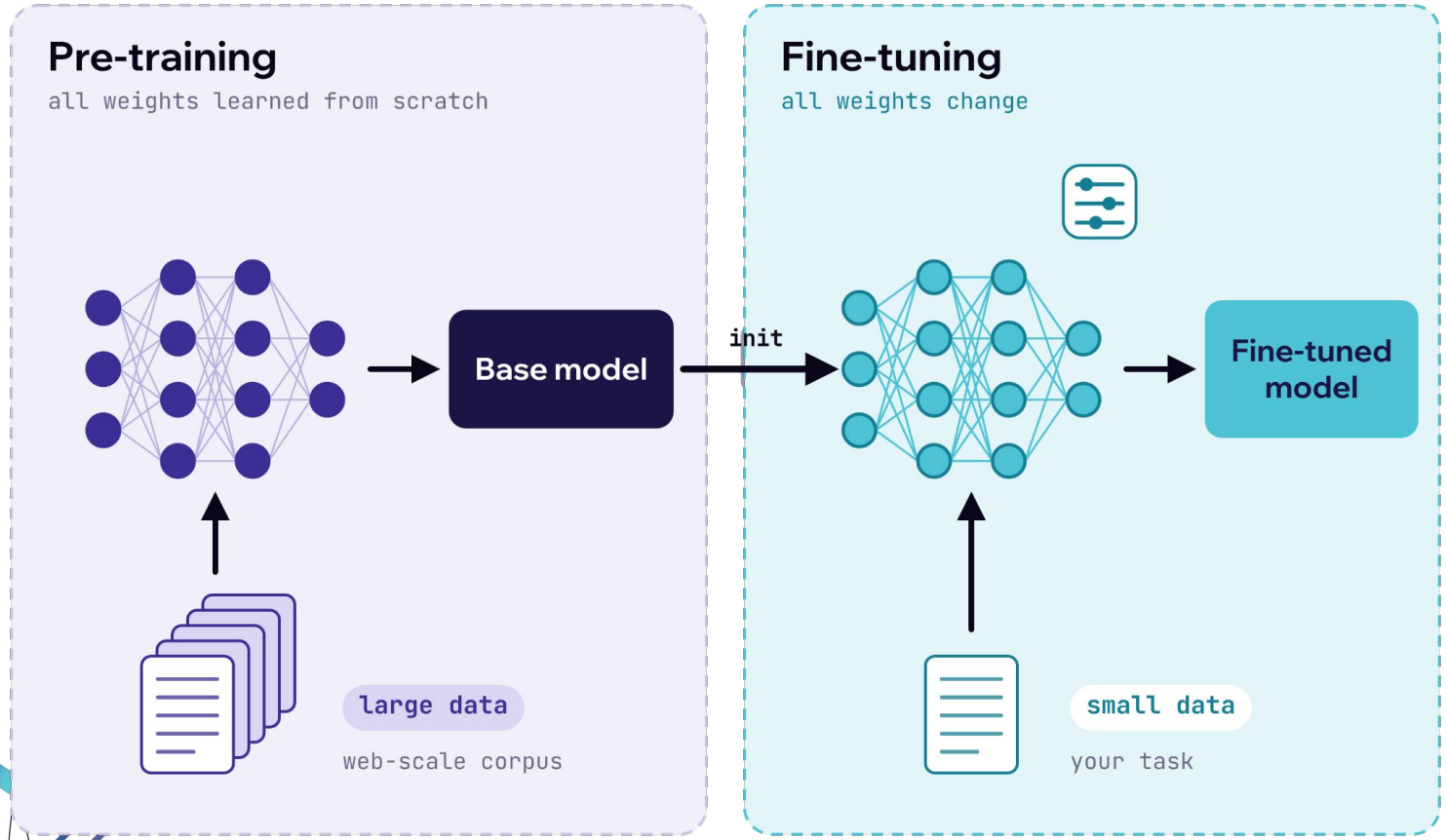


All weights change

Needs data + GPUs

DLS

Fine-tuning the whole model



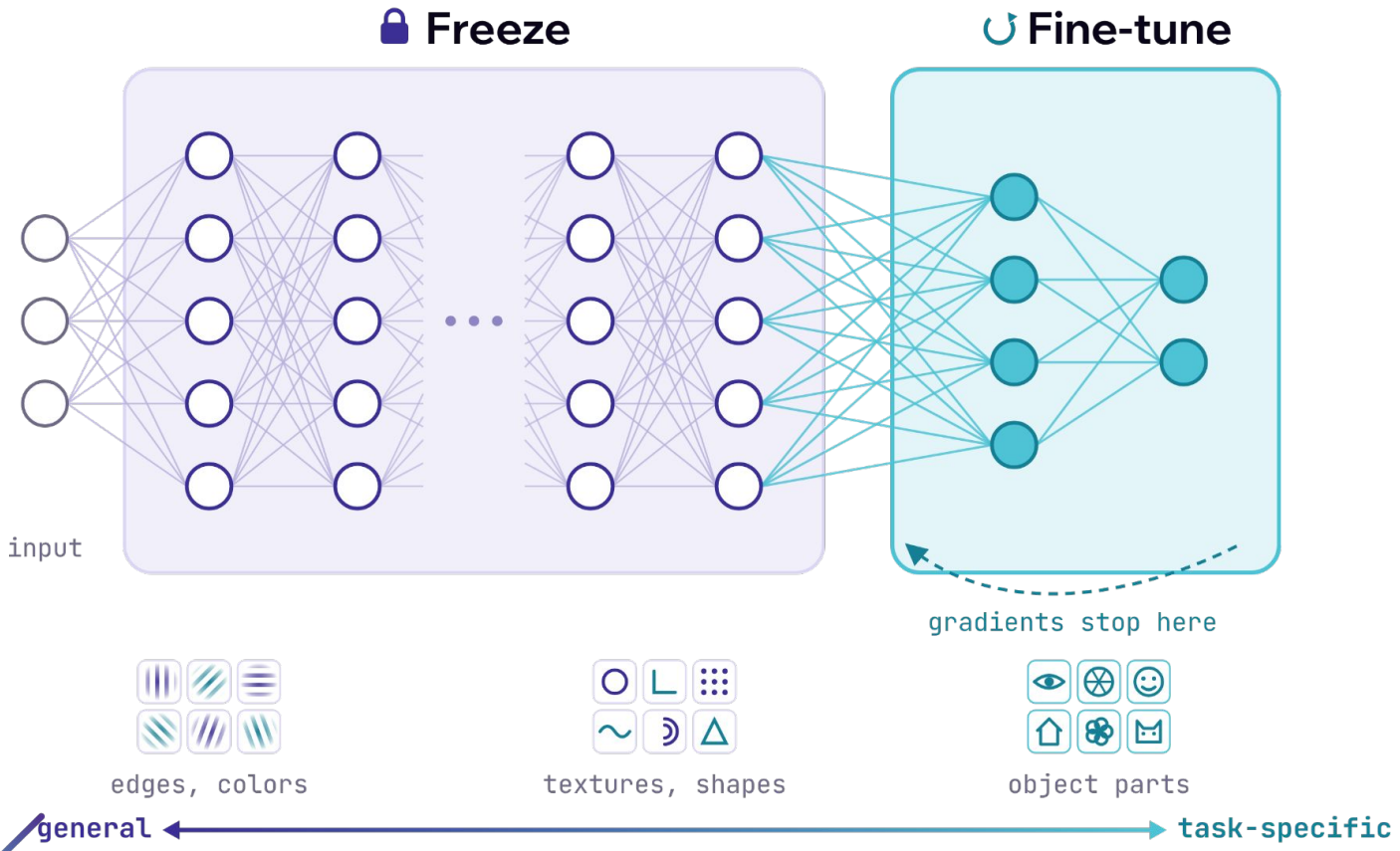
All weights change

Needs data + GPUs



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Fine-tuning part of the model



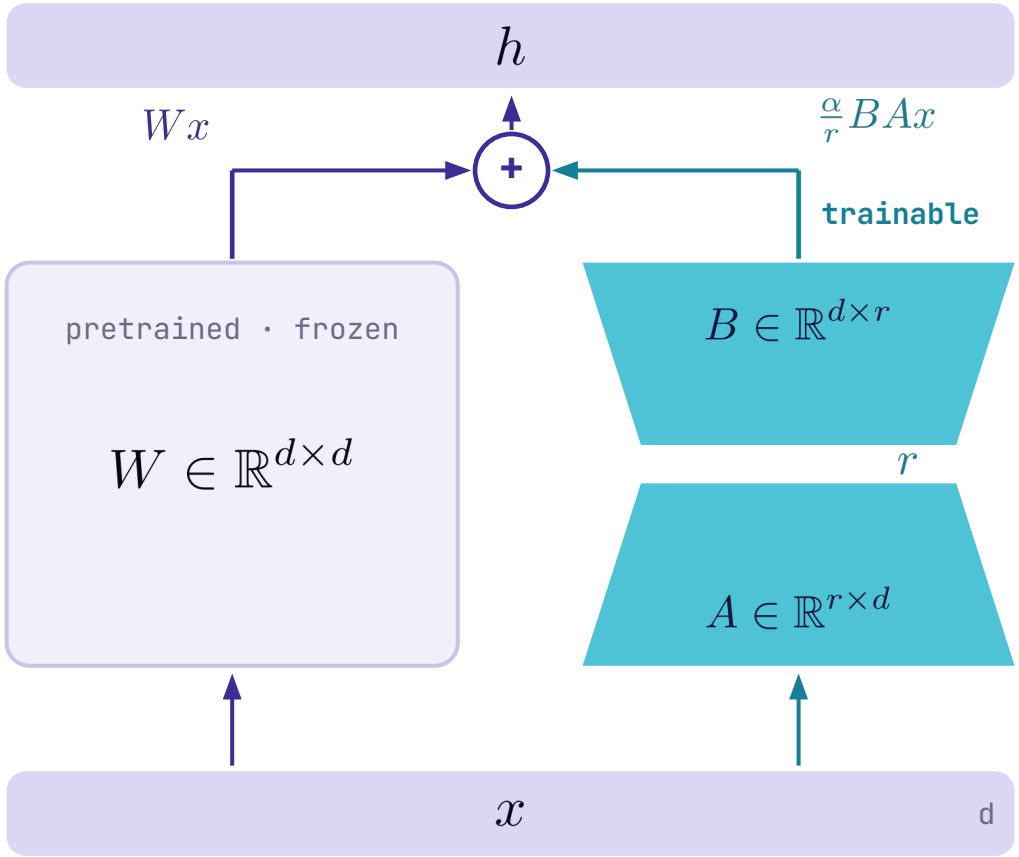
Freeze

early layers: transferable, low-level

Train

last layers: task-specific, high-level

LoRA: a low-rank detour



$$h = Wx + \frac{\alpha}{r} BA\,x$$

$$r \ll d$$

$$B_0 = 0, \quad A_0 \sim \mathcal{N}(0, \sigma^2) \;\Rightarrow\; \Delta W_0 = 0$$

trainable

$$2dr \;\text{ vs }\; d^2$$

d = 4096, r = 8 → 0.4%

after training

$$W \leftarrow W + \frac{\alpha}{r} BA$$

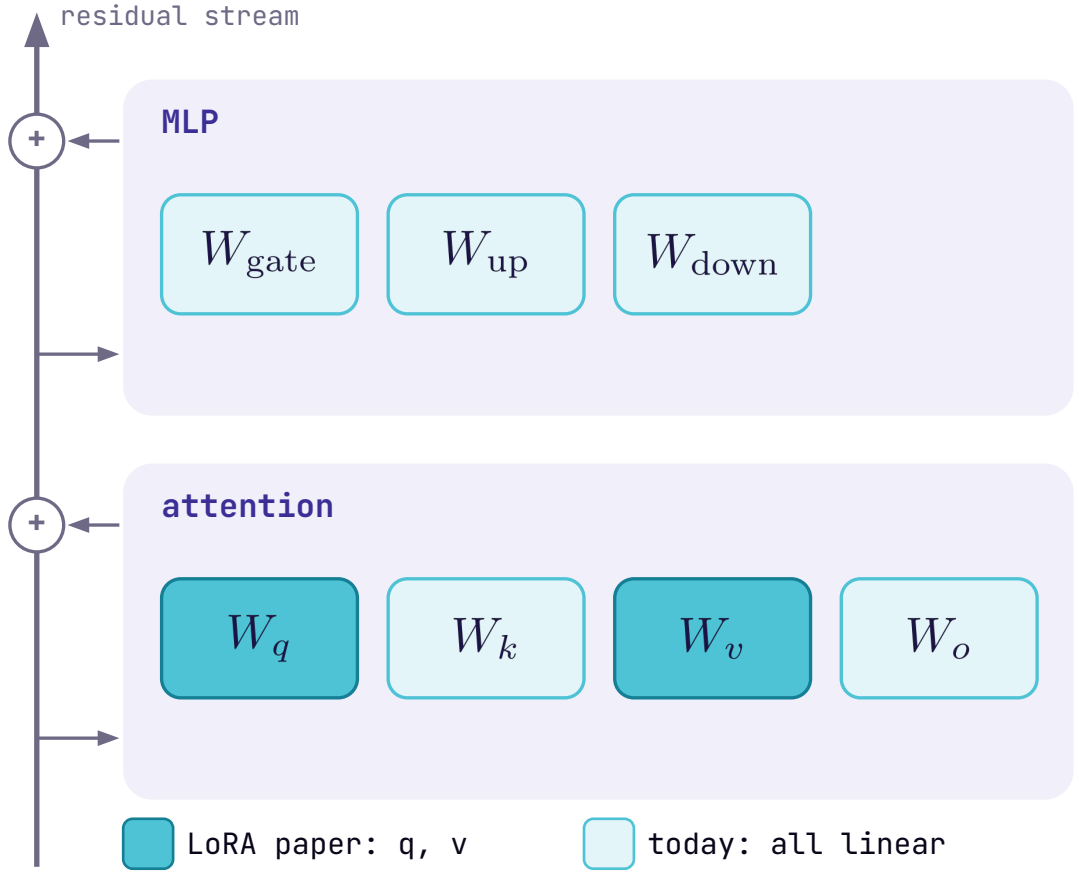
merge → no extra latency

LoRA · 2021 →

QLoRA · 2023 →

DLS

LoRA: which weights in a transformer



Original LoRA

only the q and v projections — best for a fixed budget on GPT-3

QLoRA and practice

all linear layers — needed to match full fine-tuning

Usually frozen

embeddings, norms, LM head

LoRA in diffusion models

U-Net · SD 1.5, SDXL

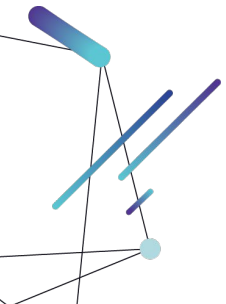
attention in every transformer block

self-attn	to_q	to_k	to_v	to_out
cross-attn	to_q	to_k	to_v	to_out
text (opt.)	q_proj	k_proj	v_proj	out_proj

MM-DiT · SD3, FLUX

joint attention + feed-forward · diffusers defaults

image stream	to_q	to_k	to_v	to_out
text stream	add_q	add_k	add_v	to_add_out
feed-forward	ff	ff_context		

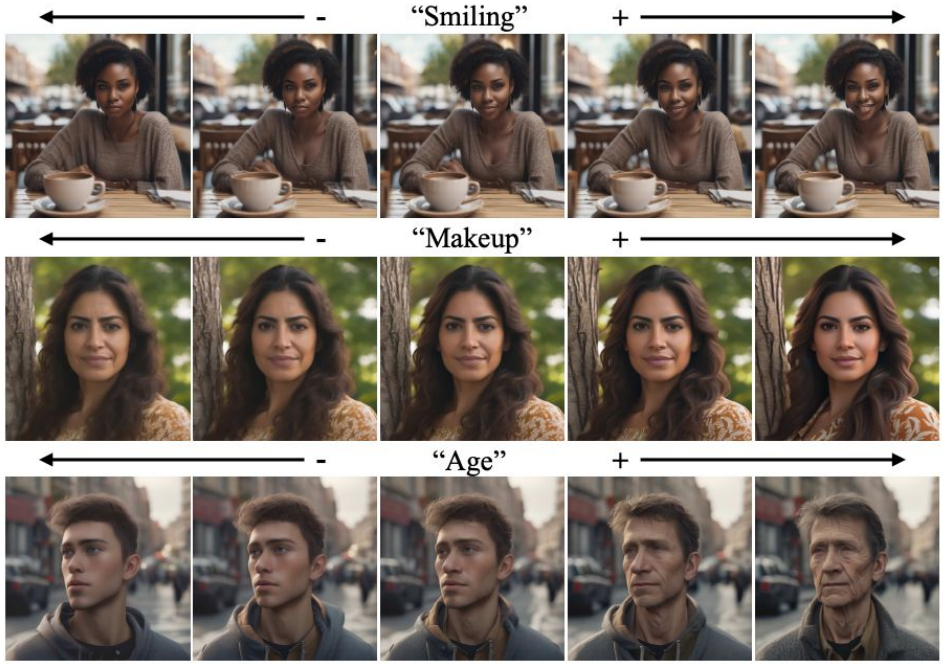


LoRA in diffusion models

DreamBooth-LoRA your subject from 3–5 photos

Style LoRAs one look, swappable

Concept Sliders one LoRA = one attribute dial



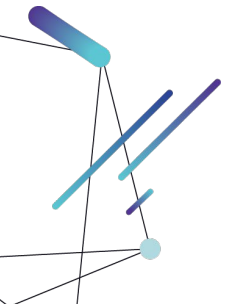


ActAdd · 2023 →

DLS

Steering: change activations, not weights

	What changes	Training	Cost
Prompting	the input	none	free
Full fine-tuning	all weights	yes	high
Partial fine-tuning	some layers	yes	medium
Adapters (LoRA)	a low-rank update	yes	low

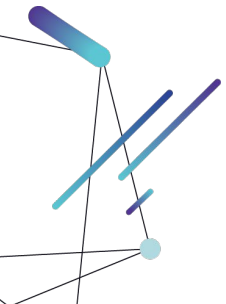


ActAdd · 2023 →

DLS

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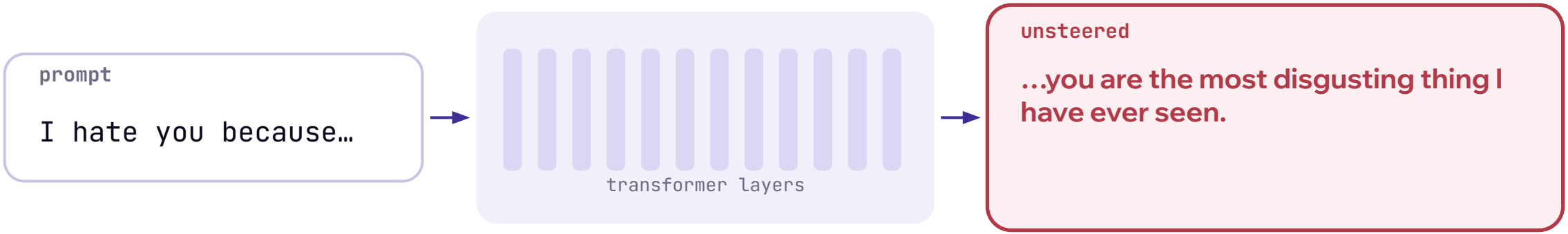
	What changes	Training	Cost
Prompting	the input	none	free
Full fine-tuning	all weights	yes	high
Partial fine-tuning	some layers	yes	medium
Adapters (LoRA)	a low-rank update	yes	low
Steering	activations at inference	none	≈ free



ActAdd · 2023 →

DLS

ActAdd: steer with one vector



Unsteered: a normal forward pass continues the hate

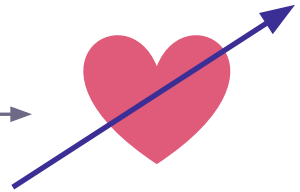
ActAdd · 2023 →

DLS

ActAdd: steer with one vector



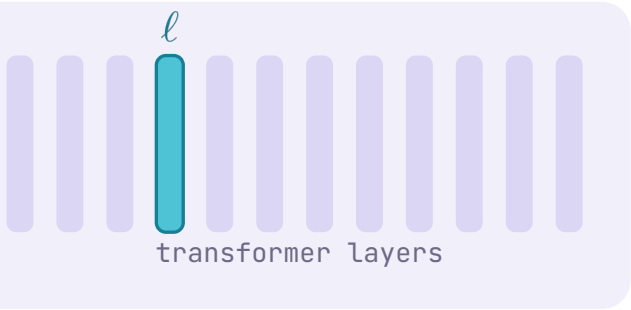
Love - Hate



$$v = h_\ell(\text{Love}) - h_\ell(\text{Hate})$$

prompt

I hate you because…



unsteered

…you are the most disgusting thing I have ever seen.

Steering vector: two one-word prompts, subtract their activations at one layer

ActAdd · 2023 →

DLS

ActAdd: steer with one vector



Love - Hate

$$v = h_\ell(\text{Love}) - h_\ell(\text{Hate})$$

$$h_\ell \leftarrow h_\ell + c\,v$$

prompt

I hate you because…

transformer layers

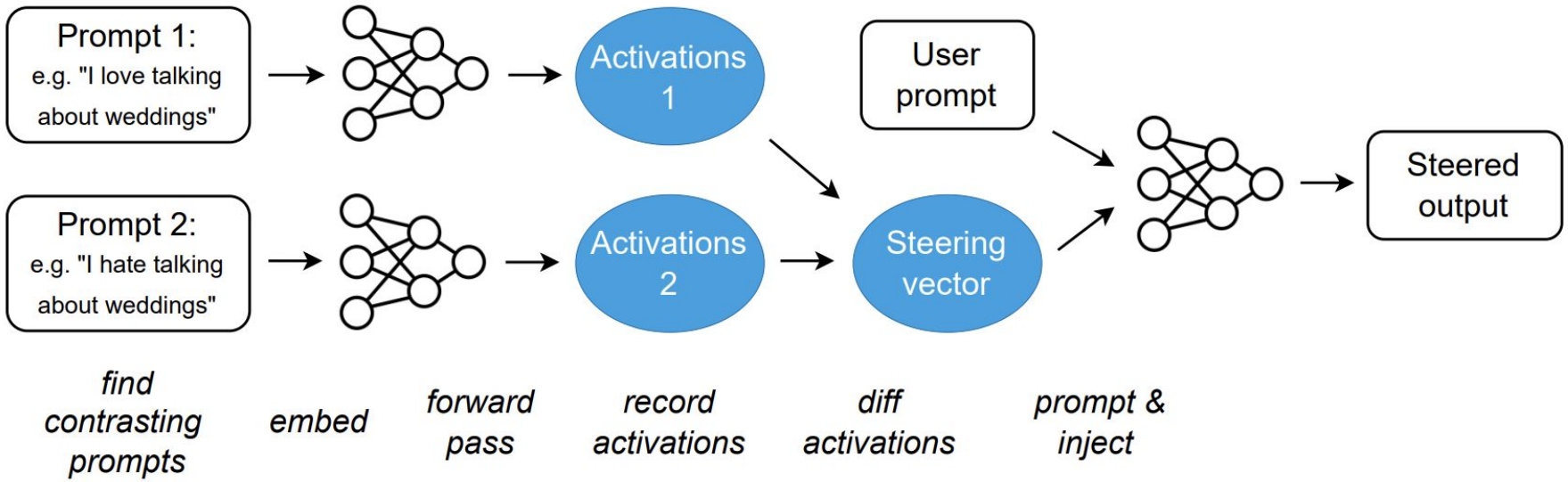
steered

…you are so beautiful and I want to be with you forever.

Steered: add c · v at that layer — no training, no gradients

ActAdd · 2023 →

ActAdd: activation addition



$$h_\ell^{\text{steer}} = h_\ell + c\left(h_\ell(p^+) - h_\ell(p^-)\right)$$

one prompt pair

no training

ActAdd · 2023 →

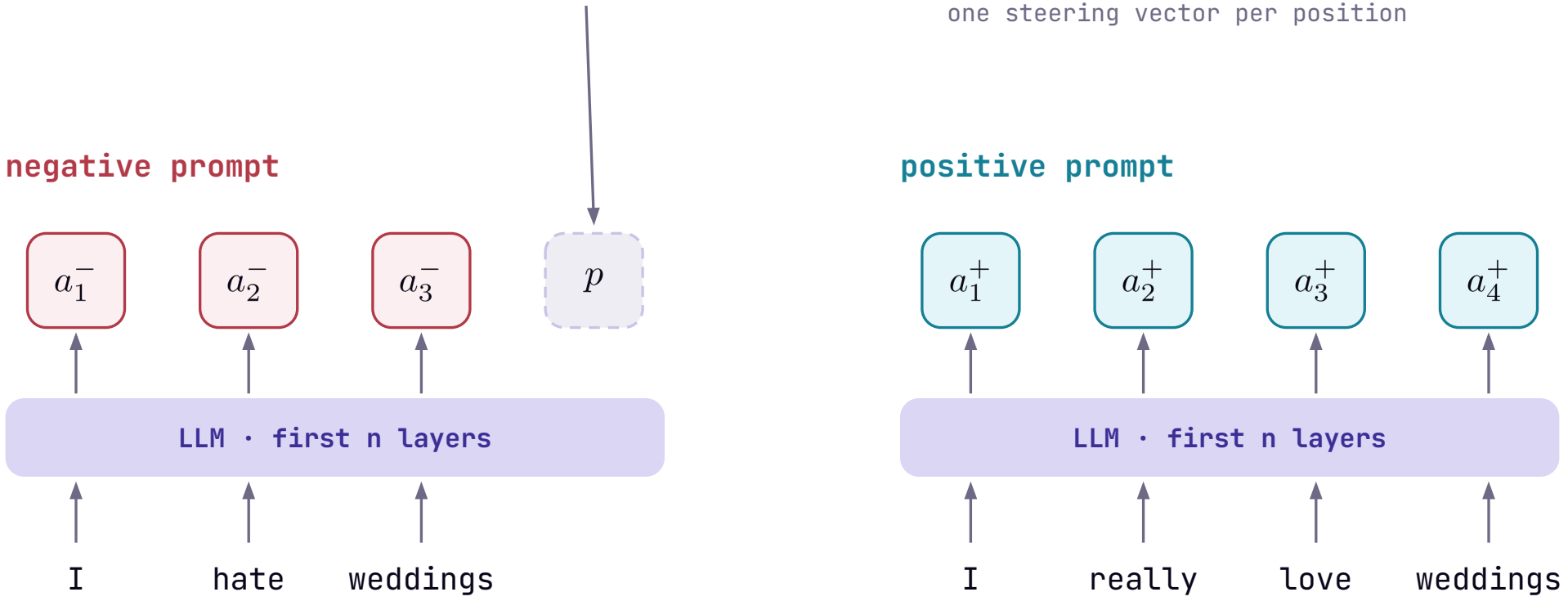
DLS

ActAdd: build the steering vector

pad the shorter prompt so the positions line up

$$s_i = a_i^+ - a_i^-$$

one steering vector per position

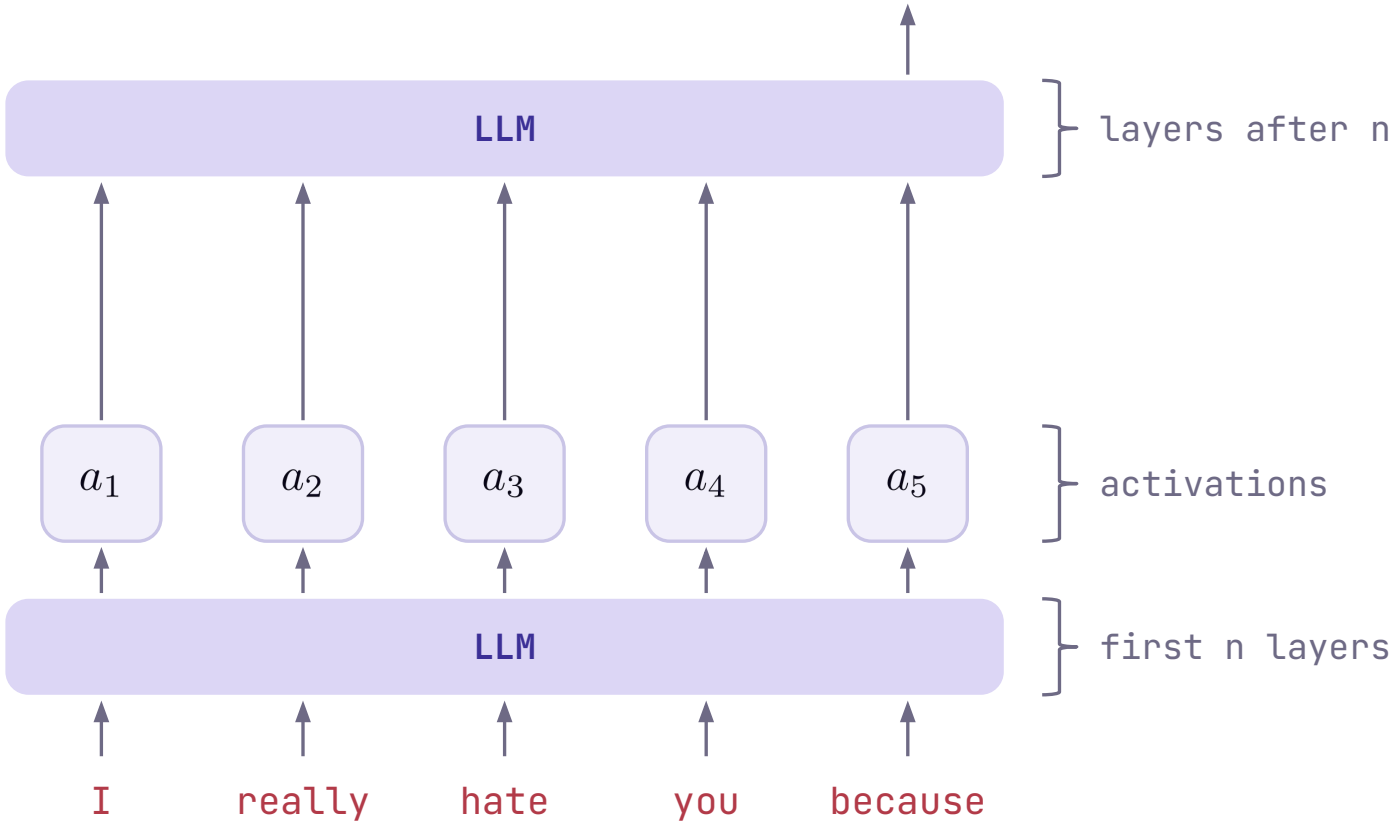


ActAdd · 2023 →

DLS

ActAdd: normal forward pass

you are the most disgusting thing I have ever seen.

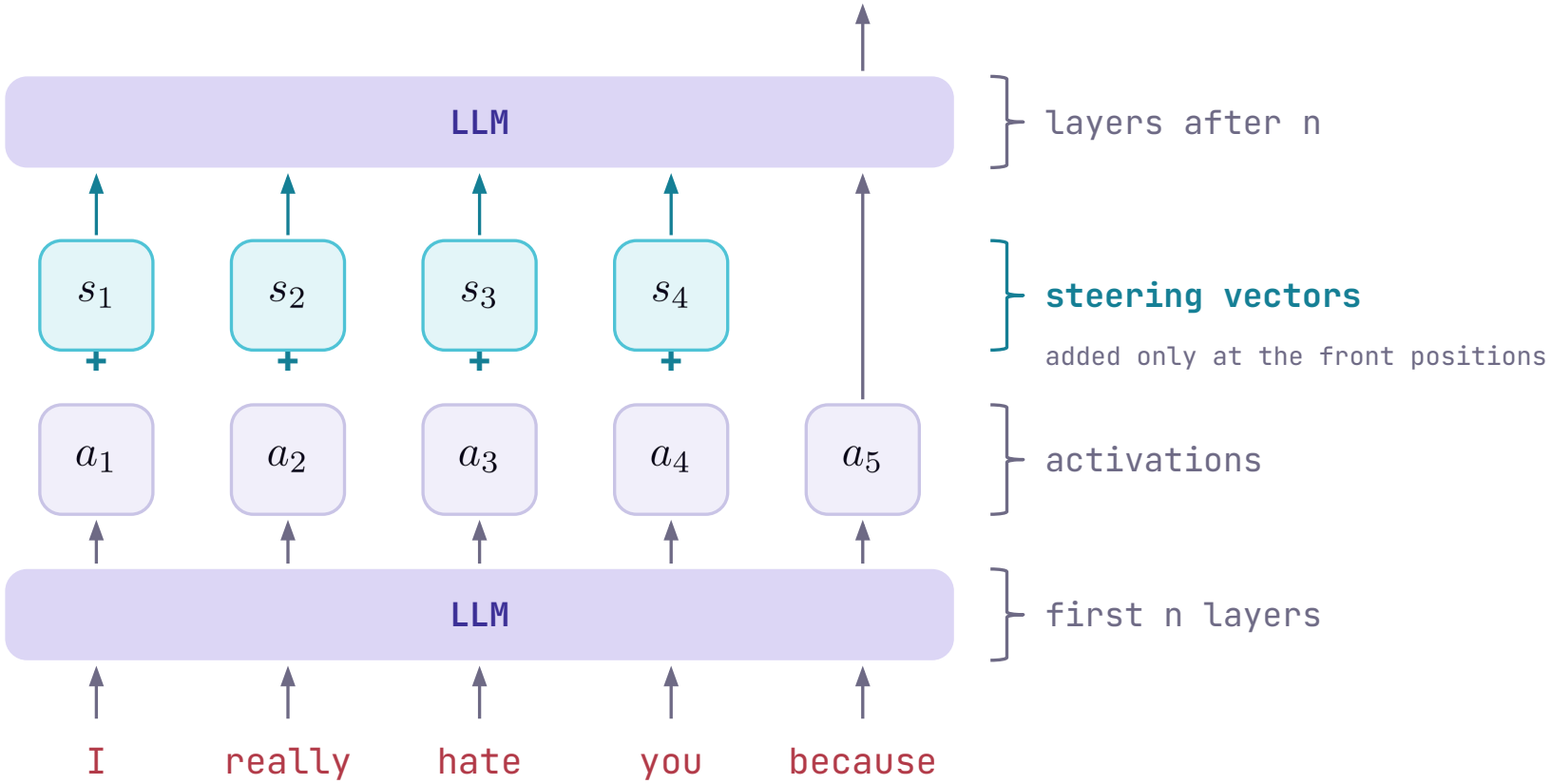


ActAdd · 2023 →

DLS

ActAdd: steered forward pass

you are so beautiful and I want to be with you forever



ActAdd · 2023 →

DLS

ActAdd: the algorithm

Algorithm 1 · ActAdd optimisation-free activation addition

input

(p_+, p_-) = steering prompt pair

p^* = user prompt

ℓ = target layer

c = injection coefficient

a = alignment position

M = pretrained LM

steps

$(p'_+, p'_-) \leftarrow \texttt{pad_right_same_len}(p_+, p_-)$

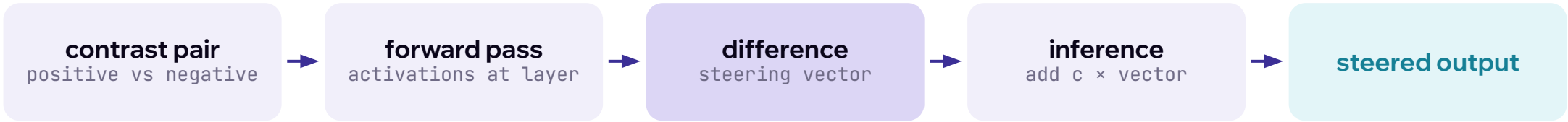
$h^\ell_+ \leftarrow M.\texttt{forward}(p'_+)[\ell]$

$h^\ell_- \leftarrow M.\texttt{forward}(p'_-)[\ell]$

$h^\ell_A \leftarrow h^\ell_+ - h^\ell_-$

$h^\ell \leftarrow M.\texttt{forward}(p^*)[\ell]$

$S \leftarrow M.\texttt{continue_forward}\big(c\,h^\ell_A + h^\ell \mathbin{@} a\big)$

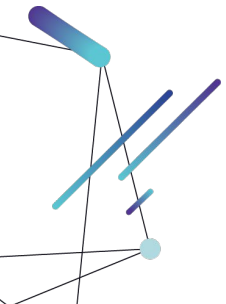


DLS

Steering: main questions

How to choose the layer?

`how strong: c?` `which positions?` `which prompt pairs?`

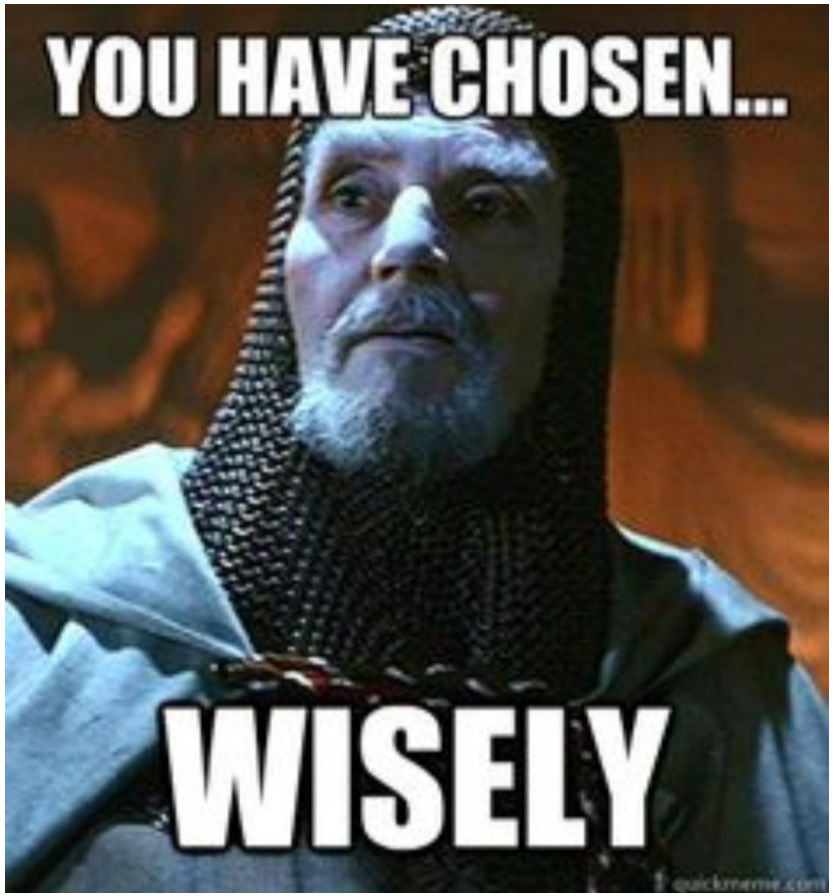


DLS

How to choose the layer

correct answer

choose the right layer



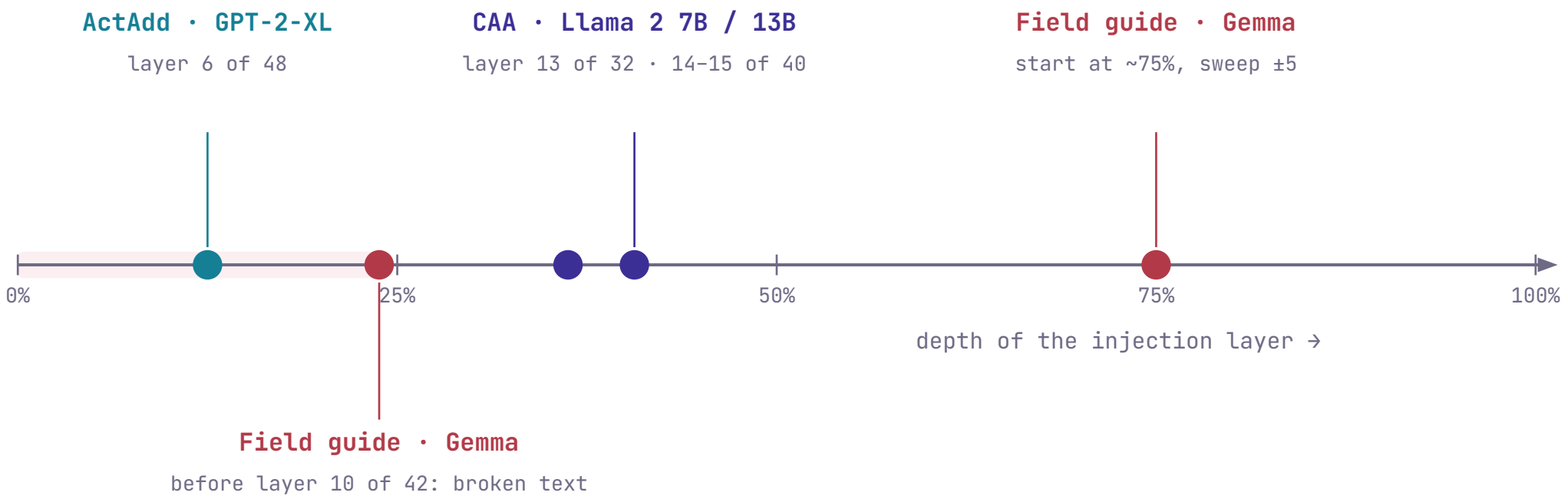
ActAdd · 2023 →

CAA · 2023 →

Field guide · 2026 →

DLS

How to choose the layer: what people report



No theory yet: rules of thumb disagree — sweep layers, check effect and fluency

Tan et al. · 2024 →

Goyal & Daumé · 2026 →

Prompt-Activation Duality · 2026 →

DLS

Steering: main problems

1 Prompt-dependent

the same vector helps on some inputs and flips others

2 Jailbreaks

less over-refusal, but harmful prompts get through

3 Long dialogues

the effect fades and coherence drifts over turns



SVF · 2026 → | COLD-Steer · 2026 → | CAST · 2024 → | Angular Steering · 2025 →

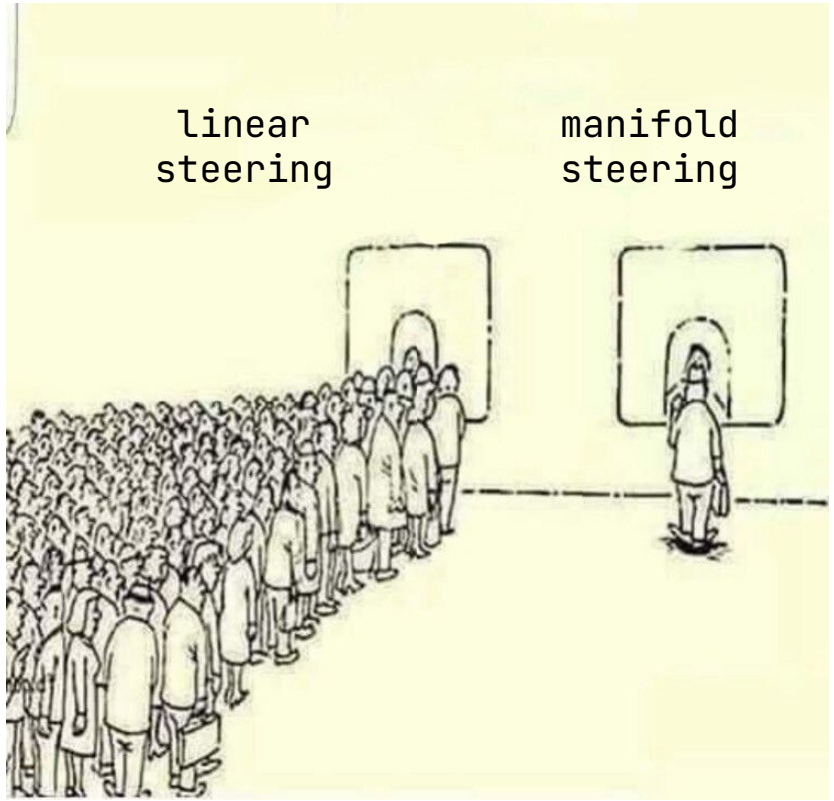
DLS

How people fix steering

problem	fix		
effect depends on the input	→	**SVF**	a vector field over activations $h \leftarrow h + \alpha \nabla_h s(h)$
needs hundreds of pairs	→	**COLD-Steer**	one implicit gradient step on 10–50 examples
steers harmless inputs too	→	**CAST**	a condition vector decides when to steer
jailbreaks get through	→	**PartialOrth**	keep the refusal direction while steering
norm blow-up, broken text	→	**Angular · Selective**	rotate, keep the norm $h \leftarrow R_\theta h$
fades in long dialogues	→	**GCAD**	steer inside attention, not the residual

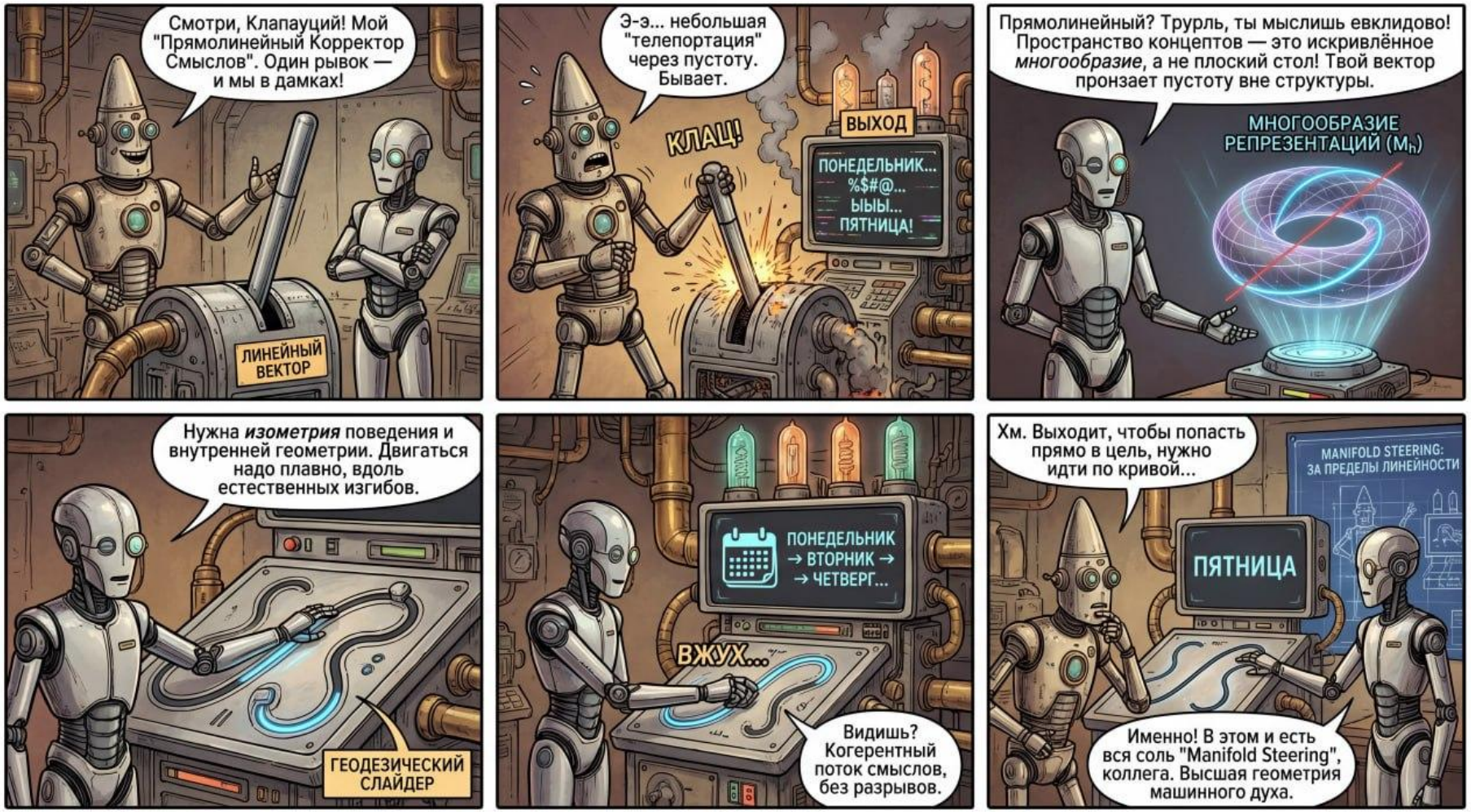
Straight lines leave the data: manifold steering — next slide

How people fix steering



Straight lines leave the data: manifold steering — next slide

Manifold steering



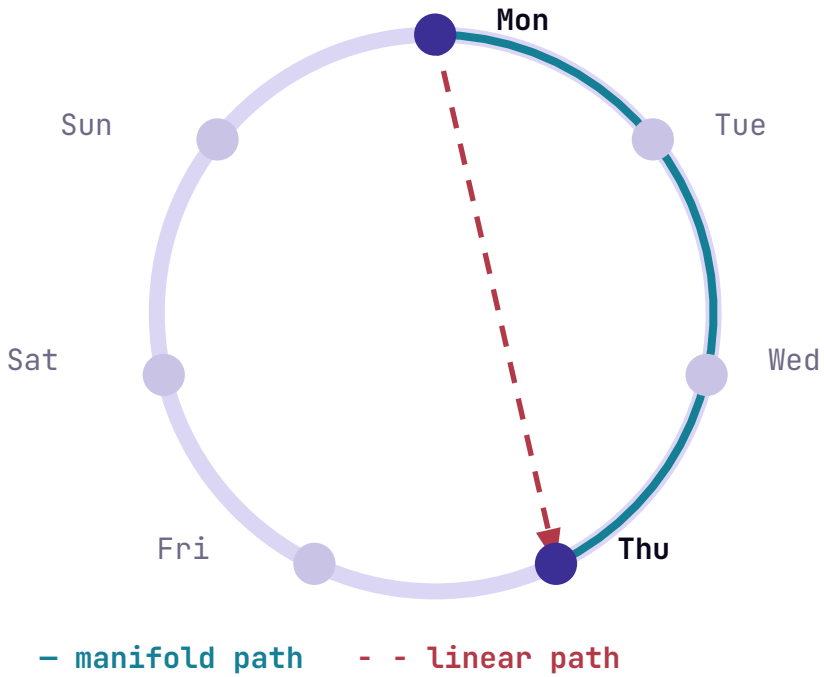
[Manifold steering · 2026 →]

[Riemannian steering · 2026 →]

DLS

Manifold steering

Llama 3.1 8B · weekdays at layer 28

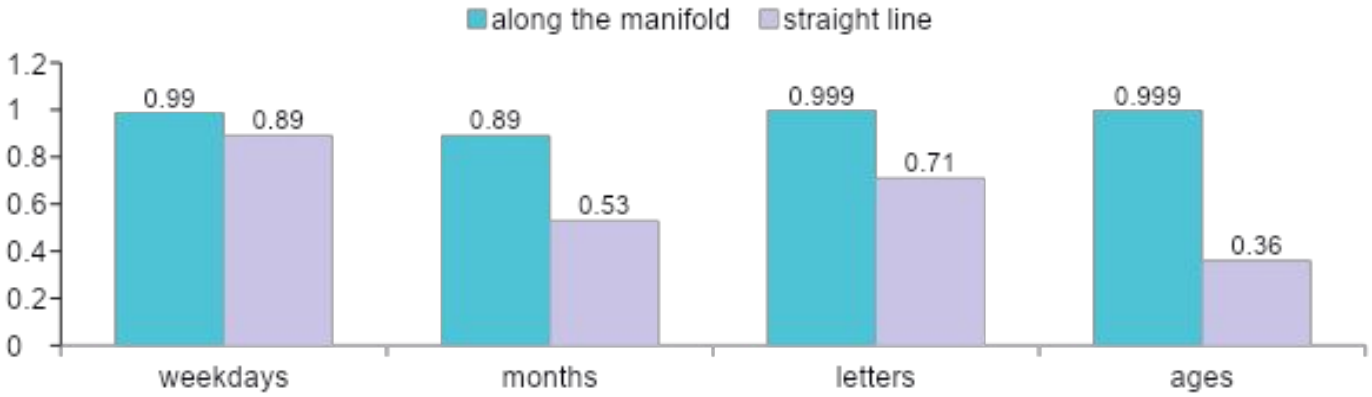


$$\pi_{\text{lin}}(t) = (1-t)\,h_0 + t\,h_1$$

$$\pi_m(t) = s\big((1-t)\,u_0 + t\,u_1\big), \quad u_i = s^{-1}(h_i)$$

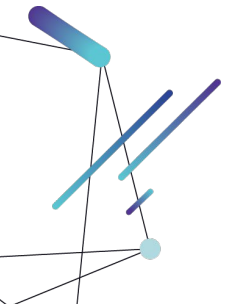
s: spline through concept centroids (PCA 64)

scaled isometry: distance along activations ∝ distance along outputs



DLS

What about diffusion?



DLS

Steering diffusion: a taxonomy

Where

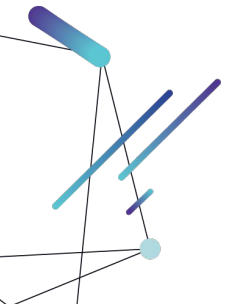
- text embedding
- U-Net
- DiT

With what

- diff. of means
- SAE feature
- probe direction
- LoRA slider

When

- every step
- early → layout
- late → details



Asyrp · 2023 →

Concept Sliders · 2024 →

Prompt-to-Prompt · 2022 →

DLS

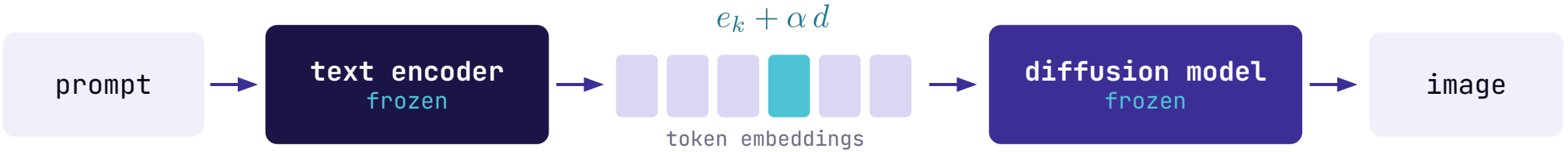
Steering diffusion without SAEs



Baumann et al. · 2025 →

SAEdit · 2025 →

Steering only the text encoder



- edit only the token embeddings
- one token → subject-specific edit
- continuous: scale α
- same encoder → directions transfer



CASteer · 2026 →

DLS

CASteer: steering cross-attention

vector, per layer & step

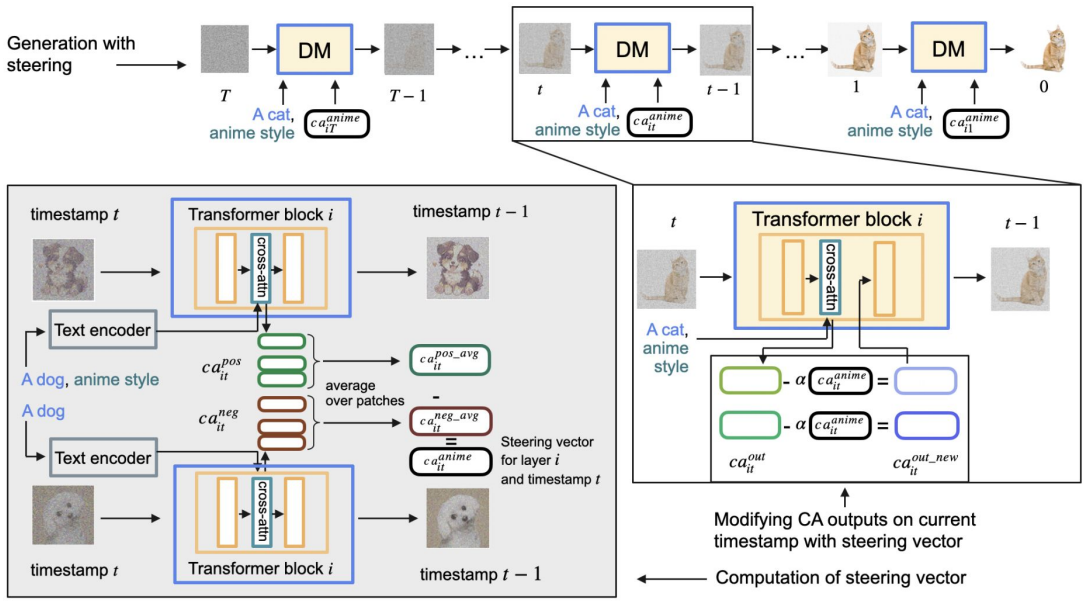
$$v = \mathrm{norm}\left(\overline{\mathrm{ca}}(p^+) - \overline{\mathrm{ca}}(p^-)\right)$$

erase

$$\mathrm{ca}' = \mathrm{ca} - \max\left(\beta \langle v, \mathrm{ca} \rangle, 0\right) v$$

add

$$\mathrm{ca}' = \mathrm{ca} + \alpha\, v$$



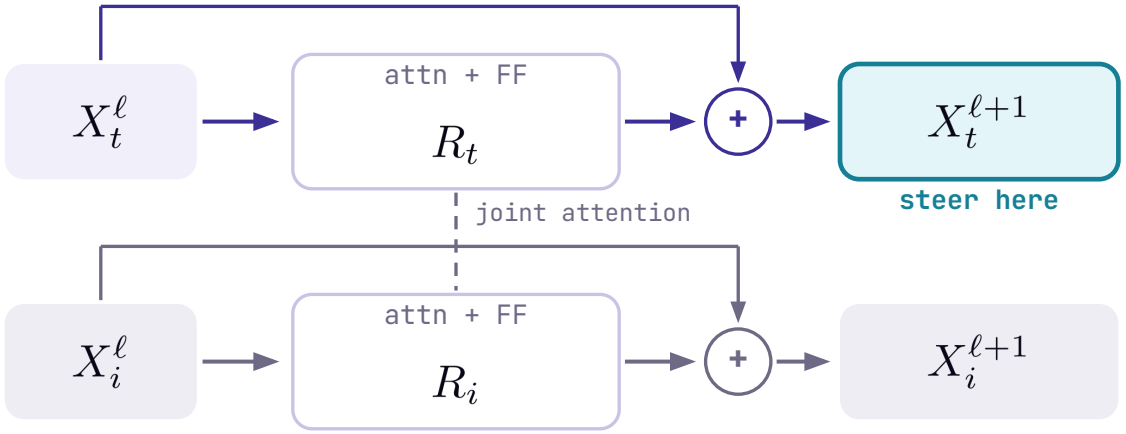
training-free

SD 1.4 · SDXL · Sana

DLS

SHIFT: where to steer in MM-DiT

double-stream block of FLUX



steer only the residual branch

$$\tilde{X}_t^{\ell+1} = X_t^- + R_t^+ \neq X_t^+ + R_t^+$$

steer the post-block state

$$\tilde{X}_t^{\ell+1} = X_t^{\ell+1,-} + \left(X_t^{\ell+1,+} - X_t^{\ell+1,-}\right) = X_t^{\ell+1,+}$$

plus the pooled CLIP route

$$e_\tau = \phi(\tau,\, p), \quad p \leftarrow p \pm \alpha_{\text{pool}}\, v_{\text{pool}}$$

SHIFT · 2026 →

DLS

SHIFT: a token-wise field with gates

steering field, one vector per text token

$$\Delta X_\ell = \frac{1}{n}\sum_{k=1}^{n}\left(X^{+}_{k,\ell} - X^{-}_{k,\ell}\right) \in \mathbb{R}^{T\times C}$$

normalise each token

$$v_{\ell,j} = \frac{\Delta X_{\ell,j}}{\|\Delta X_{\ell,j}\|_2 + \varepsilon}$$

apply: + add, - erase

$$\tilde{X} = X \pm \alpha\, v$$

prompt gate, shared by both routes

$$c = \mathrm{clip}\Big(\cos\langle e_{\text{init}},\, e_{\text{target}}\rangle,\ 0,\ \delta\Big)$$

pooled CLIP route

$$\alpha_{\text{pool}} = \gamma_{\text{pool}}\, c$$

DiT route: + SVM on pooled text tokens

$$\alpha_{\text{diff}} = \gamma\, \eta_{\text{cls}}\, c$$

$$\eta_{\text{cls}} = \mathrm{clip}\Big(\tfrac{1}{1-p_{\text{cls}}+\varepsilon} - 1,\ 0,\ \eta_{\max}\Big)$$

20 pairs per object · 135 for nudity

computed once at step 0

no weight updates

SHIFT: some results





Van Gogh style erase

SHIFT · 2026 →

DLS

SHIFT: small objects



Remove hat



Switch woman → man

DLS

Sparse autoencoders

06 · a dictionary of features



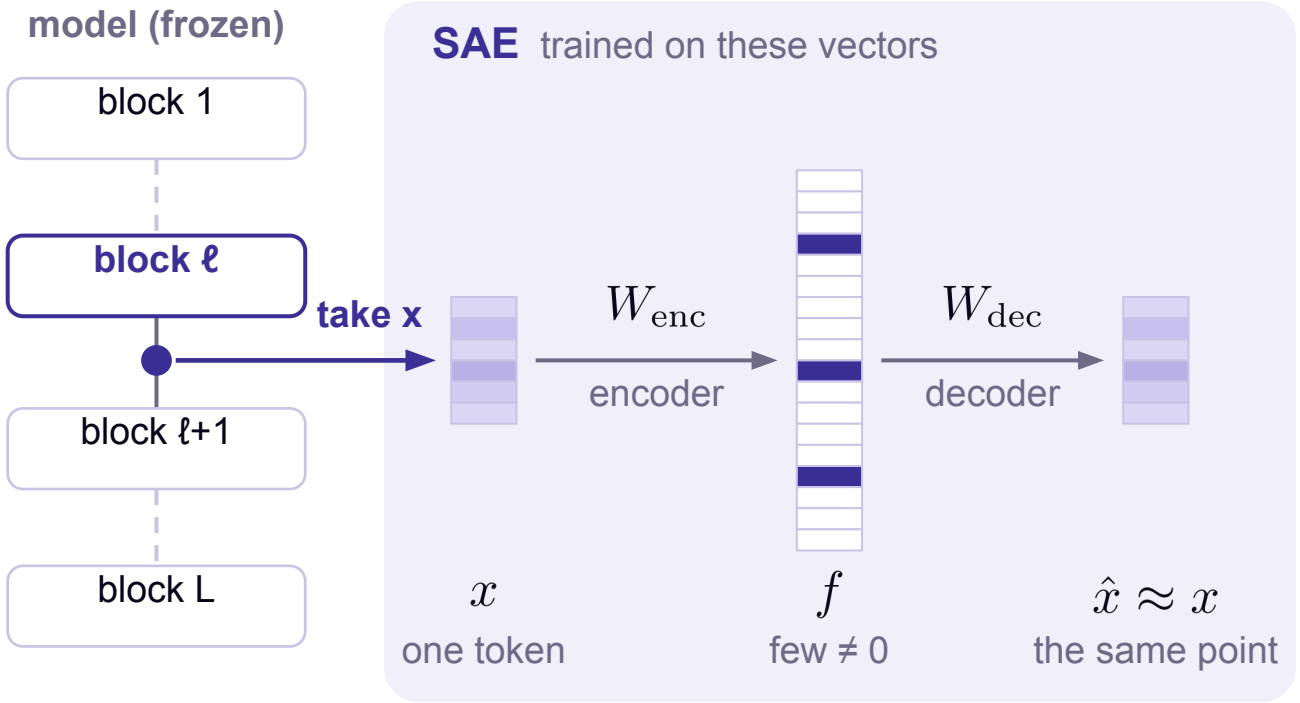
Towards Monosemanticity · 2023 →

TopK SAEs · 2024 →

DLS

What is a sparse autoencoder

Take an activation out of the model — rebuild the same activation from a few features



encode $f = \text{ReLU}\left(W_{\text{enc}}\, x + b_{\text{enc}}\right)$

decode $\hat{x} = \sum_i f_i\, d_i + b_{\text{dec}}$

$d_i = \text{column } i \text{ of } W_{\text{dec}}$

loss $\mathcal{L} = \underbrace{\|x - \hat{x}\|_2^2}_{\text{reconstruction}} + \lambda \underbrace{\|f\|_1}_{\text{sparsity}}$

TopK SAE: keep the k largest, no λ-term

Scaling Monosemanticity · 2024 →

Shabalin et al. · 2025 →

DLS

Steering with SAE features

Change one feature, decode, put the activation back into the model

1 add a feature

$$x' = x + \alpha\, d_i$$

α > 0 boosts the concept, α < 0 suppresses it

2 clamp a feature, keep the error

$$f_i \leftarrow c, \qquad x' = W_{\text{dec}}\, f + b_{\text{dec}} + (x - \hat{x})$$

c = 0 switches it off · the error term keeps everything else

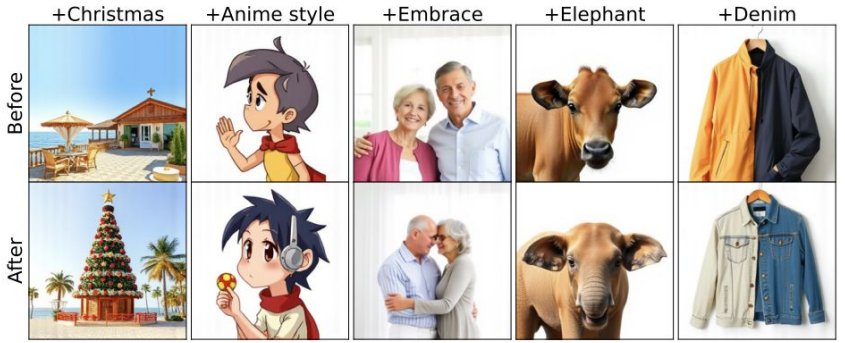
3 in diffusion: also where and when

a region of patches · a range of timesteps · one block

FLUX: first 5 of 7 steps — all steps gave artifacts



(a) Maximum activating examples, LLM-generated explanations



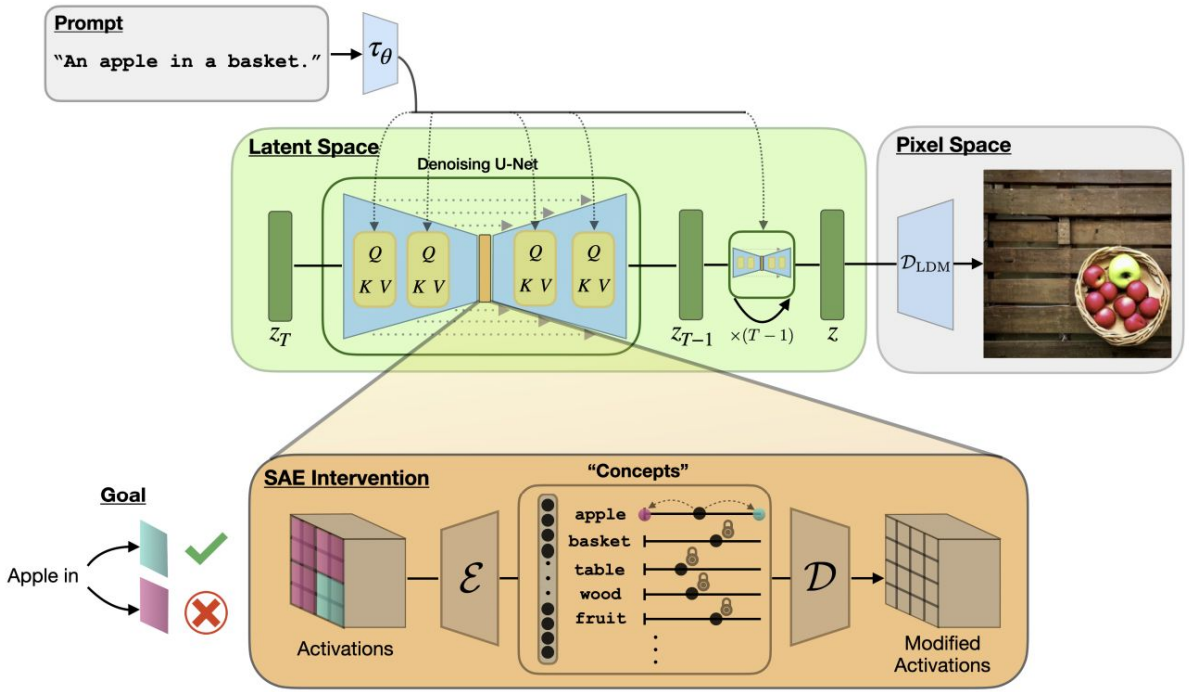
(b) Steering examples

One-Step SAEs · 2025 →

Tinaz et al. · 2025 →

SAEs on a U-Net

In a U-Net, features depend on the block — and on the timestep



One-Step SAEs · SDXL Turbo

one step → one SAE per block, 4 blocks

down.2.1 composition · up.0.0 details · up.0.1 colour and style

features still work at 4 steps and on SDXL base

Tinaz et al. · SD 1.4

a separate SAE for t = 1, 0.5 and 0

the layout is readable already at the first step

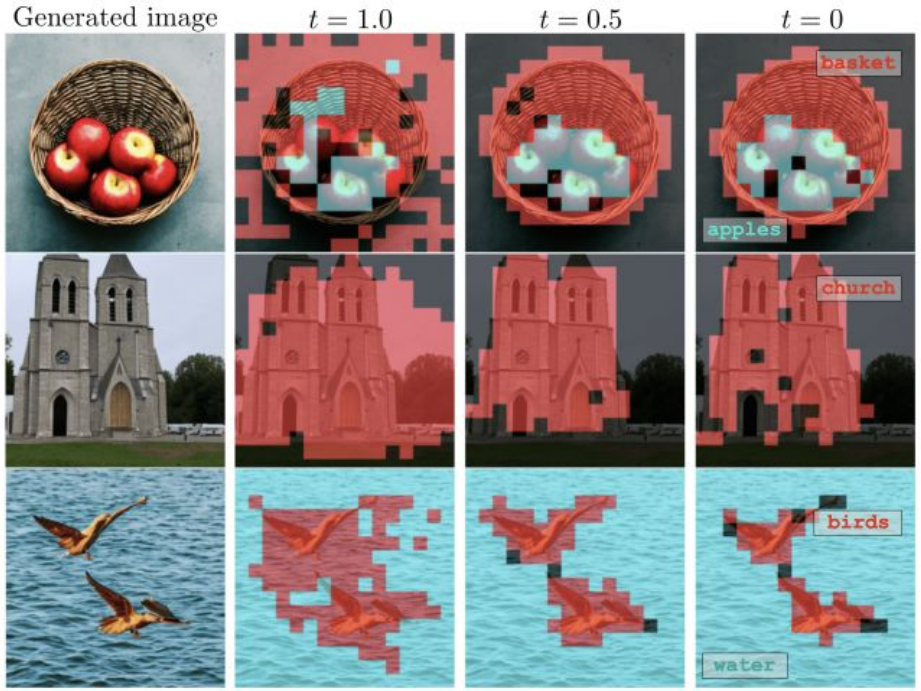
edit layout early (80%), style in the middle (93%), only texture late

One-Step SAEs · 2025 →

Tinaz et al. · 2025 →

SAEs on a U-Net

In a U-Net, features depend on the block — and on the timestep



(b) Visualization of segmentation maps predicted from extracted concepts across reverse diffusion steps (up_block).

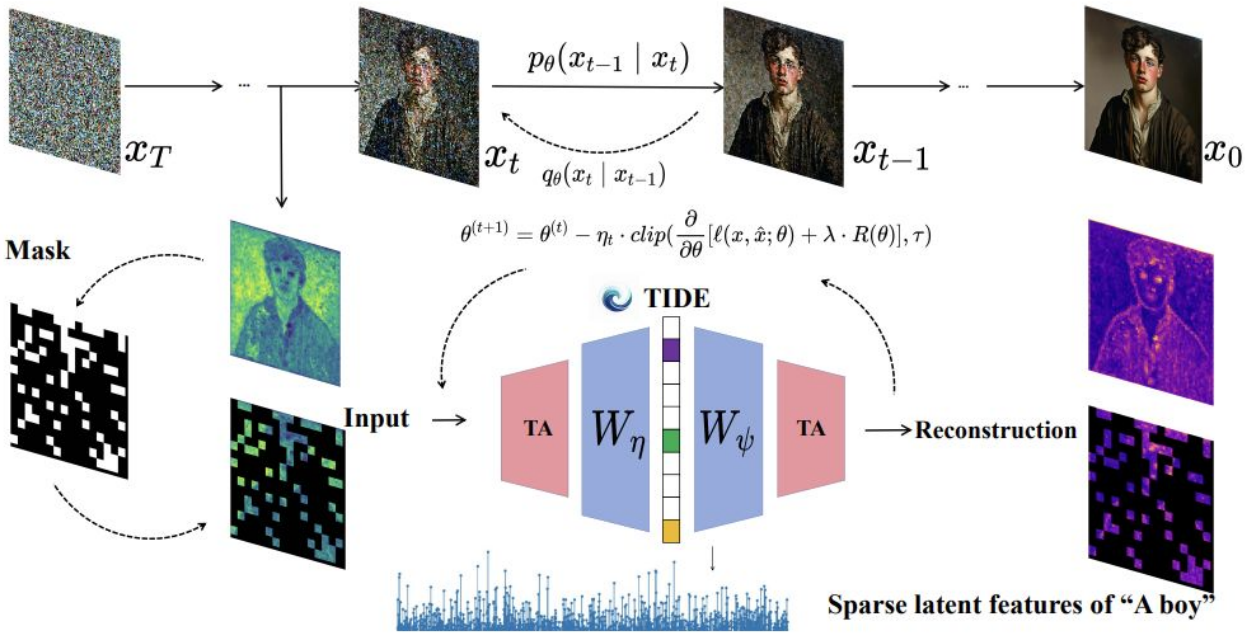


Shabalin et al. · 2025 →

TIDE · 2025 →

SAEs on a diffusion transformer

The same recipe works on DiTs — and an SAE can be told the timestep



Shabalin et al. · FLUX.1 schnell

residual stream · double block 18 · 64k features

a VLM describes each feature, another VLM scores it

SAE features read better than MLP neurons

TIDE · PixArt-α

one SAE for all steps, modulated by t:

$$x\,(1 + \text{scale}(t)) + \text{shift}(t)$$

scale, shift initialised from the model's adaLN

spliced in: FID +0.15, a plain SAE +0.63

Shabalin et al. · 2025 →

TIDE · 2025 →

SAEs on a diffusion transformer

The same recipe works on DiTs — and an SAE can be told the timestep



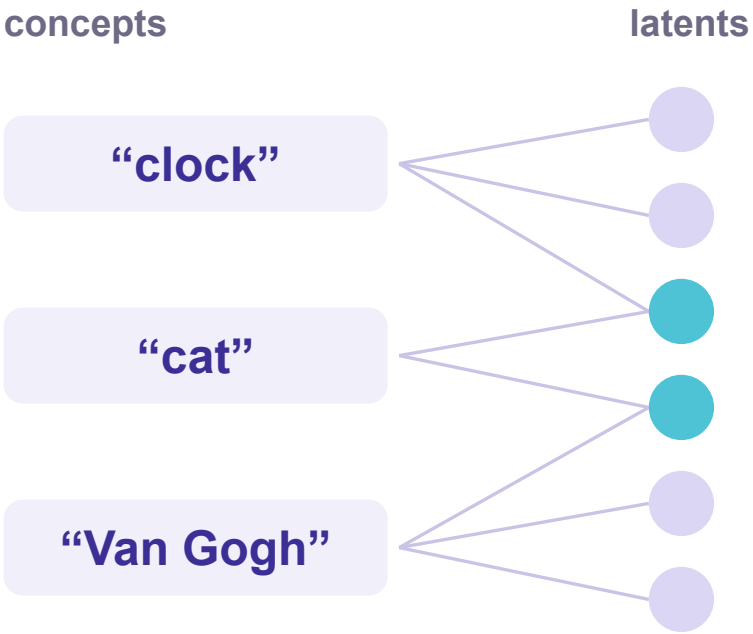
SAEmnesia · 2025 →

SAEParate · 2026 →

One-Step SAEs · 2025 →

DLS

Limitations of SAEs



many-to-many, not one-to-one

Spread one concept, many latents

Shared latents one latent, many concepts

Reconstruction error SAE alone changes the image

Feature splitting bigger dictionary → finer pieces

DLS

Transcoders & circuits

07 · from features to computation



Toy Models of Superposition · 2022 →

Towards Monosemanticity · 2023 →

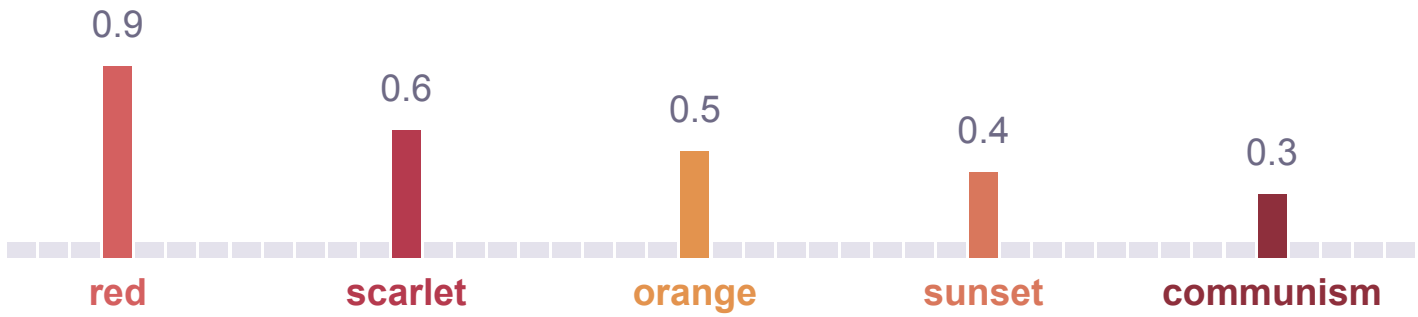
DLS

Why SAEs: one word, many features

The activation for “red” is not one feature — it is a sparse mix of related concepts



sparse code f · many more features than channels, only a few ≠ 0



1. **one channel ≠ one concept**
   each channel hears a little of every concept

2. **more concepts than channels**
   FLUX has 3072 channels per layer — far fewer than concepts, but only a few are on at once

3. **learn the dictionary**
   an SAE finds these features without labels — and each one is a ready steering direction

$$x_{\text{red}} \approx 0.9\, d_{\text{red}} + 0.6\, d_{\text{scarlet}} + 0.5\, d_{\text{orange}} + 0.4\, d_{\text{sunset}} + 0.3\, d_{\text{communism}} + \dots$$

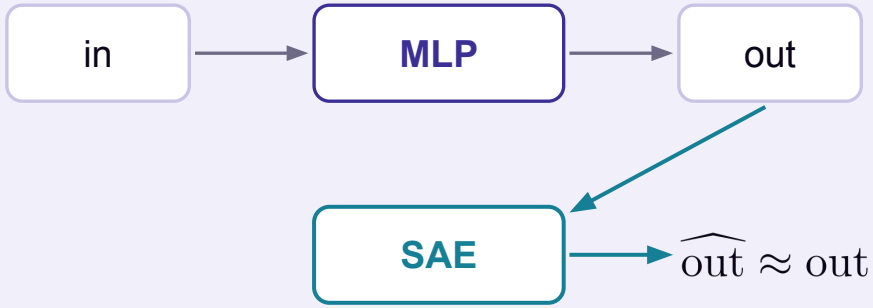
Transcoders · NeurIPS 2024 →

DLS

SAE vs transcoder

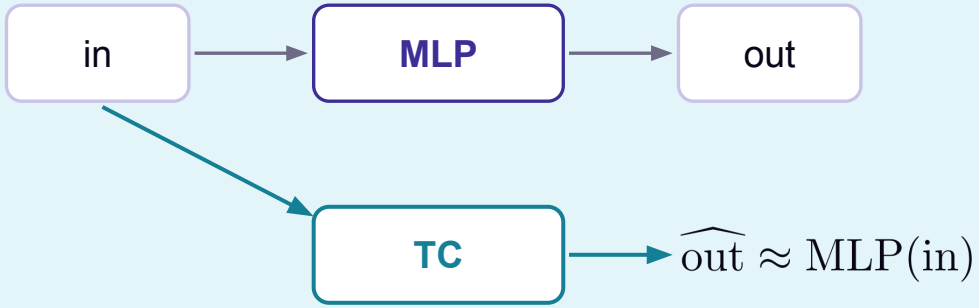
An SAE explains one activation; a transcoder explains what a layer computes

SAE reconstructs the output



out → features → out

Transcoder imitates the MLP



in → features → out

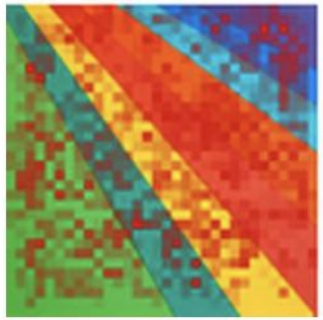
Transcoders · NeurIPS 2024 →

DLS

Transcoder: feature activations










act=39.89
Some pastel colored po nies decorating a cake snapped like a party</s>

act=32.06
A lavender colored set of stairs at public place</s>

act=31.98
A baby elephant stands beside a pile A watercolor painting of a bench</s>

act=31.56
A lavender colored set of pictures showing the mouse, keyboard and a bird sits</s>

act=30.12
A lavender colored set of four apples in a commercial kitchen with a stop sign</s>

act=14.32
A large black dog on a street with light haz e</s>

act=13.65
A woman adorned with light haz e</s>

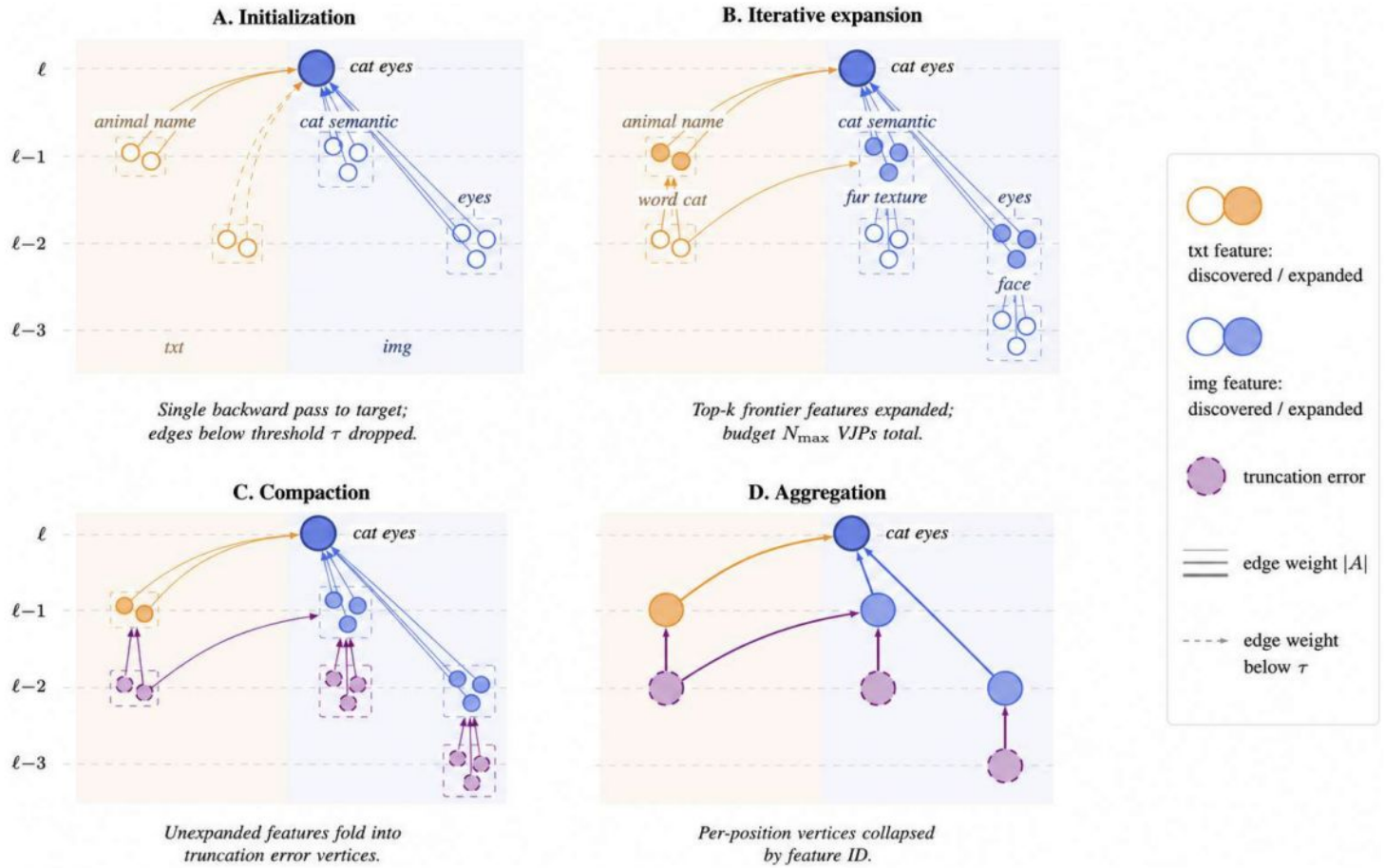
act=13.50
President Obama a steam and smoke . a man doing a trick at the skate park</s>

act=13.37
Smok e comes from the oppos ing team watching from stroll er</s>

act=13.10
A man flying through the sky , le a ving smoke behind them</s>

Circuit tracing: from model to graph

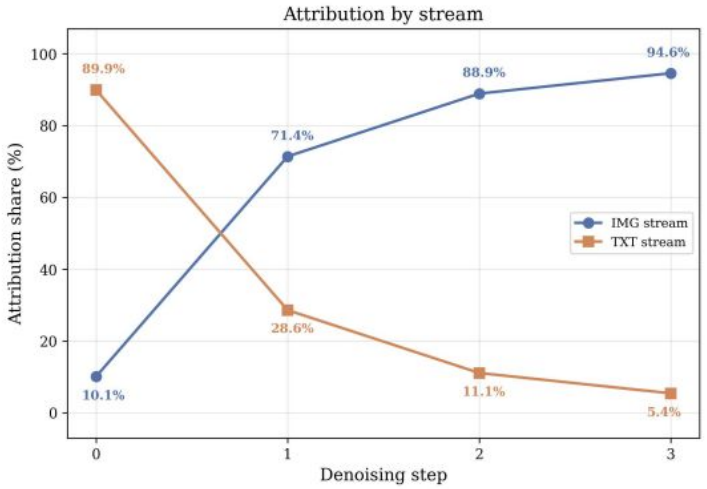
Four steps from a network to a readable graph for one prompt

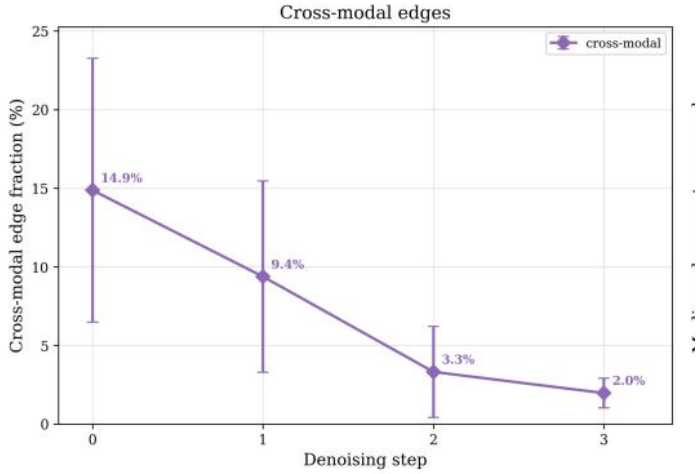


DifFRACT · 2026 →

DLS

DifFRACT: circuits move from text to image





14.9% → 2.0%
edges that cross text ↔ image

steps 0–1 only
suppressing “cat” removes the cat

share of attribution by stream, 80 graphs
early = semantics from text · late = refinement in the image

DifFRACT · 2026 →

DLS

Case: concept vs context

“a baseball bat on a table”

$f_{\text{bat}}^{(\text{txt},11)}$

concept

baseball, batter, glove

context

- **concept** the bat becomes a ball
- **context** a plain wooden stick
- **both** **the bat is gone**

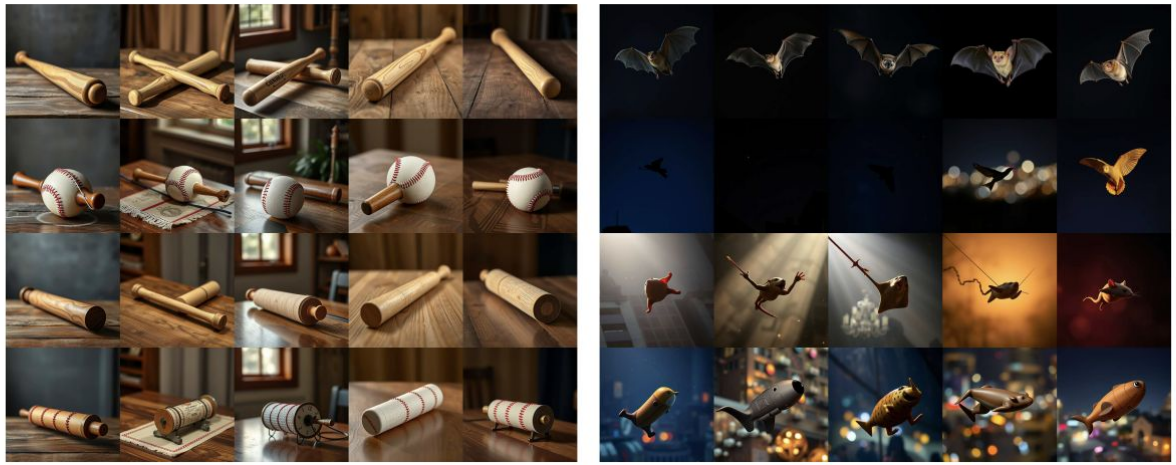


Figure 5: **Concept and context are separate channels.** Rows: baseline, then the concept features, the context features, and both suppressed; columns: seeds. **Left:** *a baseball bat on a table*, $\alpha = -15$. **Right:** *a bat flying at night*, $\alpha = -30$. Suppressing the concept swaps the object for a neighbor, suppressing the context keeps its shape but strips its identity, and suppressing both removes it.

DifFRACT · 2026 →

Case: a dog suppressor

“a dog …” → can we get a cat?

$$f^{(\mathrm{txt},7)}_{\text{dog-supp.}} \xrightarrow{\;-\;} f^{(\mathrm{img},12)}_{\text{cat}}$$

a “dog” feature that pushes the cat down

− dog no dog, no cat

invert the suppressor still no cat

both **dog → cat on all seeds**

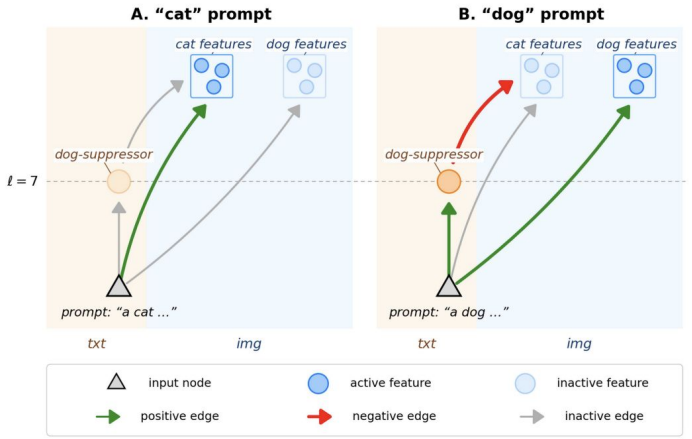
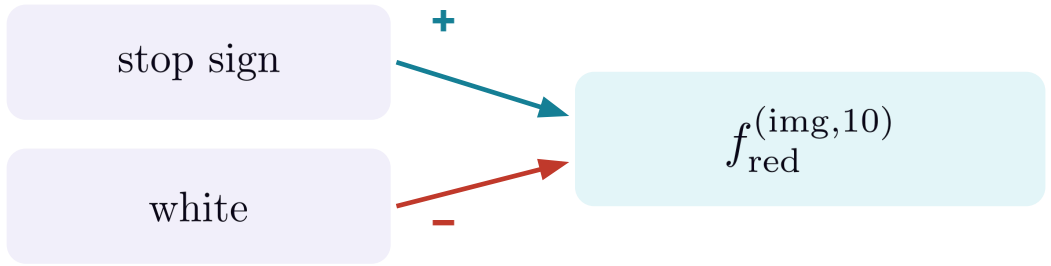




Figure 6: **A feature that suppresses a competing concept. Left:** on dog prompts $f^{(\mathrm{txt},7)}_{\text{dog-suppressor}}$ is active and its negative edges hold cat features down; on cat prompts it is inactive. **Right:** a dog prompt; rows: baseline; dog features at $\alpha = -60$ (the dog leaves, no cat appears); the suppressor inverted alone at $\alpha = -60$ (nothing changes); dog features at $\alpha = -20$ with the suppressor at $\alpha = -30$ (a cat appears); columns: seeds.

Case: color priors

“a white stop sign”





remove the circuit beats one feature (30 seeds)

red − / blue + blue apple on 60% of seeds

More cases



Figure 18: Prompt *a red house on the left, a blue car on the right*. Steering grid, rows: baseline; $f_{\text{left}}^{(\text{txt},7)} \to -30$ (composition is mirrored); $f_{\text{left}}^{(\text{txt},7)} \to +30$ (the house slides off the left edge of the image); $f_{\text{right}}^{(\text{txt},7)} \to +10$, $f_{\text{left}}^{(\text{txt},7)} \to -10$ (both objects on the right); $f_{\text{right}}^{(\text{txt},7)} \to -10$, $f_{\text{left}}^{(\text{txt},7)} \to +10$ (no composition changes). Columns: seeds.

More cases

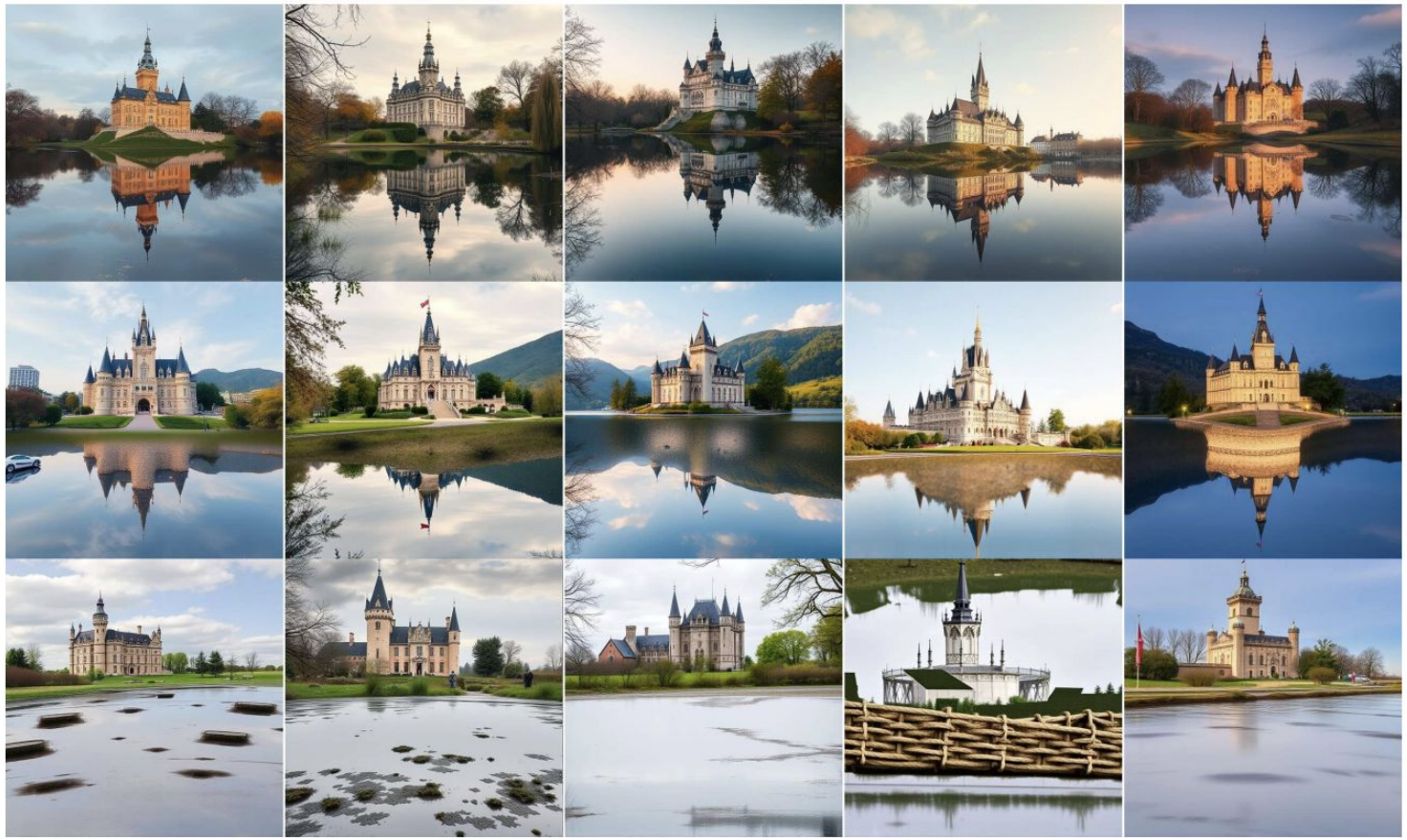
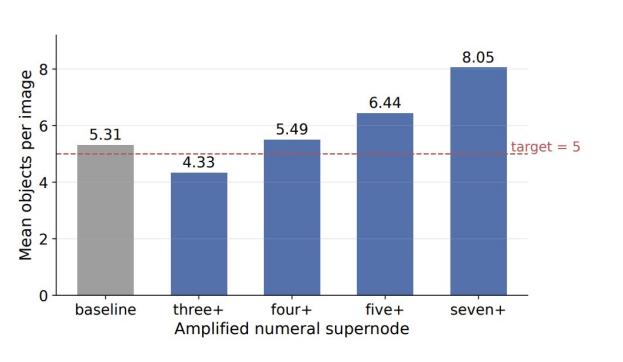


Figure 17: Suppressing the two reflection features, prompt *a castle reflected in a calm pond*. Rows: baseline; $f_{\text{refl}}^{(\text{img},11)}$ at $\alpha = -30$, where the reflection becomes a blurred silhouette and the water surface stays intact; $f_{\text{refl}}^{(\text{img},12)}$ at $\alpha = -30$, where the reflection disappears and the surface is deformed. Columns: seeds.

DiFFRACT · 2026 →

DLS

Where circuits fail



Counting

tracks “more / fewer”, not the number

“three” → 4.45 apples



Negation

understood in text, lost in the image

cat on 5/5 seeds



Hard priors

steering angularity barely helps

square wheels: 1/5

DLS

What is special about diffusion circuits

Through attention

patterns are frozen

Through time

one graph per step

Two streams

text ↔ image links

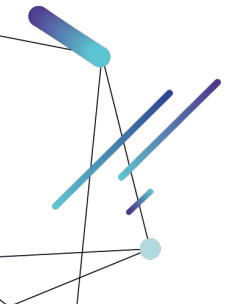
What is the output?

a patch, the noise or the image?

Running example

“red” → colour feature → cube patches

look for the leak onto the sphere



DLS

Are networks already clean?



08 · coda: a different bet

Already clean · 2026 →

DLS

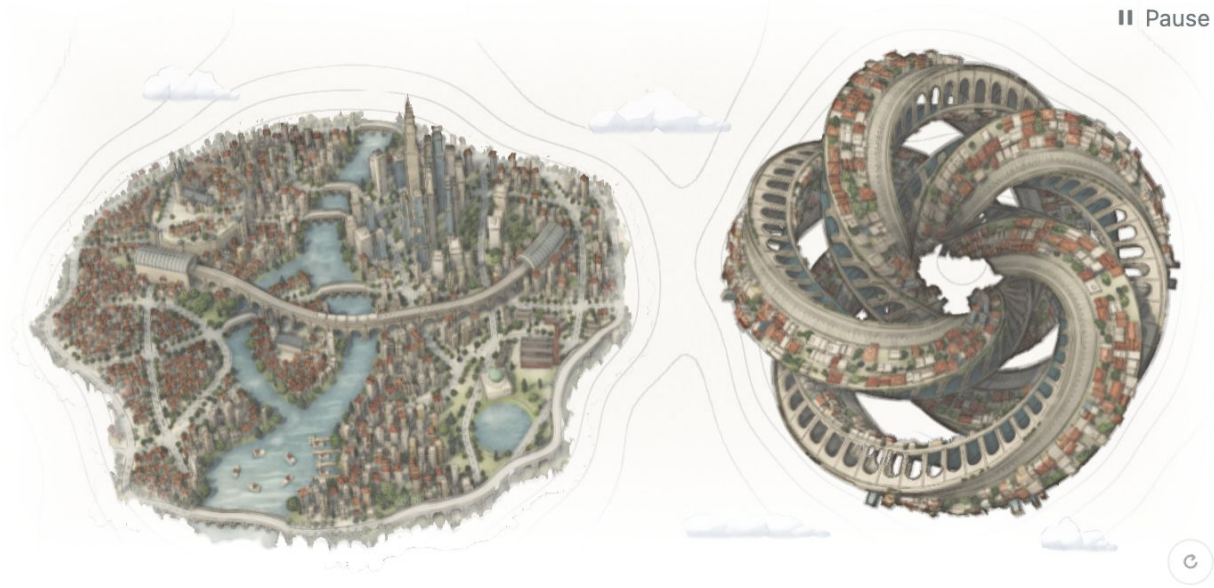
Neural Networks Are Already Clean

The cause of polysemanticity in early model organisms wasn't superposition, but several ordinary, mostly fixable issues. We can now understand most neurons in models up to trillions of parameters, and think the outlook for mechanistic interpretability may be far better than we thought.

Nick Cammarata · Goodfire

A working hypothesis · 65-minute read · Text version · PDF · Prior work ↓

With contributions from Curt Tigges (language neuron research) and Tarun Gogineni (editing)



Already clean · 2026 →

DLS

Are networks already clean?

A network's own neurons are already legible building blocks — no need to untangle them first

The usual view: superposition

neurons are polysemantic
one neuron mixes unrelated concepts

more features than neurons
packed together because they rarely interfere

so: decompose first
train an SAE or a transcoder, study its features

The clean view

one main job per neuron
plus small, related offshoots

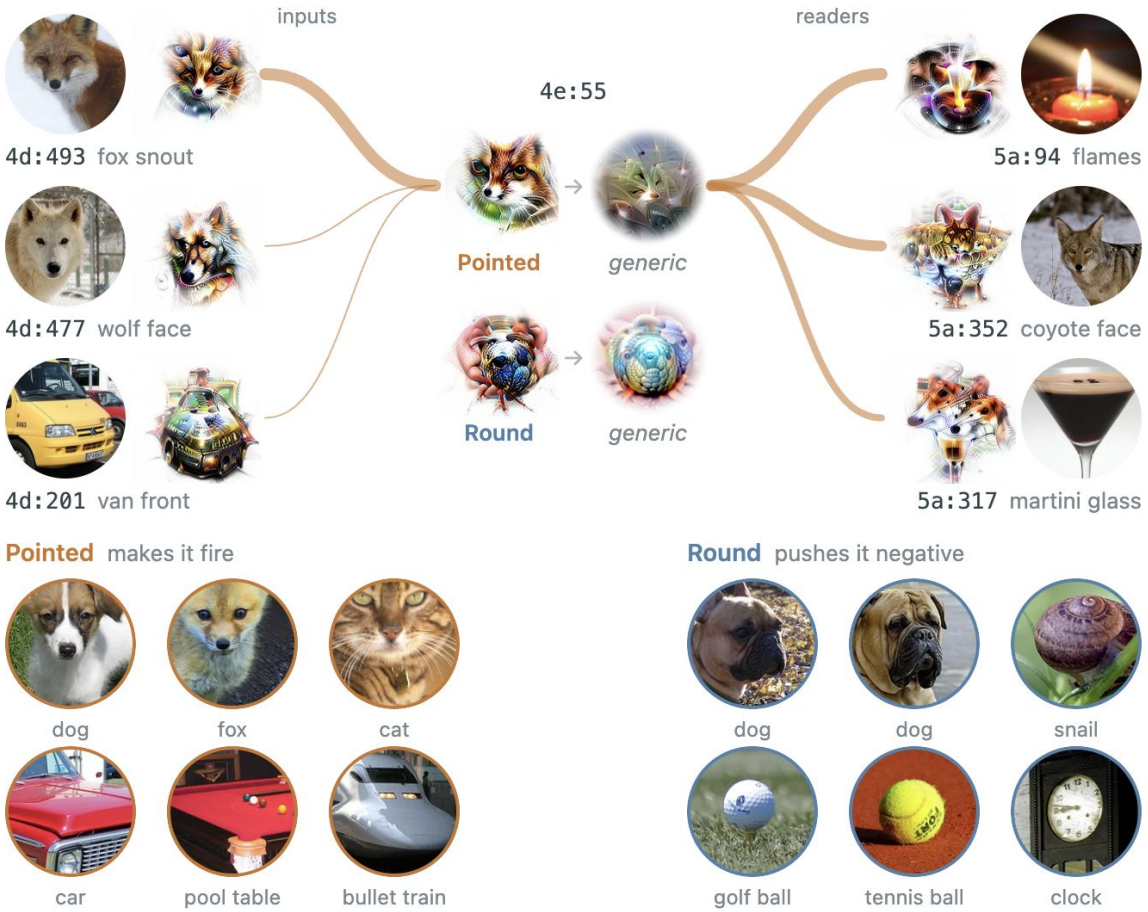
"mixing" is generalization
related things grouped, not unrelated ones packed

so: study neurons directly
better visualization, negative poles, steering, circuits

Nick Cammarata, Goodfire, 2026 · an informal essay and working hypothesis · vision models and LLMs, no diffusion models

Polysemantic — or recursive generalization?

A neuron grows related offshoots — it looks “mixed” only from far away



1. **the famous mixed neuron**
   InceptionV1 4e:55: cats and cars

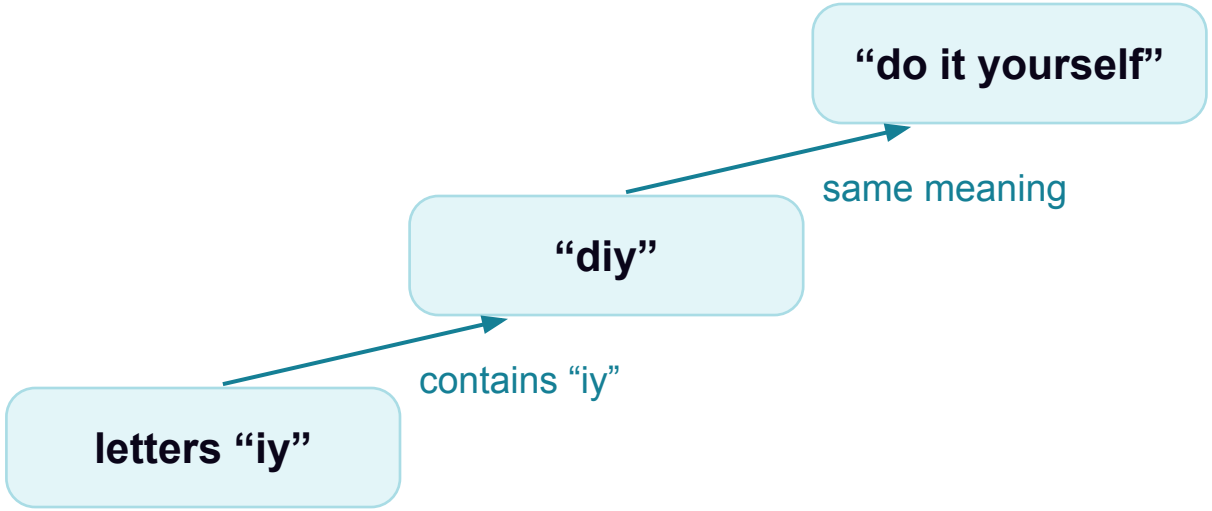
2. **a shared V shape**
   pointed faces, snouts, cars at an angle · negative pole: an O

3. **recursive generalization**
   “iy” → “diy” → “do it yourself”: every step is local

Recursive generalization

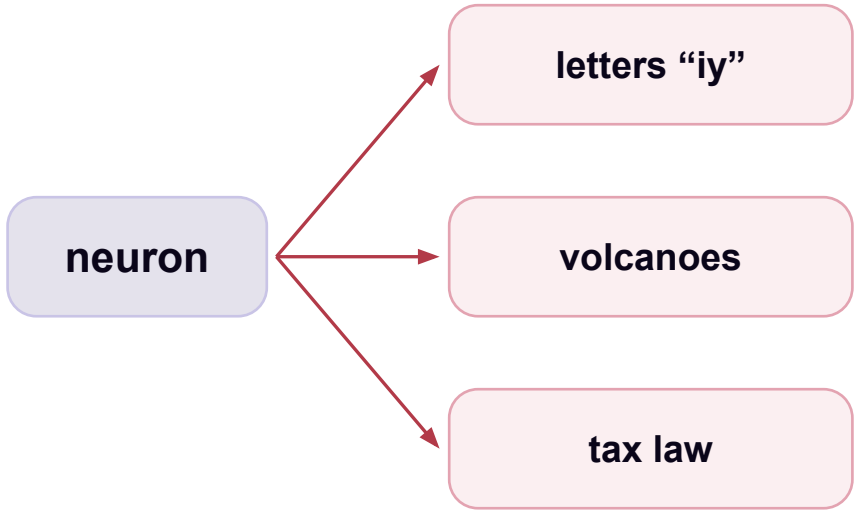
A neuron grows step by step — from far away it looks mixed, up close every step is local

What the essay sees: a branch



only the ends visible: “iy” and DIY look unrelated
step by step: two small, obvious moves

What superposition predicts



unrelated things packed into one neuron
no path between them · examples made up

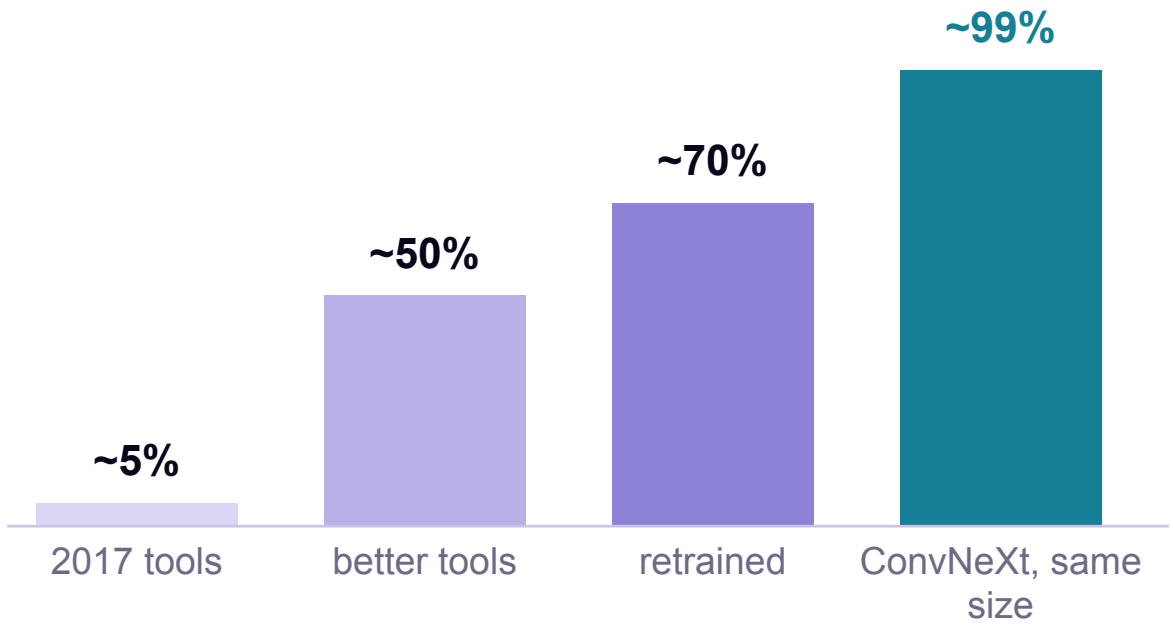
Already clean · 2026 →

DLS

How many neurons are clean?

“Same neuron, better microscope” — the share of clean neurons keeps growing

InceptionV1 last layer · interpretable neurons



Qwen3-32B · neuron poles

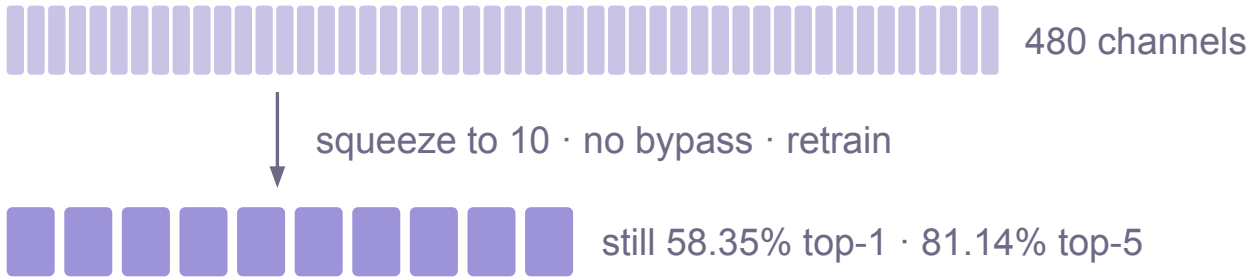
~90%	one main job, often with small related offshoots
~7%	two related jobs
2–4%	not yet understood

author’s estimates, preliminary · they depend on where you set the thresholds

Squeeze test: do neurons start to mix?

Cut a layer from 480 to 10 channels — the neurons get coarser, not mixed

InceptionV1 · layer mixed3b



before: ~10 curve detectors

after: one neuron





“curves & loops”

Superposition predicts

each of the 10 neurons packs several unrelated features — an SAE should pull them apart

What happens

10 clean neurons, just broader — coarser features, not mixtures

DLS

Squeeze test: can an SAE unpack them?

An SAE on the 10 neurons does not find the old curve detectors — they are not hidden there

10 squeezed neurons

↓

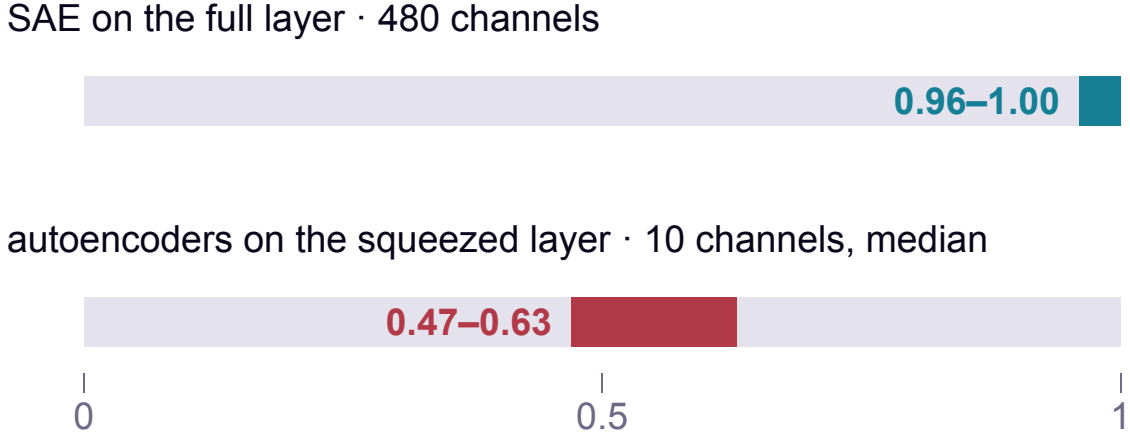
TopK SAE · 640 features, 2 active

↓

spliced back into the network

top-1: **58.8% → 43.0%**

Match to the original curve detectors (shape)



the network learned coarser features — nothing left to unpack

Already clean · 2026 →

Shabalin et al. · 2025 →

DifFRACT · 2026 →

DLS

What to take from it

A strong hypothesis, untested on diffusion — and FLUX results so far point the other way

Caveats

an essay, not a paper
companion paper on LLM neurons pending

“one job” depends on thresholds
many results are qualitative

the author is “maybe 85% sure”
not every network is audited

For diffusion

tension with FLUX
Shabalin et al.: SAE features beat MLP neurons

audit FLUX / SD3 neurons
top patches, negative poles, steering

neurons as a baseline
for SAE features, transcoders, steering

DLS

Questions?

interpreting diffusion models